Dealer Hedging and Gamma Exposure

How options market makers hedge, what their hedging does to prices, and how to read it from either side of the trade

Abstract

Options dealers hedge mechanically. Their inventory, not their opinion, sets what they must buy or sell after a price move. This chapter asks when that forced trading moves prices, and by how much.

A dealer who sells an option trades the underlying to remove its directional exposure. The option's delta changes as the price moves, so the dealer must re-hedge. Gamma measures that rate of change. A dealer who is short gamma buys after a rise and sells after a fall, which amplifies moves. A dealer who is long gamma does the reverse and damps them. A continuously hedged option earns gamma times the gap between realized and implied variance. At 20% implied volatility the break-even daily move is about 1.05%.

Options are in zero net supply, so gamma summed over all holders is zero. A claim that "the market is short gamma" is therefore a claim about who holds each side and which side hedges. Open-interest data does not record who is long. Every published gamma-exposure estimate rests on a positioning assumption, and a wrong assumption can flip the sign of the estimate. In the chapter's worked example, the estimate changes from plus 3.9 million dollars per 1% move to minus 63 million.

The direction of the hedging effect is established. Its size for broad indices is contested and appears modest. The one study that used measured dealer positions found at most 3.3 points added to S&P 500 annualized daily volatility over 2020–2023. Part of the reported link between calm markets and positive gamma runs in reverse. The best-documented effect is pinning: near expiry, single-stock prices move toward strikes with large open interest. Popular accounts hold up less well. In GameStop, market makers bought calls, which contradicts the gamma-squeeze account. In August 2024, dealers affected prices through wider quotes and thinner liquidity, not through hedging flows.

The chapter recommends using dealer gamma as a conditioning variable for volatility and intraday reversals, never as a directional signal. To be useful, it must improve on a volatility forecast that already uses implied volatility. Its main practical value is defensive: it informs position sizing and warns against supplying liquidity while dealers are forced to trade in the same direction as the move.

What this is. A from-first-principles tutorial on the business of making markets in options, on the dynamic hedging that business requires, and on the claim — now repeated daily in market commentary — that the hedging of options dealers moves the underlying market. "Dealers are short gamma", "we crossed the gamma flip", "the pin at 6,000 should hold into expiry", "vanna and charm will support the market after the Fed": these sentences compress a long chain of reasoning, and each link in the chain can be stated precisely, checked, and in places shown to be wrong. This document builds the chain one link at a time, starting from what an option is.

Who it is for. A reader who is mathematically comfortable and technically strong, but not an options specialist, and who wants to *use* this material: to build a gamma-exposure estimate, to test whether it predicts anything, or to reason about their own positions when the market is moved by someone else's hedging. Nothing is assumed about options beyond the word. Every basic concept — bid and ask, call and put, open interest, delta — is defined before it is used, and every formula is preceded by what it means and followed by a number.

How to read it. The document has four parts, and they are cumulative.

- **Part I (§1–§3) — foundations.** What a market maker does and how it earns money; what an option is; how an option is priced by replication, and the sensitivities (the "Greeks") that fall out of the price. If you already trade options, skim §1.1, §2.5 and §3.9 — the rest will be familiar.
- **Part II (§4–§5) — hedging.** How a dealer neutralises the direction of an option position, the single identity that says what a hedged option earns, and the full quadrant of worked examples: a customer buying or selling a call or a put, and exactly what the dealer on the other side then does as the price moves.
- **Part III (§6–§8) — from one book to the market.** How individual positions aggregate into a market-wide gamma exposure, why that aggregate cannot be observed and must be inferred, what the hedging does to prices (the feedback multiplier, the gamma flip, expiry, pinning, charm and vanna, 0DTE options, squeezes), and what the historical episodes do and do not show.
- **Part IV (§9–§12) — using it.** How dealers use this information, how traders on the other side use it, how to compute and validate a gamma-exposure signal without fooling yourself, and a synthesis.

§13 is the reference list, grouped by kind. Appendix A collects every concept the main text leans on without fully explaining, ordered so that it reads as a build-up.

If you read four things, read **§4.4** (the identity that explains why dealers hedge and what it costs them), **§5.7–§5.8** (the quadrant on one page, and why call-versus-put does not matter for gamma), **§6.3** (why nobody actually knows the sign of dealer gamma), and **§7.1–§7.3** (the feedback multiplier and how big it realistically is).

Relationship to the other notes. Simple and Log Returns covers the return conventions used for volatility here. Trend-Following in Financial Markets derives the convexity of trend rules; §5.10 shows that a trend follower and a short-gamma options dealer place the same trades for opposite reasons. Market Regimes and Hidden Markov Models is the right frame for treating "the dealer gamma regime" as a conditioning variable in §10 and §11. Each note stands alone.

Epistemic tags. Claims are flagged by status where the status changes what you should do:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement among people who have looked.
- **[Contested]** — documented, but with live disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. May well be right; the evidence is private or absent.

Untagged sentences are definitions, derivations, or arithmetic. Results from simulations I ran while writing are labelled **[Simulated]**; the generating code is committed alongside this document in `figures/dh_*.py`, and each script prints the numbers quoted in the text, so you can change my parameters and rerun.

Notation. The **underlying** is the asset an option is written on — a stock, an exchange-traded fund, an index, or a futures contract. S (or S_t) is its price, and a one-period **return** is $\Delta S/S$. An option has **strike** K and **expiry** T; $\tau = T - t$ is the time remaining, always measured in years. r is the risk-free interest rate and q the underlying's dividend yield; to keep the arithmetic clean, every worked example sets $r = q = 0$ unless it says otherwise.

σ is **volatility**: the annualised standard deviation of log returns (§3.1). Two versions must be kept apart. σ_i is **implied** volatility, the number that reproduces an option's market price (§3.7); σ_r is **realised** volatility, the number the price path actually delivers. Volatility is quoted in percent, and a "vol point" is one percentage point of σ.

V is the value of an option, C a call and P a put. $N(\cdot)$ is the standard normal cumulative distribution function and $\varphi(\cdot)$ its density. d_1 and d_2 are the Black–Scholes quantities of §3.3.

The option **Greeks** are partial derivatives of V, per option and per unit of the underlying:

$$\Delta = \frac{\partial V}{\partial S}, \qquad \Gamma = \frac{\partial^2 V}{\partial S^2}, \qquad \Theta = \frac{\partial V}{\partial t}, \qquad \nu = \frac{\partial V}{\partial \sigma}, \qquad \text{vanna} = \frac{\partial \Delta}{\partial \sigma}, \qquad \text{charm} = \frac{\partial \Delta}{\partial t}.$$

Θ and charm differentiate with respect to calendar time t moving forward, so $\partial/\partial t = -\partial/\partial \tau$. ν is vega; the Greek letter nu stands in because "vega" is not a Greek letter.

M is the **contract multiplier** — the number of units of the underlying one contract controls, 100 for US equity and index options. n is a signed position in contracts (positive is long); n^D is the position held by **dealers**, the hedging intermediaries of §1. OI is open interest (§2.5).

At the market level, GEX(S) is **dealer gamma exposure** in dollars per 1% move: the dollar value of the underlying that dealers must sell (if positive) or buy (if negative) to stay hedged after a 1% rise in the price, as defined in §6.1. D is **market depth** in the same units — the dollars of trading that move the price by 1% — and $\kappa = \text{GEX}/D$ is the dimensionless **feedback strength** of §7.1. Π is the profit and loss (P&L) of a hedged position, Δt a hedging interval, and N_h the number of rehedges over an option's life.

One collision is held apart by typography throughout: Δ alone is the Greek, while ΔS and Δt are changes in price and in time.

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Appendix

- A. Concepts and prerequisites — every idea the main text leans on without fully explaining, built from first principles and ordered by dependency.

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1. Market makers and dealers

1.1 The wrong intuition

The most common mental model of a market maker is a casino. You buy a call option; somebody sold it to you; that somebody is "the house", and the house is betting that the stock will not go up. If the stock rallies, the house loses and you win. On this view, the options market is a set of bets between customers and dealers, and a dealer with a large book of sold calls is a large bet against a rally.

Almost every part of that picture is wrong, and the way it is wrong is the subject of this document.

A market maker does not want your bet. What it wants is to be paid for *immediacy* — for being there, with a price, at the moment you want to trade — and then to get rid of the risk that trading with you created. Within seconds of selling you a call, a well-run options market maker has bought shares of the underlying stock (or futures on it) in the amount that makes its position indifferent to small moves in the price. It no longer cares, to first order, whether the stock goes up or down. What it now cares about is how *much* the stock moves, how its position changes as it does, and how much it will cost to keep re-neutralising that position until the option expires.

That re-neutralising is **dynamic hedging**, and it is not optional: it is the price of being in the business without being a gambler. It is also mechanical. A dealer's position, the price, the time to expiry and the volatility together determine how many shares it must hold; when any of them changes, the dealer must trade. Because the trades are forced by arithmetic rather than by opinion, and because dealers as a group can hold very large option positions, the trades can be large enough to move the market that is being hedged. That is the whole idea behind "gamma exposure". Everything after this section is about making it precise.

Two corrections to the casino picture follow immediately and are worth stating before anything else.

1. **Dealers do not "want" the price to go anywhere.** Commentary routinely says that dealers "need" the market to fall, or "defend" a strike. A hedged dealer has no directional preference to defend. Its hedging trades have predictable *side effects* on prices, and those side effects can look like intent. They are not.
2. **The option contracts themselves do nothing to prices.** Options are side bets in zero net supply (§2.5): for every contract someone holds long, someone else holds it short. What can move the underlying is the *hedging* — and only the hedging done by one side that the other side does not offset.

1.2 What a market maker sells: immediacy

Start with the simplest market maker, one that trades a single stock and holds no options.

At any moment it posts two prices. The **bid** is the price at which it will buy; the **ask** (or **offer**) is the price at which it will sell. The ask is higher than the bid, and the gap between them is the **bid–ask spread**. The average of the two is the **midpoint** or **mid**, which is the usual working estimate of what the stock is "really" worth right now. A quote of 99.98 bid, 100.02 offered is written 99.98 / 100.02; the mid is 100.00 and the spread is 4 cents, or 4 basis points (a **basis point**, "bp", is one hundredth of a percent).

A customer who wants to sell immediately sells at the bid; one who wants to buy immediately pays the ask. If one of each arrives, the market maker buys 1,000 shares at 99.98, sells 1,000 shares at 100.02, and has earned $40 without ever holding a view on the stock. That is the business: **capture the spread, many times a day, carrying as little risk in between as possible.**

The two customers did not have to arrive at the same moment, and that gap is where both of a market maker's problems live.

Problem one: inventory risk. Suppose the seller arrives first. The market maker now owns 1,000 shares that it did not want, and it owns them until a buyer turns up. If the price falls by 30 cents in the meantime, the $40 spread becomes a $260 loss. A market maker's holding of the thing it makes a market in is its **inventory**, and the risk of holding it is **inventory risk**. Ho and Stoll (1981) showed the natural defence: a dealer holding unwanted inventory shades *both* quotes downward, making its bid less attractive (so fewer people sell to it) and its offer more attractive (so more people buy from it), until the inventory is worked off. A market maker's quotes are therefore not only a price; they are an inventory-management tool. Avellaneda and Stoikov (2008) turned the same idea into the model most electronic market makers start from, in which the quotes are centred not on the mid but on a **reservation price** that moves against the inventory in proportion to the dealer's risk aversion, the variance of the price, and the time left in the trading session.

Problem two: adverse selection. Suppose instead the seller knew something — an earnings miss about to be announced — and the price drops to 99.50 a minute later. The market maker bought at 99.98 from someone who knew the stock was worth less. Trading with better-informed counterparties is **adverse selection**, and it is the reason a spread exists even for a market maker that does not mind risk: Glosten and Milgrom (1985) showed that if some fraction of order flow is informed, a

competitive market maker must set a spread wide enough that what it earns from uninformed traders covers what it loses to informed ones. Kyle (1985) added the matching idea on the price side: when market makers cannot tell informed from uninformed orders, every order moves the price a little, in proportion to its size. The slope of that relation — how far the price moves per unit traded — is **Kyle's lambda**, and it will reappear in §7 as the price impact of dealers' own hedging trades.

Put together: **a market maker earns the spread, loses on inventory it cannot offload, and loses to counterparties who know more than it does.** Everything a market maker does — how it quotes, how fast it hedges, when it widens its spread or pulls its quotes entirely — is a trade-off between those three quantities. [Fact] These are the foundational results of market microstructure, surveyed in Biais, Glosten and Spatt (2005), and the evidence that dealer inventory moves prices is direct: Hendershott and Menkveld (2014), using New York Stock Exchange intermediary data, estimate that inventory-driven price pressure averages 0.49% with a half-life of under a day.

1.3 An options market maker's version of the same problem

An options market maker faces the same three quantities, with one change that alters everything: its inventory is not shares but *options*, and the risk of an option position does not stay put.

If a stock market maker is long 1,000 shares, its exposure to a $1 move is $1,000 whether the stock is at 100 or at 110. If an options market maker sells 10 call contracts, its exposure to a $1 move is about $511 at a stock price of 100, and about $648 a day later if the price has moved to 102 (§5.3 works this out). The exposure changes *because the price changed*. Neutralising the exposure once is not enough; it must be done again every time the price moves, and every time a day passes, and every time the market's estimate of volatility changes. The quantity that governs how fast the exposure changes with the price is **gamma**, and a dealer's problem is gamma management in the way a stock market maker's problem is inventory management.

The three quantities of §1.2 survive, transformed:

Stock market maker	Options market maker
Earns the bid–ask spread on shares	Earns the bid–ask spread on options, plus or minus the difference between the volatility it sold (or bought) and the volatility that is realised (§4.4)
Inventory risk: holding shares it does not want	Delta risk, hedged continuously; gamma, vega and jump risk that hedging the underlying cannot remove (§4.9)
Adverse selection: informed traders	Informed traders in the options *and* in the underlying; the options market is a venue where informed traders act (Easley, O'Hara and Srinivas, 1998; Pan and Poteshman, 2006)
Shades quotes against inventory	Shades option quotes against its *Greeks* — against accumulated gamma and vega, not just contract counts (§9.2)

The second row carries the weight. An options market maker cannot get rid of all of its risk by trading the underlying; it can only get rid of the first-order, directional part of it. What remains is priced into option spreads — Jameson and Wilhelm (1992) found that the risks of discrete rebalancing and uncertain volatility explain a significant share of quoted spreads on Chicago Board Options Exchange options — and it is what the rest of Part II is about.

1.4 Market maker, dealer, liquidity provider: names that overlap

The words in this area are used loosely, sometimes interchangeably, sometimes with precise legal meanings. What matters for this document is behaviour, not registration, but the labels are worth sorting out.

Term	What it usually means	Relevance here
Market maker	A firm registered with an exchange that has an obligation to post two-sided quotes in some set of products, in exchange for fee rebates or priority	Options market makers ("OMMs") are the core hedgers
Dealer	A firm that trades as *principal* — for its own account — rather than as an agent for a customer; covers exchange market makers and bank derivatives desks	In gamma-exposure language, "dealer" means *whoever is on the other side of customer option trades and hedges them*. I follow that usage

Term	What it usually means	Relevance here
Liquidity provider	Anyone who posts resting orders that others trade against, whether obligated or not	Broader than market makers; includes proprietary trading firms that are not registered market makers
Specialist / designated market maker	The single firm with special obligations in a listed stock on some exchanges	Historical background; not central to options
Wholesaler / internaliser	A firm that executes retail brokerage orders against its own inventory, typically paying the broker for the flow	In US options, the same large firms are often both wholesalers and exchange market makers
Intermediary	The academic umbrella term for dealers and market makers together	Used by the demand-pressure and intermediary-constraint literature (§6.3, §9.6)
End user / customer	Everyone else: institutions, retail investors, hedge funds, corporations	The side whose positions are *not* assumed to be dynamically hedged

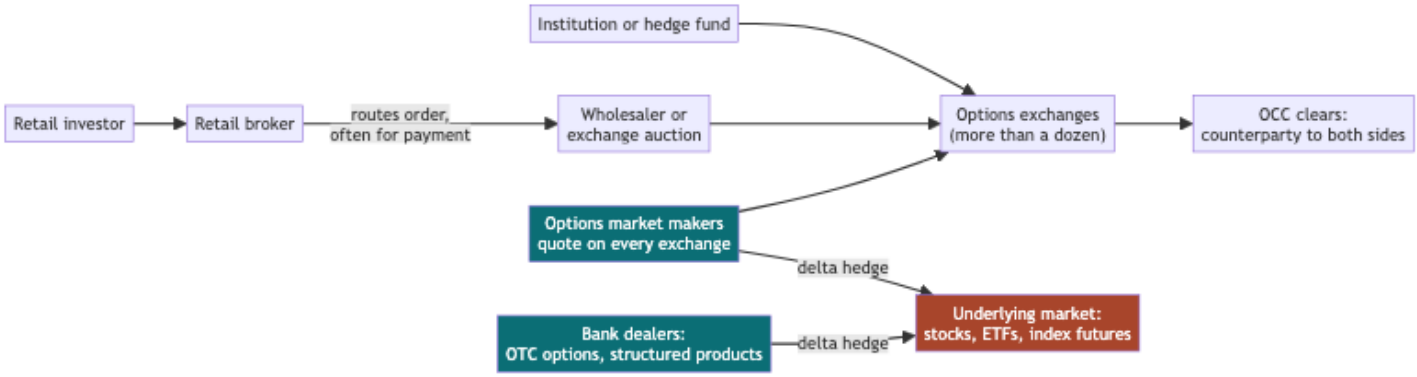
The large options market makers in the United States are a small group of specialised trading firms — names such as Citadel Securities, Susquehanna, Optiver, IMC, Jane Street and Wolverine recur — alongside the derivatives desks of the major banks, which dominate over-the-counter (off-exchange) options and structured products. The defining property for everything that follows is not the legal category. It is this: **a dealer is a participant who hedges its option position mechanically in the underlying.** An end user is one who, as a modelling assumption, does not.

1.5 Who trades with whom in listed options

A listed option in the United States is traded on one of more than a dozen competing exchanges, and every trade is **cleared** by the Options Clearing Corporation (OCC), which becomes the legal counterparty to both sides: the buyer holds a contract against the OCC, and so does the seller. That is what makes listed options safe to trade with strangers. It also means that, legally, nobody holds an option "against" a particular dealer; economically, though, every long position was created by somebody's short one, and the question "who is short?" is the question this document keeps returning to.

Each trade is tagged by the exchange with the **capacity** of each side — broadly, *customer*, *professional customer*, *broker-dealer* (a firm's own account) and *market maker* — and as **opening** (creating a new position) or **closing** (reducing an existing one). Those tags are the raw material for measuring who holds what, and they are mostly not public in real time. §6.3 and §11.2 describe what is available.

The flow of a typical trade looks like this:



Three features of this picture matter later.

- **The hedging lands in a different market from the option.** An S&P 500 index option is traded on Cboe, but the dealer who sold it hedges with S&P 500 futures on the CME, with SPY exchange-traded fund shares, or with baskets of stocks. Gamma exposure is a claim that trading in one market (options) mechanically generates trading in another (the underlying).
- **Bank dealers are invisible to exchange data.** Over-the-counter options, variance swaps and structured notes create hedging needs that appear in no public open-interest file. Any gamma-exposure estimate built from listed open interest is blind to them. [Fact]
- **Retail flow is intermediated.** Most retail option orders reach an exchange through a broker that routes them to a wholesaler or into a price-improvement auction. The SEC's staff report on the January 2021 meme-stock episode (SEC, 2021) is the best single public description of this structure, and §8.5 returns to what it found.

§1 Key takeaways

1. A market maker sells immediacy and earns the bid–ask spread; it does not want directional bets and neutralises them as fast as it can.
2. Its two enemies are inventory risk (holding what it did not want) and adverse selection (trading with people who know more). Its quotes are tools for managing both.
3. An options market maker's inventory is options, whose directional exposure changes every time the price moves. Neutralising it once is not enough; it must hedge *dynamically*, and the rate at which the exposure changes is gamma.
4. Dynamic hedging is mechanical — forced by position, price, time and volatility — and therefore predictable in direction, which is why it can be reasoned about at the market level.
5. Options are in zero net supply. The contracts themselves do nothing to the underlying; only the hedging of one side, unoffset by the other, can.
6. "Dealer" in this document means whoever is on the other side of customer option trades *and hedges them*. Bank dealers' over-the-counter positions create hedging that no public data captures.

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2. Options from zero

2.1 The contract

An **option** is a contract that gives its holder a right without an obligation.

- A **call option** gives the holder the right to **buy** the underlying at a fixed price, the **strike** K, on or before a fixed date, the **expiry** T.
- A **put option** gives the holder the right to **sell** the underlying at the strike, on or before expiry.

The holder pays for this right up front. The price is the **premium**. The holder is said to be **long** the option, or to have **bought** it; the other side, who received the premium and took on the obligation to deliver (for a call) or to take delivery (for a put) if the holder chooses, is **short** the option, or has **sold** or **written** it. Deciding to use the right is **exercising** the option; being on the short side when that happens is being **assigned**.

A holder exercises only when it pays to. A call with strike 100 is worth exercising at expiry if the underlying trades above 100, because the holder buys at 100 something worth more; otherwise it is thrown away. So the call's value at expiry — its **payoff** — is

$$C_T = \max(S_T - K,\ 0), \qquad P_T = \max(K - S_T,\ 0),$$

where S_T is the underlying's price at expiry and P_T is the matching payoff of a put. Read them as: *the call pays the amount by which the price finishes above the strike, or nothing; the put pays the amount by which it finishes below, or nothing.*

Listed equity and index options in the United States have a **contract multiplier** $M = 100$: one contract controls 100 shares (or 100 times the index level), and prices are quoted per unit. An option quoted at $2.29 therefore costs $229 per contract. Every position size in this document is stated either in contracts or in options — one option being the claim on a single unit of the underlying, so 10 contracts are 1,000 options — and it says which.

2.2 Exercise and settlement

Two contract features look like administrative detail and turn out to matter a great deal for hedging flows near expiry (§7.7).

Exercise style. An **American** option can be exercised at any time up to expiry. A **European** option can be exercised only at expiry. US single-stock options and exchange-traded fund (ETF) options such as those on SPY and QQQ are American; options on the S&P 500 index itself (ticker SPX) are European.

Settlement. A **physically settled** option delivers the underlying: exercising a stock call means actually receiving 100 shares per contract and paying the strike. A **cash-settled** option pays the difference between the settlement price and the strike in cash. Stock and ETF options are physically settled; index options, which have no deliverable, are cash-settled.

Settlement price and timing. Index options add one more wrinkle. The standard monthly SPX contracts, expiring on the third Friday of the month, are **AM-settled**: they stop trading on Thursday, and their final value is computed from the *opening* prices of the index's component stocks on Friday morning. The weekly and daily SPX contracts (ticker SPXW) are **PM-settled** at Friday's — or that day's — closing value. SPXW expirations now exist for every trading day of the week; the Tuesday and Thursday expirations were added in April and May 2022, which is what made **zero-days-to-expiry (0DTE)** trading possible every day (§7.9).

Automatic exercise and pin risk. At expiry, the OCC automatically exercises any listed option that is in the money by at least a cent, unless its holder instructs otherwise. For physically settled options, holders may submit or reverse exercise instructions for some time *after* the closing bell. If a stock closes almost exactly at a strike, a dealer who is short that strike does not know until the next morning whether it has been assigned — whether it will wake up owning, or short, a block of shares it did not plan to have. That uncertainty is **pin risk**, and it is one reason dealers manage their positions around large strikes in the final hours before expiry (§9.5).

2.3 The four basic positions

There are two kinds of option and two sides to each trade, so there are four basic positions. They are the four corners of the quadrant that organises §5, and it is worth fixing their payoffs and profits now.

Profit and loss (P&L) is the payoff minus what was paid, or plus what was received. Take a strike of 100 and a premium of $2.29 for both the call and the put (the value §3.3 derives for a 30-day option at 20% volatility). At expiry, per option:

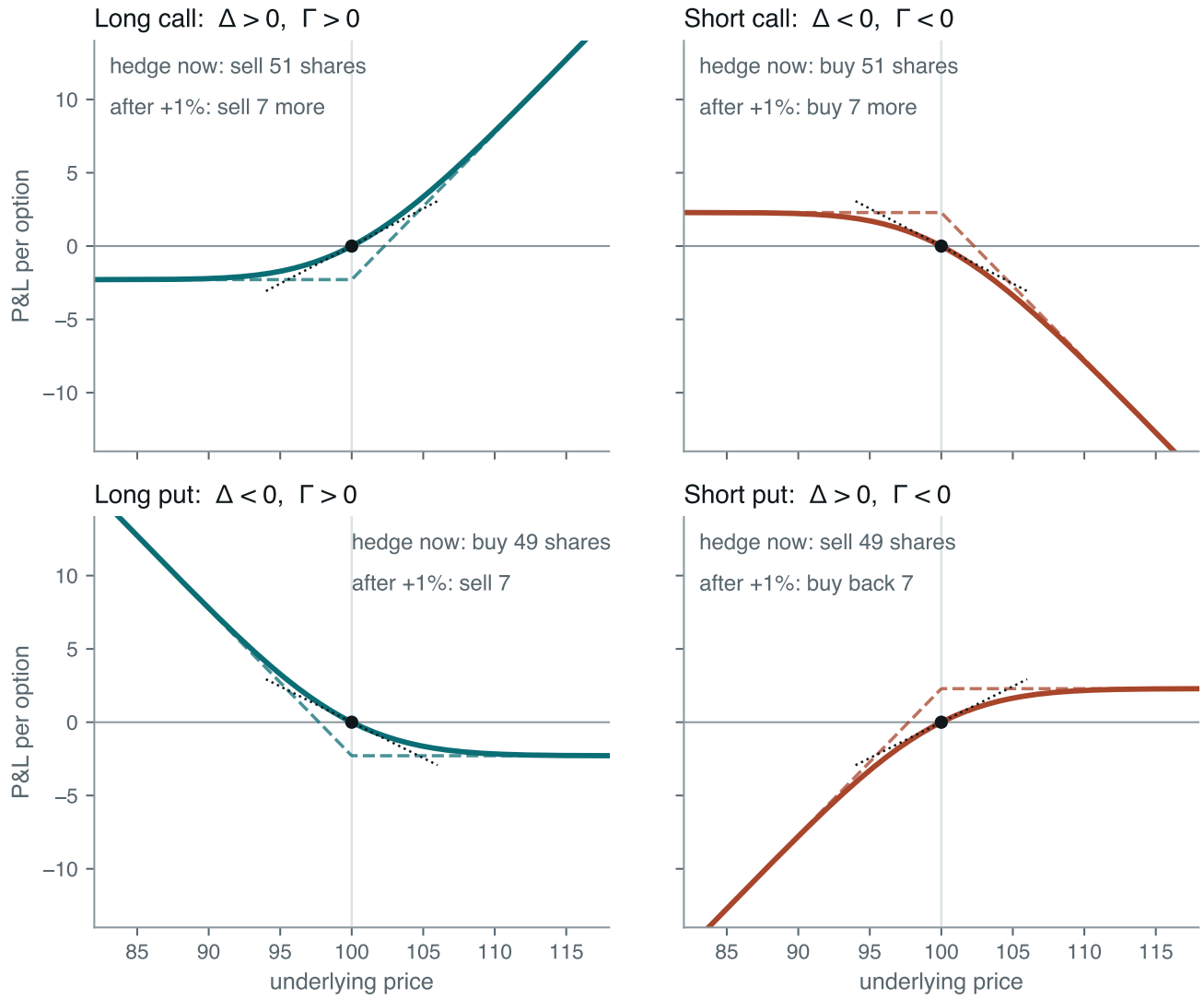
Price at expiry S_T	80	90	95	100	105	110	120
Long call: $\max(S_T - 100, 0) - 2.29$	−2.29	−2.29	−2.29	−2.29	+2.71	+7.71	+17.71
Short call: $2.29 - \max(S_T - 100, 0)$	+2.29	+2.29	+2.29	+2.29	−2.71	−7.71	−17.71
Long put: $\max(100 - S_T, 0) - 2.29$	+17.71	+7.71	+2.71	−2.29	−2.29	−2.29	−2.29
Short put: $2.29 - \max(100 - S_T, 0)$	−17.71	−7.71	−2.71	+2.29	+2.29	+2.29	+2.29

Four features of this table carry forward.

1. **Long and short are exact mirror images.** Every dollar one side makes, the other loses. Options are a zero-sum transfer between the two sides *before* counting hedging and costs.
2. **The buyer's loss is capped; the seller's is not (or barely).** A long option can lose at most its premium. A short call can lose without limit as the price rises; a short put can lose up to the strike minus the premium.
3. **Calls profit from rises and puts from falls — for the holder.** The long call breaks even at 102.29, the long put at 97.71.

4. **The kink.** Every payoff is two straight lines joined at the strike. That kink is where all of the interesting behaviour of options comes from: before expiry the kink is smoothed into a curve, and the *curvature* of that curve is gamma.

The figure shows the four positions twice: dashed at expiry, and solid as they would be valued if the price moved *now*, 30 days before expiry. The solid curves are smooth versions of the dashed ones. Their slope at today's price is **delta**, their curvature is **gamma**, and the small annotations say how many shares a holder of one contract would trade to neutralise that slope — the hedge that §4 develops.



Solid: P&L if the price moved now (30 days to expiry). Dashed: P&L at expiry. Dotted: the delta tangent.
Strike 100, volatility 20%, hedge sizes per contract of 100 options.

Look at the curvature, not the slope. The two *long* positions curve upward — a smile — whether they are calls or puts. The two *short* positions curve downward — a frown. Call versus put decides which way the curve slopes; long versus short decides which way it bends. That observation, made precise in §2.6 and §5.1, is the most important sign rule in this document.

2.4 Moneyness, intrinsic value and time value

Traders describe an option by where the underlying stands relative to the strike.

- A call is **in the money (ITM)** when $S > K$, **out of the money (OTM)** when $S < K$, and **at the money (ATM)** when $S \approx K$. For a put, the inequalities reverse.
- **Intrinsic value** is what the option would pay if exercised now: $\max(S - K, 0)$ for a call, $\max(K - S, 0)$ for a put.
- **Time value** (or **extrinsic value**) is the premium minus the intrinsic value: what the holder pays for the possibility that the price moves further in its favour before expiry.

At $S = 100$, a 95-strike call is in the money with intrinsic value 5; §3 prices it at 5.57, so it carries 0.57 of time value. The 95-strike put is out of the money, has no intrinsic value, and its entire premium of 0.57 is time value. The at-the-money 100 strike has zero intrinsic value and the largest time value of all, 2.29. **Time value is largest at the money** — where the outcome is most uncertain — and it is the part of the premium that decays to nothing by expiry (§3.6).

Practitioners also label strikes by **delta** rather than by price: a "25-delta put" is the put whose delta (§3.4) is −0.25, wherever that strike happens to lie. Delta works as a moneyness coordinate that adjusts automatically for volatility and time to expiry, which is why volatility surfaces and positioning reports are often organised by it.

2.5 Open interest: every long has a short

Two numbers describe how much an option is traded, and they are easy to confuse.

- **Volume** is the number of contracts that changed hands during a day.
- **Open interest (OI)** is the number of contracts *outstanding* — created and not yet closed — at the end of a day.

A trade can create, transfer, or destroy open interest, depending on whether each side is **opening** (establishing a position) or **closing** (unwinding one):

Day	Trade	Buyer	Seller	Volume	Change in OI	OI after
1	A buys 10 calls from B	A opens (now long 10)	B opens (now short 10)	10	+10	10
2	C buys 5 calls from A	C opens (long 5)	A closes (long 5)	5	0	10
3	B buys 5 calls from C	B closes (short 5)	C closes (flat)	5	−5	5

After day 3, A is long 5 and B is short 5. Open interest is 5, and total volume over the three days was 20.

Now the fact that the rest of this document is built on:

> **Options are in zero net supply.** Every contract of open interest is one long position and one short position. Summed over *everybody*, the net position in every option is exactly zero — and therefore so is the net delta, the net gamma and every other Greek.

It follows that "the market is long gamma" is meaningless as a statement about all holders together. A claim about gamma exposure is always a claim about **one group** of holders — the dealers — whose gamma is offset by an equal and opposite gamma held by everyone else. It can only matter for prices if the two groups **behave differently**: if dealers hedge dynamically and the other side, as a group, does not. That behavioural asymmetry is the load-bearing assumption of every gamma exposure model, and §6.3 examines it.

Open interest is published once a day, after the OCC processes the day's trades, so the figure available at the start of a trading day describes positions as of the previous close. It says how many contracts exist, never who holds which side. Both facts return in §6.3 and §11.

2.6 Put–call parity: the first equivalence

A call and a put with the same strike and expiry look like opposites. They are more closely related than that: each can be built from the other.

Compare two portfolios, holding European options on a stock that pays no dividend, with zero interest rates:

- **Portfolio A:** one call, plus K in cash.
- **Portfolio B:** one put, plus one share of the stock.

At expiry, if $S_T > K$, portfolio A exercises the call and is worth S_T; portfolio B's put expires worthless and it holds a share worth S_T. If $S_T \le K$, A lets the call lapse and holds K in cash; B exercises the put, selling its share for K. **Both portfolios are worth** $\max(S_T, K)$ **in every state of the world.** If two portfolios have identical payoffs in every state, they must cost the same today, or anyone could buy the cheap one, sell the dear one, and pocket the difference at no risk. So

$$C + K = P + S \qquad \Longleftrightarrow \qquad C - P = S - K.$$

With interest rates and dividends the same argument gives $C - P = Se^{-q\tau} - Ke^{-r\tau}$. This is **put–call parity**, stated in this form by Stoll (1969). The numbers of §2.3 obey it: the 100-strike call and put both cost 2.29 and $S - K = 0$; the 95-strike call costs 5.57 and the 95-strike put 0.57, and their difference is exactly $100 - 95 = 5$.

Differentiate parity with respect to the underlying price — once for delta, again for gamma — and with respect to volatility for vega, since the right-hand side $S - K$ has no σ in it. All three consequences for hedging appear at once:

$$\Delta_C - \Delta_P = 1, \qquad \Gamma_C = \Gamma_P, \qquad \nu_C = \nu_P.$$

A call and a put with the same strike and expiry have exactly the same gamma and the same vega. Their deltas differ by exactly one share. A long call is a long put plus one share of stock; once you hedge away the delta, the share disappears into the hedge, and **a delta-hedged call and a delta-hedged put are the same position.** §5.8 shows this numerically, trade for trade. It is the reason that, for gamma exposure, the question "calls or puts?" is secondary to "who is long?".

Parity holds exactly for European options such as SPX, up to financing and dividend assumptions. For American options it becomes a pair of inequalities, because early exercise has value, and prices can deviate from it visibly when the stock is hard to borrow. GameStop options in January 2021 are the extreme example; Hilliard and Hilliard (2023) find that the apparent violations there are largely accounted for by the costs of short selling rather than free money. The gamma equivalence survives these frictions approximately. [Fact]

§2 Key takeaways

1. A call is the right to buy at the strike, a put the right to sell. The holder is long and paid a premium; the writer is short and received it. Listed contracts control 100 units of the underlying.
2. Payoffs are kinked at the strike. Before expiry the kink is a curve: its slope is delta, its curvature is gamma.
3. Long positions curve up and short positions curve down, whether they are calls or puts. Call versus put sets the slope; long versus short sets the curvature.
4. Exercise style and settlement are not details: AM versus PM settlement, physical delivery and automatic exercise shape the flows on expiry day and create pin risk for dealers.
5. Open interest counts contracts outstanding, published once a day, and never says who is long.
6. Options are in zero net supply, so aggregate gamma across all holders is zero. Gamma exposure is only meaningful for one group — dealers — and only matters if that group hedges while the other side does not.
7. Put–call parity makes a same-strike call and put interchangeable up to a share of stock: identical gamma, identical vega, deltas one apart.

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3. What an option is worth, and how that value moves

3.1 Volatility: the input you cannot see

An option's payoff depends on where the price ends up, and nobody knows that. What turns out to matter for its value is not *which way* the price is likely to go but *how far* it is likely to wander. The measure of that wandering is volatility.

Let the **log return** over a period be $\ln(S_{t+1}/S_t)$ — for small moves, almost identical to the percentage change. If log returns over successive days are independent with standard deviation σ_{day}, the variance of the sum of n days' returns is $n\,\sigma_{\text{day}}^2$, so the standard deviation grows with the **square root** of time:

$$\sigma_{n \text{ days}} = \sigma_{\text{day}}\sqrt{n}.$$

Volatility σ is this standard deviation scaled to one year. With about 252 trading days in a year, $\sigma = \sigma_{\text{day}}\sqrt{252}$, and since $\sqrt{252} \approx 16$, practitioners use the **rule of 16**: an annual volatility of 16% means a typical daily move of about 1%. A stock with 20% volatility moves about 1.26% on a typical day, and about $20\% \times \sqrt{30/365} = 5.7\%$ over the 30 calendar days of the option priced below. Simple and Log Returns derives why log returns are the natural unit here.

Two volatilities must be kept apart from the start, and the difference between them is where options market making earns or loses its money:

- **Realised volatility** σ_r is measured after the fact from the price path.
- **Implied volatility** σ_i is the volatility that makes a pricing model reproduce an option's market price (§3.7). It is the market's price for future volatility, not a measurement of it.

3.2 Replication: why an option can be priced at all

Here is the central idea of option pricing, in a world simple enough to check by hand.

A stock is at 100. Tomorrow it will be at 110 or at 90. A call with strike 100 will pay 10 if the stock rises and 0 if it falls. What is the call worth today?

The naive answer — the expected payoff, which needs the probability of a rise — is not the answer. Instead, try to **manufacture** the call. Buy h shares and hold b dollars of cash (borrowing if b is negative; take the interest rate to be zero). Tomorrow this portfolio is worth $110h + b$ or $90h + b$. Choose h and b so it pays exactly what the call pays:

$$110h + b = 10, \qquad 90h + b = 0 \qquad \Longrightarrow \qquad h = \frac{10 - 0}{110 - 90} = 0.5, \qquad b = -45.$$

Half a share and a loan of 45 reproduce the call in both states. The portfolio costs $0.5 \times 100 - 45 = 5$ today, so the call must cost 5. If it sold for 6, you could sell the call, build the portfolio for 5, and keep 1 with no risk; if it sold for 4, you would do the reverse. **The probability of a rise never entered the calculation.** Whether the stock is 90% or 10% likely to go up, anyone can manufacture the call for 5, so nobody pays more.

The number of shares in the replicating portfolio, $h = 0.5$, is the **hedge ratio**, and it is the option's **delta**. A dealer who *sells* this call for 5 and *buys* half a share, borrowing the difference, has no risk at all: whatever happens, its shares and loan exactly cover what it owes on the call.

Now add a second day. From 110 the stock goes to 120 or 100; from 90, to 100 or 80. The call pays 20, 0 or 0. Work backwards through the tree, replicating one step at a time:

Node	Next prices	Call pays next	Hedge h	Call value
After a rise, $S = 110$	120 or 100	20 or 0	1.0	10
After a fall, $S = 90$	100 or 80	0 or 0	0.0	0
Today, $S = 100$	110 or 90	10 or 0	0.5	5

The hedge that was 0.5 today becomes **1.0 after a rise** and **0.0 after a fall**. To stay hedged, the dealer who sold the call must *buy* another half share after the price goes up, at 110, and *sell* its half share after the price goes down, at 90. It buys high and sells low, every time. Follow any path through the tree — up-then-down, down-then-up, up-up, down-down — and net the shares' gain or loss against what the dealer then owes on the call, and the result is a loss of exactly 5 on every path. (Check the extreme case: on up-up, the shares alone *gain* $0.5 \times 10 + 1.0 \times 10 = 15$, but the call finishes owing 20 — still a net of -5.) The premium it collected was 5, so the hedge exactly breaks even.

That is the whole theory of option pricing in one sentence: **an option's time value is the cost of the rebalancing its replication requires.** The rate at which the hedge must change as the price moves is **gamma**. In this toy tree the cost is the same on every path; in a real market it is not, and the gap between the rebalancing cost the premium assumed and the one actually incurred is the subject of §4.

3.3 The Black–Scholes formula, read as a recipe

Shrink the tree's steps towards zero, letting the price follow a random walk in log returns with constant volatility σ, and replication still works: at every instant there is a number of shares that makes the hedged position riskless over the next instant. The replication cost becomes a closed form, derived by Black and Scholes (1973) and Merton (1973). For a European call and put on an underlying paying dividend yield q, with interest rate r:

$$\begin{aligned} C &= Se^{-q\tau}N(d_1) - Ke^{-r\tau}N(d_2), \\ P &= Ke^{-r\tau}N(-d_2) - Se^{-q\tau}N(-d_1), \\ d_1 &= \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)\,\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}. \end{aligned}$$

Read the call formula as the recipe it is: **hold $e^{-q\tau}N(d_1)$ shares and borrow $Ke^{-r\tau}N(d_2)$ in cash.** $N(d_1)$ is the hedge ratio of §3.2, grown up. $N(d_2)$ has its own reading: it is the probability that the call finishes in the money under the **risk-neutral** probabilities — the probabilities that price assets as if no investor demanded compensation for risk, which Appendix

A explains. The quantity $\sigma\sqrt{\tau}$ is the standard deviation of the log return over the option's remaining life, and d_1 and d_2 measure the distance from the strike in units of it.

Take the example used throughout this document: $S = K = 100$, $\sigma = 20\%$, $\tau = 30/365$, $r = q = 0$. Then $\sigma\sqrt{\tau} = 0.0573$, $d_1 = 0.0287$, $d_2 = -0.0287$, $N(d_1) = 0.51144$ and $N(d_2) = 0.48856$, so

$$C = 100 \times (0.51144 - 0.48856) = 2.29.$$

By put–call parity with $S = K$, the put also costs 2.29. Those are the premiums of §2.3.

The model's assumptions are all false in some respect, and the ways they fail are the ways hedging fails (§4.9). Volatility is not constant; it varies over time and across strikes. Prices gap, especially overnight and on news, so trading is not continuous. Trading costs money. Returns have fatter tails than the lognormal distribution allows. [Fact] The formula survives anyway because markets use it as a *language*: options are quoted in the volatility that makes the formula match the price (§3.7), and dealers compute their hedges from the formula evaluated at those volatilities. [Practice]

3.4 Delta

Delta is the sensitivity of the option's value to the underlying price, $\Delta = \partial V/\partial S$. From the formula (with $q = 0$):

$$\Delta_C = N(d_1) \in (0, 1), \qquad \Delta_P = N(d_1) - 1 \in (-1, 0).$$

Across strikes, for our 30-day, 20%-volatility option on a stock at 100:

Strike	90	95	100	105	110
Call delta	0.969	0.822	0.511	0.205	0.051
Put delta	−0.031	−0.178	−0.489	−0.795	−0.949

Delta has three readings, and all three are useful.

1. **Slope.** A $1 rise in the stock raises the at-the-money call by about $0.51 and lowers the put by about $0.49.
2. **Hedge ratio.** A holder of one call is exposed like a holder of 0.511 shares; selling 0.511 shares neutralises that exposure for small moves.
3. **Rough probability.** Delta is close to the (risk-neutral) probability of finishing in the money — exactly $N(d_2)$ for that, which is near $N(d_1)$ for short-dated options. A 5-delta option is a long shot.

A **position delta** multiplies by the number of contracts and the multiplier: 10 contracts of the at-the-money call carry $10 \times 100 \times 0.5114 = 511.4$ **share-equivalents** of delta, and 10 short contracts carry −511.4. Multiplying by the price gives **dollar delta**: $511.4 \times \$100 = \$51{,}140$ of stock exposure.

3.5 Gamma

Gamma is the rate at which delta changes as the price changes, $\Gamma = \partial\Delta/\partial S = \partial^2 V/\partial S^2$ — the curvature of the solid curves in the §2.3 figure. For calls and puts alike (§2.6):

$$\Gamma = \frac{e^{-q\tau}\,\varphi(d_1)}{S\,\sigma\sqrt{\tau}}.$$

For the at-the-money 30-day option, $\Gamma = 0.0695$: when the stock rises from 100 to 101, the call's delta rises from 0.511 to about 0.580. For 10 contracts, that is $1{,}000 \times 0.0695 = 69.5$ shares of delta per $1 move — the number of shares a hedger must trade after each $1 move to stay neutral. A **long** option position has **positive** gamma; a **short** position has **negative** gamma; that is true of calls and puts equally.

The formula has $\sigma\sqrt{\tau}$ in the denominator, and three consequences follow that recur for the rest of this document.

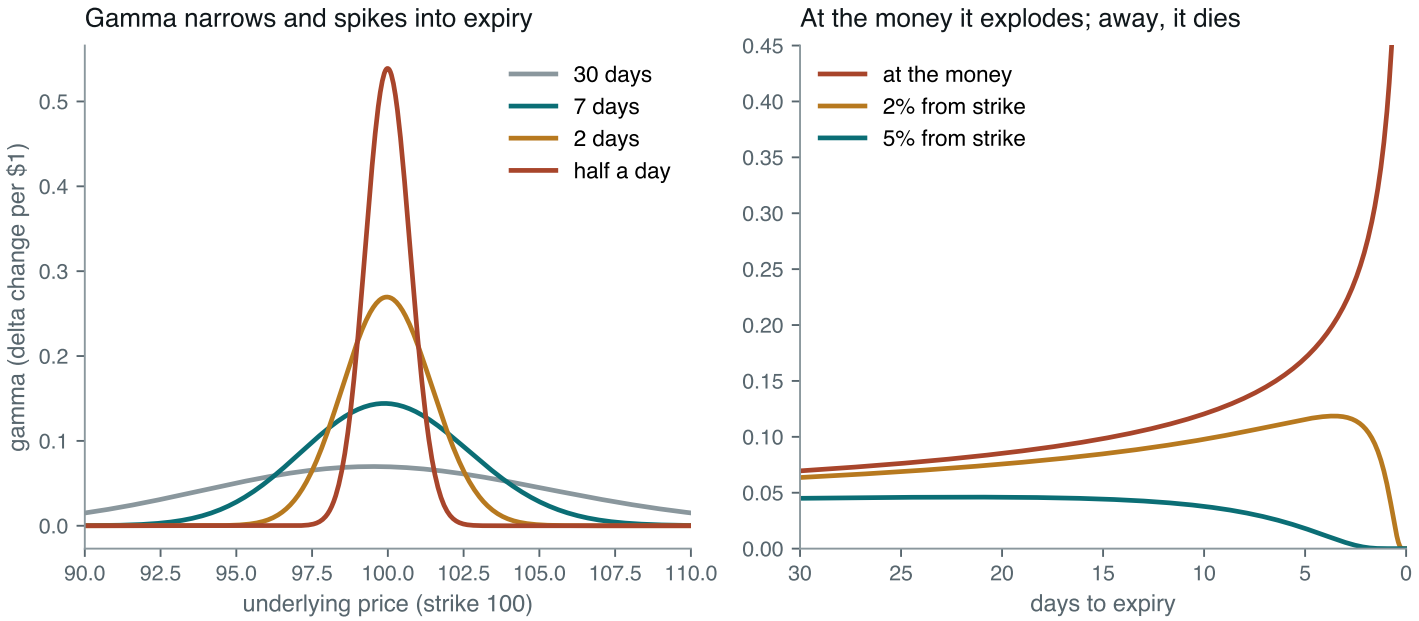
Gamma is concentrated near the strike. $\varphi(d_1)$ is largest when $d_1 = 0$, so gamma peaks close to the money and falls away on both sides.

Gamma explodes into expiry, near the strike. As $\tau \to 0$ the at-the-money gamma grows like $1/\sqrt{\tau}$, while the band of prices over which gamma is large narrows. The same at-the-money option has gamma 0.0695 with 30 days left, 0.144 with 7, 0.381 with 1 and 0.762 with a quarter of a day.

Away from the strike, gamma peaks and then dies. An option whose strike is a log distance $x = |\ln(S/K)|$ from the price has gamma that *rises* as expiry approaches only until the remaining standard deviation shrinks to that distance, and then *collapses* to zero: the option is now almost certain to expire worthless (or almost certain to be exercised), so its delta is pinned at 0 (or ±1). The peak comes where $\sigma\sqrt{\tau} = x$ — maximising $\varphi(d_1)/\sqrt{\tau}$ over τ gives

$$\tau^\star \approx \frac{x^2}{\sigma^2}.$$

For a strike 2% away at 20% volatility the peak comes 3.6 days before expiry; for 5%, 21.7 days before.



Low volatility means high, narrow gamma. Gamma at the money is inversely proportional to σ. In a calm market with implied volatility of 12%, the same open interest carries about two-thirds more at-the-money gamma than at 20%, packed into a narrower range of prices. This is purely mechanical, and it matters when interpreting data: gamma exposure estimates will read high in calm markets partly *because* the market is calm, whatever the effect of hedging on calm itself. §10.1 returns to this confound.

3.6 Theta, and why it is gamma's mirror image

Theta is the change in an option's value as time passes with everything else held fixed, $\Theta = \partial V/\partial t$. Long options lose value as time passes: the at-the-money 30-day option loses $0.0381 per day, and 10 contracts lose $38.11. The loss accelerates as expiry approaches, because it is the time value of §2.4 draining away.

Theta is not an independent quantity. The replication argument of §3.2, carried out in continuous time, produces a partial differential equation that every option price must satisfy — the **Black–Scholes equation**:

$$\Theta + \tfrac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta - rV = 0.$$

With $r = 0$ it reduces to a statement a trader can say aloud:

$$\Theta = -\tfrac{1}{2}\sigma^2 S^2 \Gamma.$$

Theta is the rent paid for gamma. An option with a lot of gamma decays fast, and an option whose value does not decay has no gamma. Check it on the example: $\frac{1}{2} \times 0.20^2 \times 100^2 \times 0.0695 = 13.91$ per year, or 0.0381 per day. The holder of gamma pays this rent every day; the seller collects it. §4.4 shows what each gets in return.

A practical wrinkle: options decay over weekends and holidays, when the underlying does not trade. The formulas above use calendar days (a year of 365), which spreads decay evenly over days that have no price moves; many desks instead measure time in trading days or in "variance time" that weights days by their expected activity. [Practice] This document uses calendar days throughout, so a day's theta is matched against a day's move.

3.7 Vega and implied volatility

Vega is the sensitivity to volatility, $\nu = \partial V/\partial\sigma = Se^{-q\tau}\varphi(d_1)\sqrt{\tau}$. The at-the-money 30-day option gains $0.114 for each percentage point of volatility; 10 contracts gain $114. Vega, like gamma, is identical for a call and a put of the same strike and expiry.

Because volatility is the only input to the formula that cannot be observed, options markets turn the formula around. Given an option's market price, the **implied volatility** is the value of σ that makes the formula return that price. Options are quoted and compared in implied volatility, the way bonds are quoted in yield.

If the model were right, every option on the same underlying would have the same implied volatility. They do not. Implied volatility varies with the strike — the **smile** or, for equity indices where it slopes down from low strikes to high, the **skew** — and with the expiry — the **term structure**. The whole map from (strike, expiry) to implied volatility is the **volatility surface**. The downward skew in equity index options, in which out-of-the-money puts are priced at much higher implied volatility than calls, has been a permanent feature of the market since the 1987 crash. [Fact] Part of the explanation is demand: end users buy index puts for protection, dealers who absorb that demand cannot hedge it perfectly, and they charge for the risk they keep (Bollen and Whaley, 2004; Gârleanu, Pedersen and Poteshman, 2009). [Contested] as to how much of the skew is explained by demand, relative to fear of crashes itself.

Two consequences for hedging:

- **Greeks depend on which volatility you use.** The same option has a different delta at 15% than at 25%. A dealer computes its hedge from its own volatility surface, and two dealers with different surfaces hedge the same option differently.
- **When the price moves, the surface moves too — somehow.** If every strike keeps its implied volatility as the price moves, the surface is **sticky strike**; if implied volatility moves with moneyness, so that an option's volatility depends on its delta rather than its strike, it is **sticky delta**. Neither is exactly right. The choice changes the gamma a model assigns to a position at a hypothetical price, and therefore where a computed "gamma flip" lies (§6.5). [Practice]

3.8 The cross-Greeks that move hedges: vanna and charm

Gamma describes how delta changes when the price moves. Delta also changes when volatility changes and when time passes, and those changes force hedging trades even on a day when the price is still.

Vanna is the change in delta per unit of volatility. With $r = q = 0$:

$$\text{vanna} = \frac{\partial\Delta}{\partial\sigma} = -\frac{\varphi(d_1)\,d_2}{\sigma}.$$

It is the same for a call and a put. Its sign follows $-d_2$: positive for strikes above the price, negative for strikes below. Intuitively, higher volatility makes an out-of-the-money option more likely to finish in the money, so its delta grows in magnitude: an out-of-the-money call's delta rises, and an out-of-the-money put's delta becomes more negative. Lower volatility does the reverse. For a 95-strike put at 30 days and 20% volatility, vanna is −0.0113 per volatility point: if implied volatility falls from 20% to 17%, the put's delta shrinks from −0.178 to about −0.141.

Charm is the change in delta as time passes, with price and volatility fixed:

$$\text{charm} = \frac{\partial\Delta}{\partial t} = \frac{\varphi(d_1)\,d_2}{2\tau},$$

again with $r = q = 0$ and again the same for calls and puts. As expiry approaches, an out-of-the-money option's delta drains towards zero and an in-the-money option's delta grows towards ±1, because the outcome is becoming certain. The 95-strike put's delta goes from −0.178 to −0.174 in one day, and to −0.148 after a week, with the price unchanged.

30 days, 20% vol, $S = 100$	95 put	100 call or put	105 call
Delta	−0.178	+0.511 / −0.489	+0.205
Gamma	0.0454	0.0695	0.0496
Vega per vol point	0.0747	0.1143	0.0816
Vanna per vol point	−0.0113	+0.0006	+0.0125
Charm per day	+0.0038	−0.0002	−0.0042

At the money, vanna and charm are nearly zero; away from the money they are substantial, and they have *opposite signs* on the two sides of the price. §7.8 turns them into flows.

3.9 Units: share, dollar and percent Greeks

Greeks per option are clean for mathematics and useless for comparing positions across underlyings trading at 100, 600 and 6,000. Every desk and every data vendor rescales them, usually without saying which way. Here are the conventions, for a position of 10 at-the-money 30-day calls (1,000 options) on a stock at 100:

Quantity	Formula	Value	What it tells you
Share delta	$nM\Delta$	511 shares	Shares to sell to hedge
Dollar delta	$nM\Delta\, S$	\$51,140	Stock-equivalent exposure
Share gamma	$nM\Gamma$	69.5 shares per \$1	Shares to trade per \$1 move
Dollar gamma per 1%	$nM\Gamma\, S^2 \times 0.01$	\$6,955	Dollars of stock to trade per 1% move
Gamma P&L of a 1% move	$\frac{1}{2} nM\Gamma\, (0.01S)^2$	\$34.78	What a long holder gains from a 1% move, before theta
Theta per day	$nM\Theta/365$	−\$38.11	Daily decay
Vega per vol point	$nM\nu \times 0.01$	\$114.33	P&L per point of implied volatility

The row that matters most later is **dollar gamma per 1%**. It is the dollar value of the underlying a hedger must trade when the price moves by 1%, and it is comparable across underlyings, price levels and contract sizes: a position in SPX options and a position in SPY options can be added once both are in dollars per 1%. Gamma exposure (§6) is this quantity summed over a dealer population. Note the relation between two rows: the gamma P&L of a 1% move equals $\frac{1}{2} \times 0.01\times$ the dollar gamma per 1%.

§3 Key takeaways

1. Volatility is the standard deviation of log returns per year; it scales with the square root of time. By the rule of 16, 16% volatility is about 1% a day.
2. An option is priced by replication, not by forecasting: the premium is what it costs to manufacture the payoff by trading the underlying, and it does not depend on the probability of a rise.
3. Replicating a call means buying after rises and selling after falls. An option's time value is the cost of that rebalancing.
4. Delta is the hedge ratio; gamma is how fast it changes. Gamma is largest near the strike, explodes into expiry at the money, dies away from the money, and is inversely proportional to volatility.
5. Theta is the rent for gamma: with zero rates, $\Theta = -\frac{1}{2}\sigma^2 S^2 \Gamma$ exactly.
6. Markets quote options in implied volatility, which varies by strike and expiry. Every Greek depends on the volatility assumed, and on how the surface is assumed to move with the price.
7. Vanna and charm move delta when volatility changes and when time passes. They are near zero at the money and opposite in sign on either side of it.
8. Dollar gamma per 1% move — the dollars a hedger must trade per 1% move — is the unit in which gamma can be compared and aggregated.

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4. Delta hedging, and the identity that governs it

4.1 Why a market maker hedges

Suppose a market maker sells a customer 10 at-the-money 30-day calls and does nothing else. It is now short 511 share-equivalents of the stock (§3.4). At 20% volatility the stock's typical daily move is about $1.05 (§3.1, but counting calendar days, as §3.6 does), so the position's daily P&L has a standard deviation of about $511 \times \$1.05 \approx \$535$. What did the market maker earn for taking that on? If it sold the calls five cents above their fair value, $50.

Unhedged, the spread is noise. The market maker has become a speculator with a tiny edge and a large bet it never meant to place. So it buys 511 shares. Now a small move up loses on the calls what it gains on the shares, and a small move down does the reverse. What is left is the curvature: over a typical day, the hedged position's P&L has a standard deviation of about $54 (§4.7) — a tenfold reduction — and that residual is governed by gamma. Hedging does not make the risk disappear. It converts a large, *directional* risk into a smaller, *non-directional* one: a bet on how much the price moves, not on which way.

4.2 Static and dynamic hedging

There are two ways to neutralise an option.

- **Static hedging** offsets it with another option: buy back the same call from someone else, or buy a similar one. A perfect static hedge removes delta, gamma and vega at once and never needs adjusting. It requires a counterparty willing to sell at a price that leaves the spread intact, which is exactly what a market maker cannot count on.
- **Dynamic hedging** offsets the option's delta by trading the underlying, and keeps doing so as the delta changes. It can always be done, in any size the underlying market can absorb, and it removes only the first-order risk.

Real books use both. A market maker's inventory nets thousands of options across strikes and expiries; it hedges the **net delta** of the whole book in the most liquid instrument available — index futures for index options, the stock or ETF for single-name options — and it manages net gamma and vega by adjusting its option quotes to attract offsetting trades (§9.2). [Practice] Gamma exposure is about the dynamic part: the net gamma that could not be offset with other options, whose ever-changing delta must therefore be hedged by trading the underlying.

4.3 A dealer sells calls and hedges for a week

The rest of this section rests on one worked example. A dealer sells 10 contracts (1,000 options) of the at-the-money 30-day call at 20% implied volatility, buys the delta hedge, and re-hedges once a day at the close. Implied volatility stays at 20% throughout, so the only things moving are the price and the calendar. To keep the arithmetic simple, the week has no weekend: five consecutive days, each carrying both a price move and a day of decay. Here are two different weeks.

A quiet week. The stock ends the week where it started, at 100, after small moves.

Day	Price	Call delta	Hedge trade (shares)	Day P&L	Identity (§4.4)	Cumulative
0	100.00	0.511	buy 511.4			
1	100.40	0.539	+28.0	+32.86	+32.54	+32.86
2	100.10	0.518	−21.2	+35.95	+35.58	+68.81
3	99.70	0.489	−29.4	+33.93	+33.71	+102.74
4	100.20	0.526	+36.7	+31.39	+30.86	+134.13
5	100.00	0.510	−15.1	+39.85	+39.46	+173.97

A wild week. The stock also ends the week at 100, after large swings.

Day	Price	Call delta	Hedge trade (shares)	Day P&L	Identity (§4.4)	Cumulative
0	100.00	0.511	buy 511.4			
1	102.00	0.648	+136.4	−99.84	−100.99	−99.84
2	99.60	0.482	−165.6	−155.89	−149.13	−255.73
3	101.90	0.646	+163.3	−150.22	−151.80	−405.94
4	98.80	0.421	−224.6	−301.22	−284.37	−707.16

Day	Price	Call delta	Hedge trade (shares)	Day P&L	Identity (§4.4)	Cumulative
5	100.00	0.510	+89.5	−12.42	−13.73	−719.58

"Hedge trade" is the number of shares bought (+) or sold (−) at that day's close to restore the hedge. "Day P&L" is exact: the change in the value of the 1,000 short calls (repriced at 20% volatility with one day less to run) plus the gain or loss on the shares held over the day. "Identity" is the one-line approximation derived next.

Read the two tables side by side.

- **Same start, same end, opposite result.** Both weeks start and finish at 100. The calls are worth $2.088 at the end of both, down from $2.287: $199 of time decay, collected by the dealer either way. In the quiet week the dealer keeps most of it and makes $174. In the wild week it loses $720. The difference is entirely in the *path*: annualised, the quiet week realised 7.1% volatility and the wild week 43%.
- **Every hedge trade in the wild week chases the move.** The dealer buys 136 shares after the rise to 102, sells 166 after the fall to 99.60, buys 163 after the rise to 101.90, sells 225 after the fall to 98.80. A dealer short options, hedging, **buys after rises and sells after falls**, and the bigger the move, the bigger the trade. That is the replication cost of §3.2, now incurred at real prices on a real path.
- **The trades are large relative to the move that caused them.** Day 4's move of 3% forced the dealer to sell 225 shares — 44% of its initial hedge — at the low. If many dealers hold the same position, their combined selling lands in the market at the same moment. That is the seed of §7.

4.4 The identity: hedged P&L is gamma times the variance surprise

Hold one long option worth $V(S,t)$ and short Δ shares against it, with zero interest rates. Over a short interval in which the price changes by ΔS, the hedged position's P&L is

$$\Delta\Pi = \Delta V - \Delta \cdot \Delta S.$$

Expand the option's value to second order in the price move and first order in time:

$$\Delta V \approx \Theta\,\Delta t + \Delta \cdot \Delta S + \tfrac{1}{2}\Gamma\,(\Delta S)^2.$$

The delta terms cancel — that is what hedging is for — leaving

$$\Delta\Pi \approx \Theta\,\Delta t + \tfrac{1}{2}\Gamma\,(\Delta S)^2.$$

Now use §3.6: an option marked at implied volatility σ_i has $\Theta = -\frac{1}{2}\sigma_i^2 S^2\Gamma$. Substituting,

$$\Delta\Pi \;\approx\; -\tfrac{1}{2}\sigma_i^2 S^2\Gamma\,\Delta t + \tfrac{1}{2}\Gamma(\Delta S)^2 \;=\; \tfrac{1}{2}\,\Gamma\left[(\Delta S)^2 - \sigma_i^2 S^2 \Delta t\right],$$

and factoring S^2 out of both terms inside the brackets:

$$\boxed{\Delta\Pi \;\approx\; \tfrac{1}{2}\,\Gamma S^2\left[\left(\frac{\Delta S}{S}\right)^2 - \sigma_i^2\,\Delta t\right]}$$

For a **short** option, flip the sign. This is the most useful equation in this document, and it says, in words:

> **The P&L of a delta-hedged option over any interval is one half of its dollar gamma, times the difference between the squared return that actually happened and the variance that implied volatility charged for the interval.**

The "Identity" column of the tables above is this formula, and it tracks the exact P&L to within a few dollars a day. Its error grows with the size of the move (day 4 of the wild week: −$301 exact against −$284), because it ignores the third-order change in gamma with the price. It also leaves out P&L from changes in implied volatility itself, which is vega's job. With non-zero interest rates, the financing of the option and of the hedge contributes terms that cancel, and the identity is unchanged.

Five things follow.

1. There is a break-even move. The bracket is zero when $|\Delta S/S| = \sigma_i\sqrt{\Delta t}$. At 20% implied volatility and one calendar day, that is a move of 1.047%. A dealer short gamma makes money on any day that moves less than that, and loses — quadratically — on any day that moves more. For the 1,000 short calls of §4.3, a 1% day is almost exactly break-even; a 2% day loses about $101, nearly three days' worth of time decay; a 3% day loses about $275, seven days' worth.

2. Gamma is a position in variance, and theta is its price. Long gamma is a purchase of realised variance at the implied price; short gamma is a sale. The position's direction is irrelevant — the squared return in the bracket does not care about sign. That is why a hedged dealer does not "want" the price to go anywhere, and why the only thing it wants from the path is that it be quieter (if short gamma) or wilder (if long gamma) than implied volatility charged for.

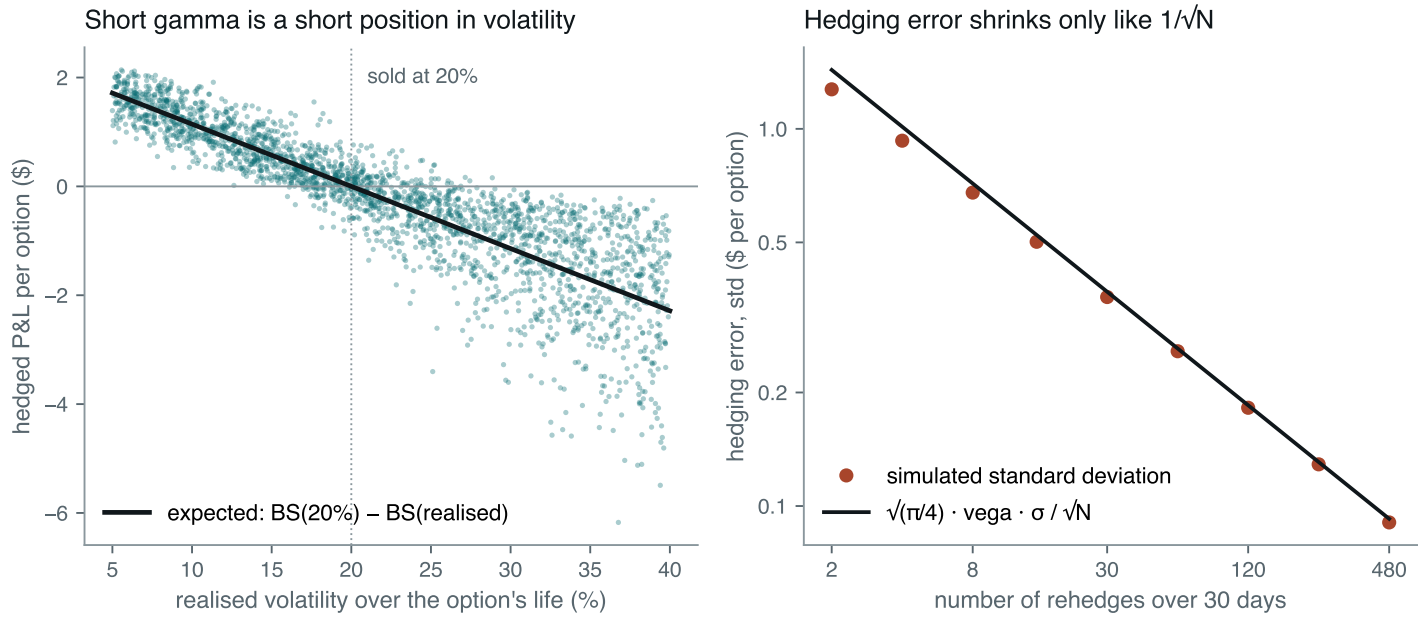
3. Summed over the option's life, the P&L is a weighted variance surprise. Adding up the intervals,

$$\Pi_{\text{total}} \approx \tfrac{1}{2}\sum_t \Gamma_t S_t^2 \left[\left(\frac{\Delta S_t}{S_t}\right)^2 - \sigma_i^2 \Delta t\right],$$

and in continuous time this becomes the robustness result of El Karoui, Jeanblanc-Picqué and Shreve (1998): for options with convex payoffs such as calls and puts, a seller who hedges with a volatility at or above the true one at every moment cannot lose on the hedged position, a seller who hedges below it cannot win, and the size of the gain or loss is the integral of dollar gamma against the gap between the two variances.

4. The expected P&L is a volatility spread. For an option sold at σ_i, hedged at σ_i, and held to expiry while the underlying actually moves with volatility σ_r, the expected P&L (with no drift and zero rates) is exactly the difference in the option's price at the two volatilities, $V(\sigma_i) - V(\sigma_r)$, which is close to $\nu \times (\sigma_i - \sigma_r)$. The figure below simulates it. For our option, realised volatility of 10% earns $1.14 per option on average — half the premium — and 30% loses $1.14. **[Simulated]**

5. The P&L is not deterministic even when volatility is right. The scatter around the line in the figure is hedging error, the subject of §4.7: with realised volatility between 15% and 25%, the dealer lost money on 46% of paths, and with realised volatility between 25% and 30%, on 96% of them. **[Simulated]**



4.5 Long gamma and short gamma, as businesses

The identity turns "long gamma" and "short gamma" from jargon into two businesses with opposite economics.

	Long gamma (owns options, hedges)	**Short gamma** (sold options, hedges)
Hedge trade after a rise	Sells	Buys
Hedge trade after a fall	Buys	Sells
Daily cash flow from time	Pays theta	Collects theta

	Long gamma (owns options, hedges)	**Short gamma** (sold options, hedges)
Makes money when	Realised volatility exceeds implied	Realised volatility is below implied
Typical P&L pattern	Many small losses, occasional large gains	Many small gains, occasional large losses
Effect of its hedging on the price	Leans against moves: stabilising	Chases moves: destabilising
Practitioner name	"Gamma scalping"	"Selling premium", "short vol"

A long-gamma hedger sells into every rally and buys every dip, and each of those round trips locks in a small gain; whether the gains outrun the rent is a question about realised volatility. A short-gamma hedger does the opposite, collects the rent, and pays out when the market moves. Neither is a directional bet. **Both are volatility bets, and their hedging has opposite effects on the market they trade in** — the observation that §7 turns into a model.

Which side do dealers end up on? That depends on what customers want, and §6.3 shows the answer differs by product and era. What *is* established is that selling equity index volatility has been compensated on average: delta-hedged positions in S&P 500 index options have lost money for buyers (Bakshi and Kapadia, 2003), and implied variance on the index has exceeded subsequently realised variance on average, by a margin that constitutes a large **variance risk premium** (Carr and Wu, 2009). [Fact] That premium is not free money. It is the price of the rare large losses in the right-hand column, and it is earned by whoever holds short index gamma through the crash that eventually comes.

4.6 Which volatility to hedge at

The dealer in §4.3 computed its hedge from the implied volatility of 20%, even though the path it faced realised something else. It could not have done otherwise, since realised volatility is only known afterwards. But the choice has consequences worth knowing, set out by Carr and Madan (1998) and Ahmad and Wilmott (2005).

- **Hedging at implied volatility** makes each day's P&L the clean identity of §4.4: smooth and explainable, and consistent with marking the option to market. The *total* P&L, however, is path-dependent. It is the dollar-gamma-weighted sum of variance surprises, so variance that arrives when the price sits near the strike late in the option's life counts heavily, and variance that arrives far from the strike counts for almost nothing. Two paths with identical realised volatility can produce very different results.
- **Hedging at the (known) realised volatility** would lock the total P&L at $V(\sigma_i) - V(\sigma_r)$ regardless of path, but the daily mark-to-market would swing, because the hedge no longer matches the option's market sensitivity.

Dealers overwhelmingly hedge with deltas from their implied volatility surface, because their risk limits, capital and P&L are measured on marked-to-market positions. [Practice] For gamma exposure, this matters in one way: the hedge flows that the market sees are computed from *implied* volatilities, so a model of those flows should use implied volatilities too (§11.3).

4.7 Discrete hedging: the error that does not diversify

The identity holds interval by interval. A dealer that re-hedges once a day faces, each day, a squared return that is a random draw around the variance implied volatility charged for. Even if implied volatility is exactly right on average, a day's squared return is not its average. Writing the day's return as $\sigma\sqrt{\Delta t}\,Z$ with Z standard normal, each interval contributes $\frac{1}{2}\Gamma S^2\sigma^2\Delta t\,(Z^2-1)$: mean zero, standard deviation proportional to gamma. Boyle and Emanuel (1980) derived this distribution, and Bertsimas, Kogan and Lo (2000) showed how the accumulated error behaves for general payoffs and processes.

For an at-the-money option hedged N_h times over its life, with volatility exactly as priced, the standard deviation of the total hedging error is approximately

$$\mathrm{sd}(\Pi) \approx \sqrt{\frac{\pi}{4}} \cdot \nu \cdot \frac{\sigma}{\sqrt{N_h}},$$

a rule of thumb due to Derman (1999), whose simulations put the error at 16% of the premium for daily rehedging. The right-hand panel of the figure reproduces it, and the numbers are sobering. **[Simulated]**

Rehedges over 30 days	Error sd per option	As % of premium
2	\$1.27	55.7%
8	\$0.68	29.6%

Rehedges over 30 days	Error sd per option	As % of premium
30 (daily)	\$0.36	15.7%
120 (four times a day)	\$0.18	8.0%
480 (sixteen times a day)	\$0.09	4.0%

Halving the error takes four times as many trades. And two facts make it worse than a per-option number suggests.

- **The error does not diversify within an underlying.** Every option on the S&P 500 is hedged against the *same* price path. A dealer with ten thousand index options does not have ten thousand independent hedging errors; it has one path, weighted by its net gamma. [Fact] Diversification comes only from books on unrelated underlyings.
- **Real hedging is not free.** Each extra rehedge pays a spread and moves the price. That trade-off is the next subsection.

4.8 Transaction costs and hedging bands

Leland (1985) showed that proportional trading costs can be folded into the replication argument as a change of volatility. If each round trip in the underlying costs a fraction k of the price and the hedger rebalances every Δt, the hedger of a *short* option should price and hedge as if volatility were

$$\tilde{\sigma} = \sigma\sqrt{1 + \mathrm{Le}}, \qquad \mathrm{Le} = \sqrt{\frac{2}{\pi}}\,\frac{k}{\sigma\sqrt{\Delta t}}.$$

The **Leland number** Le grows as the hedging interval shrinks, because more frequent trading means more costs. With a round-trip cost of 10 bp and 20% volatility, daily hedging calls for pricing at 20.75%, four hedges a day at 21.5%, and hourly hedging around the clock at 23.4%. A market maker selling options at 20% and hedging aggressively is giving away its edge in costs.

The better answer is not to hedge on a clock at all. Hodges and Neuberger (1989) posed the problem as optimal hedging under costs for a utility-maximising hedger, and Whalley and Wilmott (1997) solved the small-cost limit: **do nothing while the position's delta stays within a band around the Black–Scholes delta, and trade back to the edge of the band when it leaves.** The band's half-width, in units of delta per option, is

$$w = \left(\frac{3\,c\,\Gamma^2 S}{2\,a}\right)^{1/3},$$

where c is again the proportional cost (the same role k played above) and a the hedger's risk aversion. Two features matter more than the constants. The band scales as the **cube root** of cost, so doubling costs widens it by only 26%. And it scales as $\Gamma^{2/3}$: high-gamma positions get wider bands in delta terms, but narrower ones in terms of how far the price can move before a trade ($w/\Gamma \propto \Gamma^{-1/3}$). Near expiry and near the strike, where gamma is large, hedgers trade more often.

In practice desks combine clocks, price-move triggers and delta bands, hedge the net delta of the book rather than of each option, and execute the hedge with algorithms that spread it over time rather than hitting the market at once. [Practice] The consequence for gamma exposure is important and often forgotten: **the hedge flow a dealer population actually sends to the market is lumpier, later and smaller than "dollar gamma times the move"**, which is best read as an upper bound on the immediate flow and a fair description only on average over many moves.

4.9 What dynamic hedging cannot fix

The identity of §4.4 is an expansion for *small* moves over *short* intervals in a market that is *always open* and *liquid*. Each qualifier is a way it fails.

Risk	Why trading the underlying cannot remove it	What dealers do instead
Gaps and jumps	The hedge cannot be adjusted during an overnight gap or a news jump; the whole move is paid at $\frac{1}{2}\Gamma(\Delta S)^2$	Hold less net short gamma into events; buy wings (§9.4)
Volatility changes	A jump in implied volatility reprices every option at once (vega), independently of the price	Vega limits; offsetting option trades

Risk	Why trading the underlying cannot remove it	What dealers do instead
Liquidity	Hedging costs spike exactly when moves are large and many hedgers trade the same way	Wider quotes, smaller size, pre-hedging
Model and skew	A wrong volatility surface gives a wrong delta, and the hedge is wrong before the price moves	Surface calibration; conservative marks
Basis	Index options hedged with futures or ETFs, or the market exposure of single-stock books hedged with index futures: the hedge does not track the option's underlying exactly	Basis limits; direct hedges near expiry
Pin and assignment	Physical settlement at a strike leaves the post-expiry share position unknown (§2.2)	Reduce positions at large strikes into expiry (§9.5)

Gaps deserve a number. Take the 1,000 short at-the-money calls of §4.3 and let the stock open 5% higher tomorrow. The hedged position loses $773 — twenty days of time decay — and a 5% gap down loses $812. A 10% gap loses $2,709, more than the total time decay the dealer could have collected over the option's remaining month. No rehedging frequency helps: the move happened while the market was closed.

The last risk is not in the table because it is not a risk to one dealer. It is a risk to the market: if dealers as a group hold large net short gamma, their hedge trades are large, simultaneous and in the direction of the move. That is where Part III begins.

§4 Key takeaways

1. Hedging converts a large directional risk into a smaller bet on the size of moves; it does not remove risk.
2. The hedged P&L over an interval is $\frac{1}{2}\Gamma S^2[(\Delta S/S)^2 - \sigma_i^2 \Delta t]$: gamma times the surprise in realised variance relative to implied.
3. There is a break-even move, $\sigma_i\sqrt{\Delta t}$ — about 1.05% a day at 20% volatility. Short gamma loses quadratically beyond it.
4. A short-gamma hedger buys after rises and sells after falls; a long-gamma hedger does the reverse. Both are volatility bets; their hedging has opposite effects on prices.
5. Expected hedged P&L is the volatility spread $V(\sigma_i) - V(\sigma_r)$, but the outcome is path-dependent: variance realised near the strike and near expiry counts most.
6. Discrete hedging error falls only like $1/\sqrt{N_h}$ and does not diversify across options on the same underlying.
7. Costs favour hedging bands whose width scales with the cube root of cost; real hedge flows are lumpier, later and smaller than dollar gamma times the move.
8. Gaps, volatility shocks and liquidity crises are unhedgeable by trading the underlying, and they are what short gamma is really paid to bear.

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5. The gamma quadrant

This section is the heart of Part II. A customer can buy or sell, and the option can be a call or a put, so there are four trades a dealer can end up on the other side of. For each one we fix the same numbers, put the dealer into the position, hedge it, and then move the price up and down to see exactly what the dealer's hedging does.

5.1 Four sign rules

Everything in the quadrant follows from four rules, three of which you have already met.

Rule 0 — the dealer holds the opposite of the customer. If a customer buys an option from a dealer, the dealer is short it; if a customer sells, the dealer is long.

Rule 1 — gamma depends only on long or short. Long options have positive gamma and short options negative gamma, for calls and puts alike (§2.6, §3.5).

Rule 2 — delta depends on both. Long call: positive delta. Short call: negative. Long put: negative. Short put: positive.

Rule 3 — the initial hedge follows delta; every later hedge trade follows gamma. To neutralise a position with delta Δ_{pos}, a hedger takes $-\Delta_{\text{pos}}$ shares. After the price moves by ΔS, the position's delta has changed by about $\Gamma_{\text{pos}}\,\Delta S$, so the hedger must trade

$$\text{hedge trade} \approx -\,\Gamma_{\text{pos}}\,\Delta S \quad \text{shares.}$$

With positive gamma, a rise ($\Delta S > 0$) means selling and a fall means buying. With negative gamma, a rise means buying and a fall means selling.

The consequence is worth saying before the examples prove it: **which way the dealer trades *after* the price moves has nothing to do with whether the option is a call or a put.** Calls and puts only decide the sign of the initial hedge.

5.2 The set-up

All four cases use the example of §3: a stock at 100; a strike of 100; 30 days to expiry; 20% implied volatility, unchanged throughout; zero rates and dividends. The trade is 10 contracts, or 1,000 options. Per option:

	Call	Put
Premium	2.2872	2.2872
Delta	+0.5114	−0.4886
Gamma	0.06955	0.06955
Theta per day	−0.0381	−0.0381
Vega per vol point	0.1143	0.1143

In each case, the dealer hedges at the moment of the trade, a day passes, the price is somewhere between 98 and 102, and the dealer re-hedges. The tables show the dealer's P&L on the options and on the hedge, the total, the position's new delta, and the hedge trade required to become neutral again (positive means buy).

5.3 Case 1 — a customer buys calls; the dealer is short calls

Who and why. Buying calls is how an investor bets on a rise with limited downside and leverage: $2,287 controls exposure to about 51,000 dollars' worth of stock. Call buyers include retail speculators, investors expecting news, and short sellers capping their risk. In single stocks, waves of call buying are what "gamma squeeze" stories are about (§7.10).

The dealer's position. Short 1,000 calls: delta −511 shares, gamma −69.5 shares per $1, theta +$38.11 a day (it collects decay), vega −$114 per volatility point.

The hedge. The dealer is short 511 shares' worth of exposure, so it **buys 511 shares**.

One day later, price at	98	99	100	101	102
P&L on short calls	+919.52	+514.22	+38.43	−507.91	−1,122.71
P&L on 511 shares	−1,022.87	−511.44	0.00	+511.44	+1,022.87
Total	**−103.35**	**+2.79**	**+38.43**	**+3.52**	**−99.84**
New position delta	−370.6	−440.3	−511.2	−581.1	−647.8
Re-hedge trade	**sell 140.8**	**sell 71.1**	0.2	**buy 69.7**	**buy 136.4**

What happened. If the stock rises to 101, the calls are more likely to finish in the money; the short position's delta falls to −581, and the dealer must **buy** another 70 shares — at the higher price. If it falls to 99, the delta shrinks to −440 and the dealer must **sell** 71 shares — at the lower price. The dealer buys strength and sells weakness. On a quiet day it keeps the $38 of decay; a $1 move roughly breaks even; a $2 move in either direction costs about $100.

Effect on the market. The dealer's re-hedging adds buying to rallies and selling to declines: **it amplifies moves.** The more calls customers have bought, the stronger the effect.

5.4 Case 2 — a customer sells calls; the dealer is long calls

Who and why. Selling calls against a stock one owns — **covered call writing** or **overwriting** — is how investors earn income from holdings they would be willing to sell at a higher price. It is done by individuals, pension funds, and a large family of income-oriented exchange-traded funds. Lakonishok, Lee, Pearson and Poteshman (2007) found that a large share of call writing by non-market-maker investors was part of covered positions. Selling the upside is also one leg of a **collar** (§5.9).

The dealer's position. Long 1,000 calls: delta +511, gamma +69.5 shares per $1, theta −$38.11 a day (it pays decay), vega +$114 per volatility point.

The hedge. The dealer is long 511 shares' worth of exposure, so it **sells 511 shares** (short, or against other inventory).

One day later, price at	98	99	100	101	102
P&L on long calls	−919.52	−514.22	−38.43	+507.91	+1,122.71
P&L on short 511 shares	+1,022.87	+511.44	0.00	−511.44	−1,022.87
Total	**+103.35**	**−2.79**	**−38.43**	**−3.52**	**+99.84**
New position delta	+370.6	+440.3	+511.2	+581.1	+647.8
Re-hedge trade	**buy 140.8**	**buy 71.1**	−0.2	**sell 69.7**	**sell 136.4**

What happened. Every number of Case 1 with the sign reversed. When the stock rises, the long calls' delta grows and the dealer **sells** more shares — into the rally. When it falls, the dealer **buys** — into the decline. The dealer pays $38 a day in decay and earns it back only on days that move more than about 1%.

Effect on the market. The dealer sells rallies and buys dips: **it dampens moves.** Large customer call overwriting, by leaving dealers long calls, tends to calm the price around the strikes being sold. At the market level this is [Hypothesis]; §7.4 weighs the evidence.

5.5 Case 3 — a customer buys puts; the dealer is short puts

Who and why. Buying puts is **portfolio insurance**: an investor who owns stocks, or an index fund, pays a premium to cap losses below the strike — a **protective put**. It is also how bearish speculators bet on a fall with limited risk. In equity index options, persistent institutional demand for out-of-the-money puts is the leading explanation of the index skew (§3.7).

The dealer's position. Short 1,000 puts: delta **+489** (a short put gains when the price rises), gamma −69.5 shares per $1, theta +$38.11 a day, vega −$114 per point.

The hedge. The dealer is long 489 shares' worth of exposure, so it **sells 489 shares**.

One day later, price at	98	99	100	101	102
P&L on short puts	−1,080.48	−485.78	+38.43	+492.09	+877.29
P&L on short 489 shares	+977.13	+488.56	0.00	−488.56	−977.13
Total	**−103.35**	**+2.79**	**+38.43**	**+3.52**	**−99.84**
New position delta	+629.4	+559.7	+488.8	+418.9	+352.2
Re-hedge trade	**sell 140.8**	**sell 71.1**	0.2	**buy 69.7**	**buy 136.4**

What happened. When the price falls, the puts move towards the money, the short puts' delta *grows* from +489 to +560, and the dealer must **sell** 71 more shares — into the decline. When the price rises, the delta shrinks and the dealer **buys** back shares — into the rally. Look at the last two rows of Cases 1 and 3: **the totals and the re-hedge trades are identical, to the cent and to the share.** §5.8 explains why.

Effect on the market. Selling into declines and buying into rallies: **it amplifies moves** — and it does so most violently in a selloff, which is exactly when customers buy the most puts, and when implied volatility rises and inflicts a vega loss on the dealer at the same time. This is the case behind every story of dealer hedging accelerating a crash. [Hypothesis]; §7.5 weighs the evidence.

5.6 Case 4 — a customer sells puts; the dealer is long puts

Who and why. Selling puts earns the premium in exchange for an obligation to buy the stock at the strike if it falls — **cash-secured put writing**, sometimes described as getting paid to place a limit order below the market. Income strategies, systematic put-writing indices and funds, and retail investors "selling premium" all do it.

The dealer's position. Long 1,000 puts: delta −489, gamma +69.5 shares per $1, theta −$38.11 a day, vega +$114 per point.

The hedge. The dealer is short 489 shares' worth of exposure, so it **buys 489 shares**.

One day later, price at	98	99	100	101	102
P&L on long puts	+1,080.48	+485.78	−38.43	−492.09	−877.29
P&L on 489 shares	−977.13	−488.56	0.00	+488.56	+977.13
Total	**+103.35**	**−2.79**	**−38.43**	**−3.52**	**+99.84**
New position delta	−629.4	−559.7	−488.8	−418.9	−352.2
Re-hedge trade	**buy 140.8**	**buy 71.1**	−0.2	**sell 69.7**	**sell 136.4**

What happened. The mirror of Case 3, and — in its last two rows — an exact copy of Case 2. When the price falls, the long puts' delta becomes more negative and the dealer **buys** more shares into the decline; when it rises, the dealer **sells** into the rally.

Effect on the market. It dampens moves. Widespread customer put selling leaves dealers long gamma below the market, cushioning declines towards the strikes that were sold. [Hypothesis]

5.7 The quadrant on one page

The four cases collapse into a two-by-two grid in which **only the columns matter**:

	Customer **buys** — dealer is **short**	Customer **sells** — dealer is **long**
Calls	Dealer short gamma. Hedge: buy. Then buys rallies, sells dips. **Amplifies.**	Dealer long gamma. Hedge: sell. Then sells rallies, buys dips. **Dampens.**
Puts	Dealer short gamma. Hedge: sell. Then buys rallies, sells dips. **Amplifies.**	Dealer long gamma. Hedge: buy. Then sells rallies, buys dips. **Dampens.**

The same information in full, for reference:

Customer trade	Typical customer motive	Dealer holds	Dealer delta	Dealer gamma	Dealer theta
Buys calls	Upside speculation, leverage	Short calls	−	−	+
Sells calls	Overwriting, income, collars	Long calls	+	+	−
Buys puts	Protection, bearish speculation	Short puts	+	−	+
Sells puts	Income, "get paid to buy the dip"	Long puts	−	+	−

Dealer holds	Initial hedge	Hedge trade after a rise	Hedge trade after a fall	Effect on moves
Short calls	Buy shares	Buy more	Sell	Amplifies
Long calls	Sell shares	Sell more	Buy	Dampens
Short puts	Sell shares	Buy back	Sell more	Amplifies
Long puts	Buy shares	Sell	Buy more	Dampens

Three sentences summarise the section:

1. **Customers buying options — calls or puts — leave dealers short gamma, and dealer hedging then chases the price.**
2. **Customers selling options — calls or puts — leave dealers long gamma, and dealer hedging then leans against the price.**
3. **The initial hedge depends on call versus put; everything after it depends only on who is long.**

5.8 Why call versus put does not matter: parity, trade by trade

Cases 1 and 3 produced the same P&L and the same re-hedge trades in every scenario, as did Cases 2 and 4. This is not a coincidence of the numbers chosen; it is put–call parity (§2.6).

Write the dealer's hedged short put as *short one put and short 0.4886 shares* (the put's delta hedge). Substitute parity, $P = C - S + K$:

$$-P - 0.4886\,S \;=\; -(C - S + K) - 0.4886\,S \;=\; \underbrace{-C + 0.5114\,S}_{\text{hedged short call}} - K.$$

The hedged short put *is* a hedged short call plus a fixed amount of cash, $-K$, and cash does not move. Every future price, every P&L, every hedge trade is the same. In words: a put is a call plus a short position in the stock; once the dealer has hedged, the short stock has been absorbed into the hedge, and what is left is the same curvature.

The equivalence is exact for European options with zero rates and dividends, and very close otherwise. It has a practical consequence that reaches all the way to §6: **a gamma exposure model that gives calls a positive sign and puts a negative sign is not making a claim about calls and puts.** It is making an assumption about *who is long each*: that customers mostly sell calls and buy puts. Change that assumption and the sign of the answer changes with it.

5.9 Combinations, collars, and a dealer's book

Customers trade combinations, and dealers hold books. The quadrant extends by adding.

- **A straddle** is a call and a put at the same strike. A customer who buys a straddle leaves the dealer short both: gamma −139 shares per $1 for 10 of each, twice Case 1, with almost no initial delta (the dealer buys about 23 shares, the difference between the two deltas). Twice the amplifying flow.
- **A call spread** — buying the 100-strike call and selling the 105 — leaves the dealer short the 100 and long the 105. With the stock at 100 the dealer's gamma is $-69.5 + 49.6 = -19.9$ shares per $1 per 1,000 options: short gamma, but a little over a quarter of Case 1's. As the price climbs towards 105, the long 105s dominate and the dealer's gamma turns positive. **One position, opposite hedging behaviour at different prices.**
- **A collar** is what an investor who owns stock does to protect it cheaply: buy an out-of-the-money put and pay for it by selling an out-of-the-money call. For 10,000 contracts of a 95-strike put and a 105-strike call, the dealer ends up long the 105 calls and short the 95 puts. Its gamma at 100 is small and positive (+4,186 shares per $1), but at 95 it is short $5.1 million of dollar gamma per 1% move, and at 105 it is long $5.8 million. **Below the market the dealer amplifies; above it, it dampens.** Some large funds run systematic index collars that reset on a fixed calendar, and their strikes are closely watched for exactly this reason. [Practice]

A dealer's book is the sum of thousands of such positions, and its gamma is a function of the price: a **gamma profile**, positive at some prices and negative at others. The collar book above is the simplest interesting case — short gamma below, long gamma above — and it is, not by accident, the shape usually attributed to the whole index options market: the aggregate of many customers who own stock, buy downside protection and sell upside calls. §6 builds that aggregate.

5.10 Others who trade like hedgers

The flows of §5.3–§5.6 are not unique to options dealers. Any strategy that **manufactures an option by trading** places the same trades as a dealer hedging the opposite option, because replicating a long option means buying after rises and selling after falls (§3.2), and replicating a short option means the reverse.

Participant	Trades after a rise	Trades after a fall	What it is replicating	Effect on moves
Dealer short options, hedging	Buys	Sells	A long option, against its short one	Amplifies
Portfolio insurer	Buys	Sells	A long put on its portfolio	Amplifies
Leveraged or inverse ETF	Buys	Sells	Constant daily leverage	Amplifies
Trend follower	Buys	Sells	A long, straddle-like payoff	Amplifies
Dealer long options, hedging	Sells	Buys	A short option, against its long one	Dampens
Fixed-weight rebalancer	Sells	Buys	A short, option-like payoff	Dampens

Three of these rows are worth a sentence each.

Portfolio insurance in the 1980s was exactly Case 3 without the dealer: instead of buying puts, institutions replicated them by selling index futures as prices fell, following the recipe of Rubinstein and Leland (1981). The replication required selling into the decline, and on 19 October 1987 it did. §8.1 tells that story.

Leveraged and inverse ETFs must rebalance every day to keep their leverage constant. A fund with leverage L and assets A whose index returns ρ must trade

$$\text{rebalance} = L(L-1)\,A\,\rho$$

in the index at the close, which is positive — buying after a rise — for $L > 1$ and, perhaps surprisingly, also for inverse funds with $L < 0$ (Cheng and Madhavan, 2009). $10 billion in 3× funds on an index that rises 2% must buy $1.2 billion at the close: a short-gamma position of $600 million per 1% move. Whether this measurably moves prices is [Contested]: Ivanov and Lenkey (2018) find that investor flows into and out of the funds offset much of the rebalancing, while Barbon, Beckmeyer, Buraschi and Moerke (2021) find end-of-day effects from both leveraged-ETF rebalancing and options delta hedging, with the ETF effect fading over time.

Trend followers buy strength and sell weakness at horizons of weeks to months, and their returns resemble long straddles; the companion note Trend-Following in Financial Markets shows in its fifth section that a trend rule is long the energy of the filtered trend and short realised variance, and that its convexity is synthetic and exposed to gaps. Those are the two defining properties of a short-gamma dealer's *hedge*. A trend follower is, in trading terms, a hedger without the option.

The table has a lesson for anyone reading "dealer gamma" commentary: dealers are not the only source of move-amplifying flow, and a model that attributes every accelerating selloff to options hedging has left out several neighbours who trade the same way.

§5 Key takeaways

1. The dealer holds the opposite of the customer. Customers buying options leave dealers short gamma; customers selling options leave dealers long gamma.
2. A short-gamma dealer's hedging buys rallies and sells dips and amplifies moves; a long-gamma dealer's hedging sells rallies and buys dips and dampens them.
3. Call versus put determines only the initial hedge. After that, a dealer short calls and a dealer short puts make identical trades and identical P&L — exactly, by put–call parity.
4. For 1,000 at-the-money options, each $1 move forces about 70 shares of re-hedging; each $2 move, about 140. The dealer collects or pays $38 a day in decay for the privilege.
5. A "calls positive, puts negative" gamma convention is an assumption about who is long, not a property of the contracts.
6. Combinations have gamma that changes sign with the price. A collar leaves the dealer short gamma below the market and long above it — the shape usually attributed to the whole index market.
7. Anyone who manufactures an option by trading — portfolio insurers, leveraged ETFs, trend followers — places the same trades as a dealer hedging the opposite option.

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6. Gamma exposure

6.1 From one book to a market

§5 followed one dealer and one position. The claim that dealer hedging moves markets is a claim about all dealers and all positions on an underlying at once. The construction is the obvious one: add up.

For each option series i on the underlying — each combination of strike, expiry and call or put — let n_i^D be the dealers' net position in contracts (positive if dealers are net long). The dealers' combined gamma, in shares of the underlying per $1 move, is $\sum_i n_i^D M\,\Gamma_i(S)$. Converting to dollars per 1% move as in §3.9 gives the central quantity of Part III:

$$\boxed{\text{GEX}(S) \;=\; \sum_i n_i^D\, M\,\Gamma_i(S)\; S^2 \times 0.01}$$

Read it as an order that will be placed if the price moves. If GEX is positive, then after a 1% rise dealers must *sell* GEX dollars of the underlying to stay hedged, and after a 1% fall they must *buy* it. If GEX is negative, after a 1% rise they must *buy* |GEX| dollars, and after a 1% fall *sell* it. Positive GEX is the collective version of Cases 2 and 4; negative GEX is Cases 1 and 3.

The formula looks innocent. It rests on six assumptions, and most of this section is about how far each holds.

1. **The dealers' positions n_i^D are known.** They are not; they must be inferred (§6.3). This is by far the weakest link.
2. **Dealers hedge fully, to the model delta.** In practice they hedge in bands, net across books, and spread execution over time (§4.8).
3. **Each option's gamma is the Black–Scholes gamma at its implied volatility.** Dealers' own models differ, and gamma at a hypothetical price depends on how the volatility surface is assumed to move (§3.7).
4. **The move is small.** GEX is a slope; for large moves the whole profile matters (§6.5).
5. **Every product on the same underlying risk is included.** For the S&P 500 that means index options (SPX and SPXW), options on the SPY and other ETFs, options on E-mini futures, and ideally over-the-counter positions — the last of which are invisible.
6. **Nobody else offsets the hedging.** If the other side of the dealers' options also hedges dynamically, the two hedges cancel (§2.5).

SqueezeMetrics (2017), the practitioner white paper that popularised the term, states four assumptions of its own, and they map onto these: every option is facilitated by a delta hedger; calls are sold by investors and bought by market makers; puts are bought by investors and sold by market makers; and market makers hedge precisely to the option's delta — adopted, the paper says, because hedging bands cannot be observed.

6.2 Units and normalisations

Gamma exposure numbers quoted in commentary differ by factors of 100 or more for the same market, because they use different units. Before comparing any two numbers, find out which of these they are.

Convention	Formula per series	Units	Comparable across price levels?
Share gamma	$nM\Gamma$	Shares (or index units) per 1-point move	No
Dollar gamma per point	$nM\Gamma\, S$	Dollars of delta per 1-point move	No
Dollar gamma per 1%	$nM\Gamma\, S^2 \times 0.01$	Dollars to trade per 1% move	**Yes**
Gamma P&L per 1%	$\frac{1}{2}nM\Gamma\,(0.01S)^2$	Dollars of P&L from a 1% move	Yes
Normalised by liquidity	GEX divided by daily dollar volume or depth	Dimensionless	Yes, across assets and time

The SqueezeMetrics formula, for example, is $\Gamma \times \text{OI} \times 100$ — share gamma — with the note that for the index it is expressed in dollars. This document uses dollar gamma per 1% throughout.

To calibrate intuition for the size of these numbers, take the S&P 500 at 6,000 with 15% implied volatility. **One at-the-money SPX contract carries about $55,600 of dollar gamma per 1% with 30 days to expiry.** A hundred thousand such contracts — a large but ordinary concentration of open interest — carry $5.6 billion. The same hundred thousand contracts carry $30.5 billion with one day left and $61 billion with a quarter of a day left, because of the $1/\sqrt{\tau}$ scaling of §3.5; at 30% implied volatility with 30 days left, they carry only $2.8 billion.

Products on the same underlying combine once each is in dollars per 1%: an SPY option contract (100 shares of an ETF priced near a tenth of the index) carries about a tenth of the dollar gamma of the matching SPX contract, and E-mini futures options have a multiplier of 50 rather than 100.

Raw dollars are still not the right unit for asking *whether hedging matters*, because $5 billion of hedging means something different in a market that trades $400 billion a day than in a stock that trades $400 million. §7.1 introduces the ratio that does answer that question.

6.3 The positioning problem: who is actually long?

Open interest says how many contracts exist. It never says who holds which side (§2.5). The dealers' net position n_i^D in every series must therefore be assumed or estimated, and there are five ways to do it.

Approach	How dealer positions are assigned	Strengths	Weaknesses
Sign convention	Dealers long all call OI and short all put OI	Needs only public open interest	Wrong whenever customers buy calls or sell puts; wrong for many single stocks; cannot adapt
All short	Dealers short all OI	A worst-case bound	Contradicted by the evidence below for single stocks
Flow accumulation	Sum signed opening and closing volume by participant type from exchange open–close data	Uses actual capacity codes; adapts to behaviour	Needs long history and every exchange; assumes which participants hedge
Trade-side inference	Classify each trade as a customer buy or sell by comparing its price to the quotes	Public tick data; captures intraday flow	Many option trades print at the midpoint, in auctions or as multi-leg orders, where the side is ambiguous
Audit-trail data	Exchange or regulatory records with each side's identity	Closest to the truth	Not available in real time; mostly academic

Vendors blend these in undisclosed proportions. The trade-side approach relies on quote rules of the kind Lee and Ready (1991) evaluated for stocks; in options, Savickas and Wilson (2003) found that the standard rules signed only 59% to 83% of classifiable trades correctly. A dealer's net position is a small difference between large gross flows, so misclassifying a fifth of trades can reverse its sign. Exchange open–close files — Cboe's covers its four options exchanges, tags each side as customer, professional customer, broker-dealer or market maker, and marks it as opening or closing — are the best public input, and SPX options, which trade only on Cboe, are the rare product where one exchange's file covers the whole market.

What does the measured evidence say? It says the sign convention is right in some places, wrong in others, and that the answer changes over time.

Study	Market and period	What it found about dealers' net position
Lakonishok, Lee, Pearson and Poteshman (2007)	CBOE single-stock options, 1990–2001	Non-market-makers held more written than purchased open interest in *both* calls and puts: market makers net long both
Gârleanu, Pedersen and Poteshman (2009)	CBOE index and single-stock options, 1996–2001	End users net long S&P 500 index options, with large net long positions in out-of-the-money puts: dealers net short index options. End users net *short* single-stock options
SqueezeMetrics (2017)	SPX options	*Assumes* dealers long calls, short puts — a convention, not a measurement
SEC staff (SEC, 2021)	GameStop options, January 2021	Market makers were buying, not writing, calls; customer volume was driven more by put buying than call buying
Cboe (2023)	SPX 0DTE options, 2023	Market makers' net 0DTE gamma small and balanced through the day — from the exchange that lists the product
Adams, Fontaine and Ornthanalai (2024)	SPX 0DTE options	Market makers match most 0DTE order flow, and absorb positions opened earlier that become 0DTE
Amaya, Garcia-Ares, Pearson and Vasquez (2025)	SPX and SPXW, Cboe trade records, July 2020 – June 2023	Aggregate market-maker gamma typically positive, but often negative

Four lessons follow, and they should shape any gamma-exposure number you build or buy.

- **Index and single-stock markets differ.** The classic evidence has dealers short index options — especially downside puts — and long single-stock options. [Fact] for the samples studied. A convention calibrated to one is wrong for the other.
- **Behaviour changes.** The 2020–2021 surge in retail call buying, the growth of covered-call income funds, and the rise of 0DTE trading each changed who holds what. A fixed convention cannot see any of it.
- **Sign errors are not noise.** In the worked example below, flipping the convention changes the answer from +$3.9 million to −$63 million per 1%. A wrong positioning assumption does not make a GEX estimate imprecise; it makes it *wrong-signed*.

- **"Dealers hedge, customers don't" is itself an approximation.** Proprietary trading firms and volatility hedge funds hedge too, while many customers hold static positions. Ni, Pearson, Poteshman and White (2021) sort holders into those "likely to hedge" — market makers and firm proprietary traders — and those not, rather than by the dealer label.

Interested parties deserve a flag. SqueezeMetrics and other vendors sell gamma data; Cboe earns fees on the options whose market impact its research finds to be modest; and the academic committee that criticised the SEC's GameStop report (§8.5) included the founder of a gamma-analytics firm. None of that makes their work wrong. It is a reason to weight independent replications more heavily.

6.4 A worked calculation

Here is the whole computation on a chain small enough to check by hand: a stock at 100, one expiry 30 days away, five strikes, and the sign convention (dealers long calls, short puts). Implied volatility slopes down across strikes, as index skew does.

Strike	Implied vol	Call OI	Put OI	Gamma per option	$ per 1% per contract
90	24%	2,000	25,000	0.01702	170.2
95	22%	5,000	30,000	0.04427	442.7
100	20%	20,000	15,000	0.06955	695.5
105	18%	30,000	3,000	0.05065	506.5
110	17%	15,000	1,000	0.01268	126.8

The last column is $\Gamma \times S^2 \times 0.01 \times M$ (§6.2) with $S = 100$ and $M = 100$ substituted — $\Gamma \times 100^2 \times 0.01 \times 100$: the dollar gamma per 1% of one contract. Multiply by open interest and apply the signs:

Strike	From dealers' long calls $(m)\|From dealers' short puts$(m)	Net ($m per 1%)	
90	+0.34	−4.26	−3.92
95	+2.21	−13.28	−11.07
100	+13.91	−10.43	+3.48
105	+15.19	−1.52	+13.68
110	+1.90	−0.13	+1.78
Total	**+33.56**	**−29.62**	**+3.95**

At a price of 100, dealers are net long $3.95 million of gamma per 1%: after a 1% rise they sell about $4 million of stock, after a 1% fall they buy it. Three things are visible even in five rows.

- **The total is a small difference between two large numbers.** $33.6 million of call gamma against $29.6 million of put gamma. A modest error in either side's positioning swamps the net.
- **The contributions are local.** The 105 strike contributes most at a price of 100 because it combines heavy call open interest with substantial gamma; the 90 strike contributes little despite large put open interest, because a 90-strike option on a stock at 100 has little gamma with 30 days left.
- **The convention decides the sign.** If dealers were instead short everything, the total at 100 would be **−$63.2 million** per 1%.

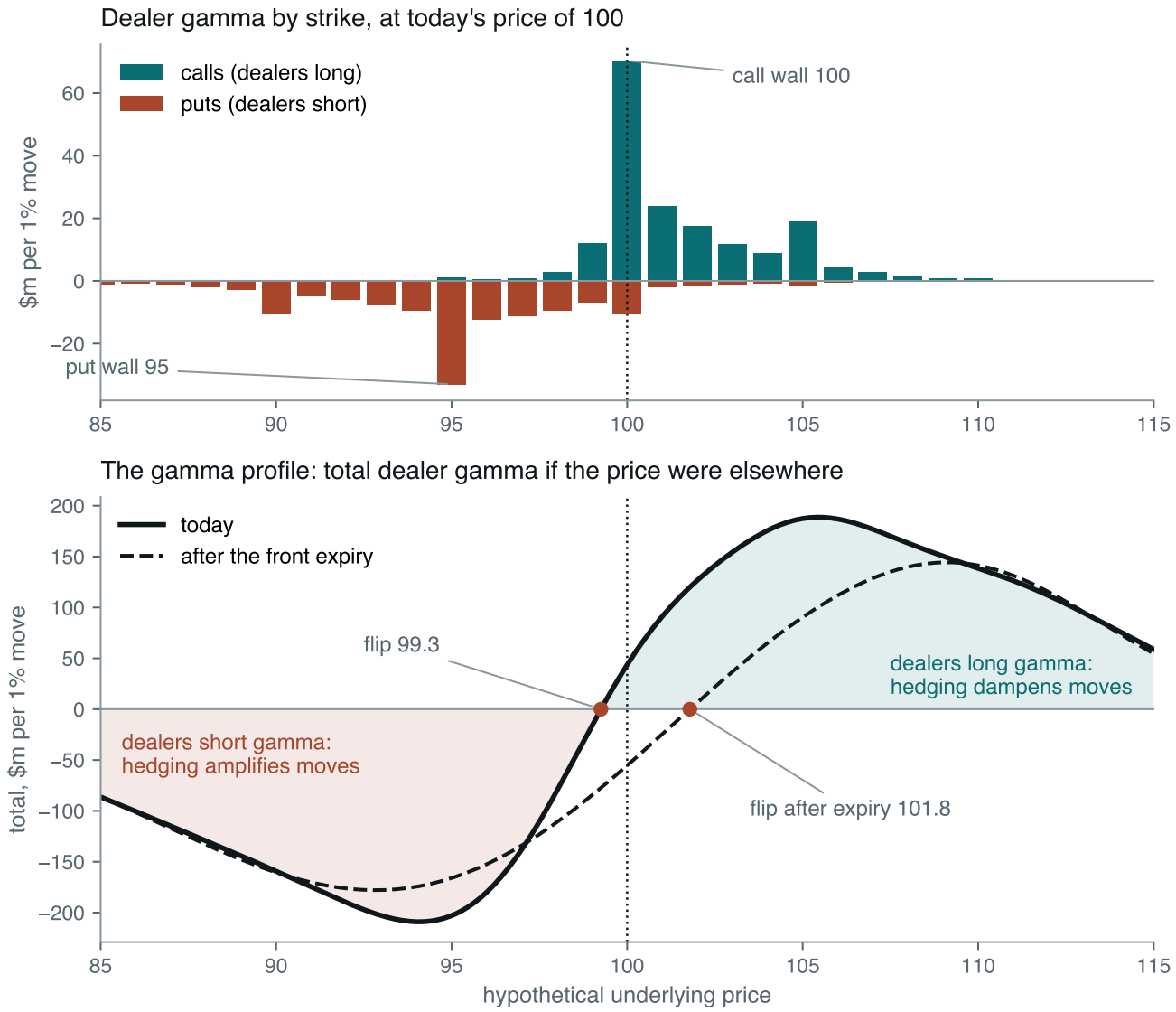
Now recompute the total at other prices, holding each strike's implied volatility fixed — the sticky-strike convention of §3.7:

Price	94	96	97	98	99	100	102	104	106
GEX ($m per 1%)	−20.7	−15.3	−11.4	−6.7	−1.5	+3.9	+14.7	+23.5	+28.8

The total crosses zero at **99.28**, only 0.7% below the current price. Above it, dealers' hedging leans against moves; below it, the heavy put open interest at 95 and 90 comes into range and dealers' hedging increasingly chases them.

6.5 The gamma profile and the gamma flip

The row you just computed is a **gamma profile**: total dealer gamma exposure as a function of a hypothetical underlying price. The price at which it crosses zero is the **gamma flip**, also called the **zero-gamma level**. The figure shows the profile for a larger synthetic index chain with two expiries — a front month with 7 days left and a back month with 35 — under the same convention.



Today the flip is at 99.3 and, at the current price of 100, dealers are long $43.7 million of gamma per 1%. **[Simulated]** One week later — the same open interest in the back month, the same implied volatilities, *the same price* — the front month has expired, and with it the long gamma dealers held in heavy near-the-money call open interest. The flip has moved to 101.8, and at a price of 100 dealers are now *short* $55.3 million per 1%. Nothing happened to the price. The regime changed because contracts expired. §7.7 returns to this.

Four things move the profile and its flip, and it is worth separating them.

1. **Time.** Near-dated gamma concentrates around its strikes and then disappears at expiry (§3.5). The flip can move a long way on expiry days with no price change at all.
2. **Implied volatility.** Higher implied volatility spreads each option's gamma over a wider range of prices and lowers it at the money, flattening the profile; lower implied volatility sharpens it.
3. **New positions.** Every day's trading changes open interest, and customers' buying and selling changes the dealers' side of it.
4. **The model itself.** The sign convention, the products included, and whether the surface is treated as sticky-strike or sticky-delta each move the computed flip.

The price itself moves *along* the profile rather than moving it, but it changes which part of the profile is relevant, and under a sticky-delta assumption it reshapes the profile too.

The flip is a model output, not an observable. It inherits every uncertainty of §6.1 and §6.3, and a profile with a shallow slope near zero can put the flip anywhere within a range of several percent. A published flip level quoted to the index point conveys a precision that does not exist. [Practice] — vendors publish point estimates; I would treat any flip as a band whose width is at least the change produced by switching between two reasonable positioning assumptions.

6.6 Walls, pins, and other strike-level features

Traders name several features of the by-strike picture.

- **Call wall** — the strike with the largest dealer call gamma (under the convention, where call open interest and gamma combine most heavily). In the synthetic chain, 100.
- **Put wall** — the strike with the largest dealer put gamma. In the synthetic chain, 95.
- **High-gamma strikes** — strikes carrying large gamma of either kind, especially in an expiry that is close.

The practitioner reading is that a call wall acts as resistance and a put wall as support. [Practice] It is worth being precise about what the mechanics do and do not support.

Near a **call wall held long by dealers**, dealer gamma is positive and large, so hedging leans against moves through the strike: a mechanical reason for rallies to slow there. But if customers are *long* those calls — speculative call buying rather than overwriting — the same strike is where dealer hedging *accelerates* a rally. The wall's effect depends entirely on the positioning assumption behind it.

Near a **put wall held short by dealers**, dealer gamma is at its most *negative*, so hedging amplifies moves towards the strike — the opposite of support. The mechanical case for "support" is subtler: once the price falls well through a large put strike, those puts are deep in the money, their gamma fades (their delta saturates at −1), and the selling pressure from re-hedging them *diminishes*. The hedging pressure peaks around the wall and weakens beyond it. That can slow a decline below the wall; it does not make the wall hold.

Max pain is a different idea that often travels with these. It is the strike at which the total intrinsic value paid to option holders at expiry would be smallest, computed from open interest alone, and the folklore says prices gravitate towards it into expiry. It contains no hedging mechanism. Prices do cluster at strikes with large open interest on expiration days (§7.7), but I am not aware of peer-reviewed evidence that they gravitate to the max-pain strike as such. [Practice]

6.7 Same name, different thing

Term	What it usually means	What to check
GEX (SqueezeMetrics)	$\Gamma \times \text{OI} \times 100$, calls positive, puts negative, summed over all strikes and expiries	A convention, not a measurement; share units
Net gamma, **dealer gamma** (vendors)	Proprietary estimates of dealers' gamma, often in dollars per 1%	Undisclosed positioning model; units
Market-maker gamma (academic)	Gamma of positions measured from capacity-coded or audit-trail data	Historical only; which participants are counted
Gamma flip, **zero gamma**	Price at which a modelled profile crosses zero	Moves with time, volatility and positions; wide error band
Vendor trigger levels	Proprietary levels below which dealer hedging is said to add volatility	Definitions differ from zero gamma
Call wall, **put wall**	Strike with the largest dealer call or put gamma	By open interest, by gamma, or by dollar gamma?
Delta exposure (DEX)	Dealers' net option delta before hedging	If dealers are hedged, it describes their hedge book, not a directional exposure
Vanna and charm exposure	Aggregate dealer vanna (per volatility point) or charm (per day)	Same positioning problem, plus surface dynamics
Gamma squeeze	Short-gamma hedging feeding a rally in a single stock	Often applied to any rally with call buying
Max pain	Strike minimising holders' payout at expiry	Not a hedging concept

§6 Key takeaways

1. Gamma exposure is dealers' aggregate dollar gamma per 1% move: the dollars of the underlying dealers must sell (if positive) or buy (if negative) after a 1% rise.
2. It rests on six assumptions, and the weakest is that dealers' positions are known. Open interest never says who is long.
3. Measured positions differ by market and era: dealers have historically been short index options (especially puts) and long single-stock options, and 0DTE positions appear largely balanced. A fixed sign convention cannot see any of this.
4. Positioning errors flip signs rather than adding noise: in the worked example, +$3.9 million per 1% becomes −$63 million.
5. Quote gamma exposure in dollars per 1% and compare it to liquidity; raw numbers across sources differ by orders of magnitude because of units alone.
6. The gamma profile shows gamma exposure at hypothetical prices; the flip is where it crosses zero. The flip is a model output with a wide band, and it moves with time, volatility and positions — sometimes a long way on expiry days, with no price change.
7. Walls and pins describe where hedging pressure concentrates, not levels that must hold; their meaning depends on the positioning assumption behind them. Max pain is not a hedging concept.

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7. What hedging does to prices

7.1 The feedback multiplier

Everything so far has described what dealers *must trade*. Whether that trading moves the price depends on a second quantity: how much trading the market can absorb. Here is the simplest model that contains both, and it is worth deriving because every more elaborate model in the literature is a version of it.

Suppose news, or other traders, would move the price by x_f percent if dealers did nothing: the **fundamental move**. Let the actual move be x percent. Dealers holding gamma exposure GEX must trade $-\text{GEX} \times x$ dollars to stay hedged (selling after a rise if GEX is positive). Let the market's **depth** D be the dollars of one-sided trading that move the price by 1%, so that dealers' hedging adds $-(\text{GEX}/D)\,x$ percent to the move. The actual move must then satisfy

$$x = x_f - \kappa\, x, \qquad \kappa = \frac{\text{GEX}}{D},$$

and solving,

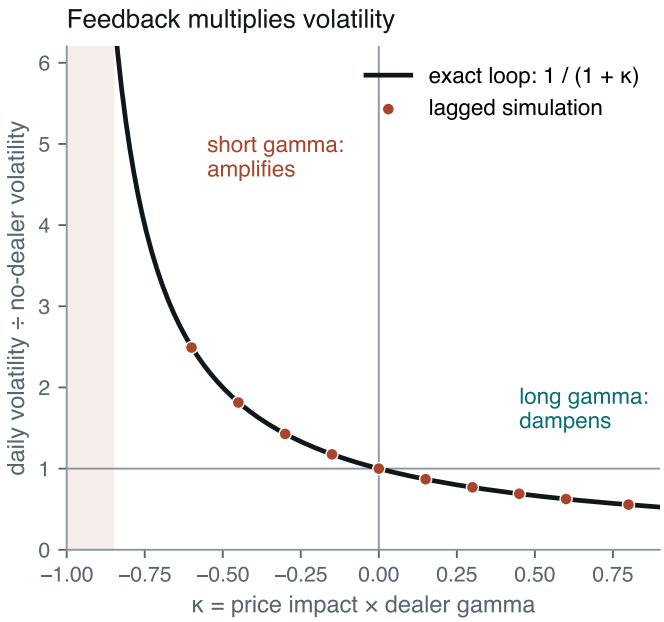
$$\boxed{x = \frac{x_f}{1+\kappa}}$$

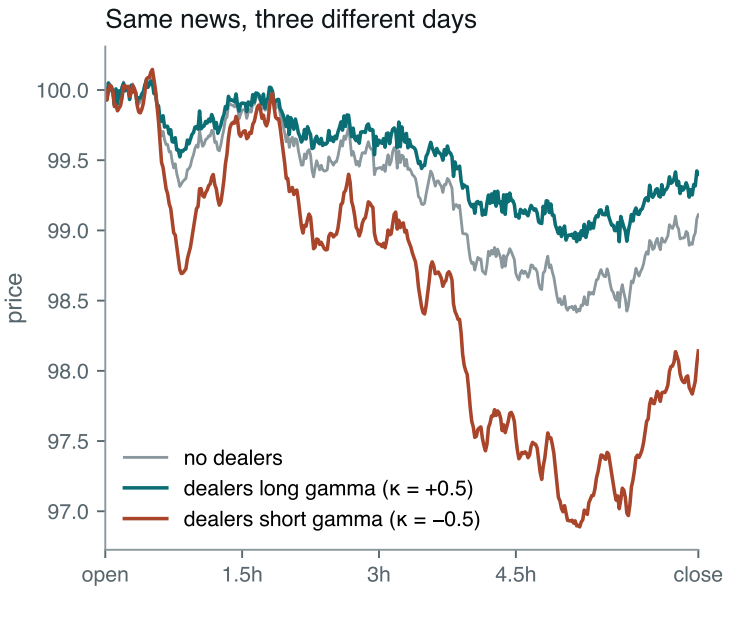
Every move — and therefore realised volatility — is multiplied by $1/(1+\kappa)$**.** The dimensionless ratio κ is dealer gamma measured in units of the market's capacity to absorb it.

- **Dealers long gamma,** $\kappa > 0$**:** the multiplier is below one. Hedging absorbs part of every move. With $\kappa = 0.25$, a 1% fundamental move becomes 0.8%.
- **Dealers short gamma,** $\kappa < 0$**:** the multiplier exceeds one. With $\kappa = -0.2$, a 1% fundamental move becomes 1.25%, and 20% volatility becomes 25%.
- **As** $\kappa \to -1$ the multiplier grows without bound. Dealers' hedging demand for any move exceeds what the market can absorb at that move, and the loop has no stable solution: the price must gap until something else — new sellers of the options, dealers abandoning their hedges, a trading halt — breaks the loop.

This one line is the skeleton of a mature theoretical literature. Grossman (1988) argued that replacing traded options with synthetic ones — portfolio insurance — removes information the market needs to supply liquidity for the replication trades. Gennotte and Leland (1990) showed that when hedging demand is underestimated and liquidity is thin, a small amount of hedging can produce a price *discontinuity*: a crash. Frey and Stremme (1997) derived how volatility is transformed by the demand of dynamic hedgers, finding that it rises and becomes price- and time-dependent; Sircar and Papanicolaou (1998), Platen and Schweizer (1998) and Schönbucher and Wilmott (2000) built the resulting nonlinear Black–Scholes models, in which hedgers' own aggregate gamma, scaled by illiquidity, enters the volatility of the price they are hedging. In each of

them the effective volatility has a denominator that approaches zero as liquidity-scaled hedging demand grows, and that singularity is the model's crash. κ is the same object with the stochastic calculus removed.





7.2 The same fact at two horizons: autocorrelation and volatility

The loop above is solved instantaneously, as if dealers hedged at the same moment the price moved. They do not. They hedge a little later — at the next rebalance, the next band crossing, the close. Let each bar's return be a fundamental shock e_t plus the price impact of dealers hedging the *previous* bar's move:

$$r_t = e_t - \kappa\, r_{t-1}.$$

This small change makes the effect visible in a second place.

- **Autocorrelation.** Consecutive returns now have correlation $-\kappa$. Long-gamma hedging turns each move into a partial reversal (negative autocorrelation); short-gamma hedging turns it into a partial continuation (positive autocorrelation). In the simulation behind the figure, $\kappa = \pm 0.3$ produces lag-one autocorrelations of ∓0.30. **[Simulated]**
- **Volatility over longer windows.** Summing the returns over a day, the fundamental shocks are scaled by $1/(1+\kappa)$ — the same multiplier as the instantaneous loop: the recursion telescopes when summed, leaving $(1+\kappa)\sum r_t \approx \sum e_t$ once the leftover endpoint terms are negligible. The simulation gives a daily volatility ratio of 0.770 for $\kappa = 0.3$ (theory 0.769) and 1.427 for $\kappa = -0.3$ (theory 1.429). **[Simulated]**
- **Volatility at the bar level barely changes.** Bar-by-bar volatility rises by only about 5% at $|\kappa| = 0.3$ in either direction. A market in which dealers are long gamma can look just as noisy tick by tick while being markedly quieter over a day.

So the two empirical signatures of dealer hedging are the same fact seen at two horizons: **intraday autocorrelation of returns, and realised volatility over the day.** The right-hand panel of the figure shows both on one simulated day built from identical shocks: the long-gamma path chops back and forth and goes nowhere; the short-gamma path trends.

The empirical literature has looked in both places.

- Barbon and Buraschi (2020) link large aggregate imbalances in dealers' gamma to intraday momentum when dealers are short and reversal when they are long, and argue the channel depends on limited liquidity rather than on information or funding frictions.
- Baltussen, Da, Lammers and Martens (2021) find, in more than 60 futures markets from 1974 to 2020, that the return over the last 30 minutes of trading is predicted by the return over the rest of the day, that the effect reverses over the following days, and that it is linked to the gamma hedging demand of options market makers and leveraged ETFs. The reversal matters: price pressure reverses, information does not.
- Gao, Han, Li and Zhou (2018) documented the same intraday momentum in the S&P 500 ETF but attributed it to infrequent portfolio rebalancing and late-informed trading — alternative mechanisms that produce the same signature. [Contested] as to how much of intraday momentum hedging explains.

- Ni, Pearson, Poteshman and White (2021), across individual stocks, find that stock volatility is lower when investors likely to hedge — market makers and firm proprietary traders — hold net purchased option positions, and higher, with more large moves, when they hold net written ones.

7.3 How big is it?

The multiplier is only as interesting as κ is large, and κ needs a number for market depth D. The best-established empirical regularity for that is the **square-root law of market impact**: a large order of Q dollars, executed over the day in a market that trades V dollars a day with daily volatility σ_d, moves the price by about

$$\text{impact} \approx Y\,\sigma_d\sqrt{Q/V},$$

with Y a constant of order one, a relation that has held across assets and decades of data (Tóth and co-authors, 2011; Bouchaud, Bonart, Donier and Gould, 2018). [Fact] This is the price impact promised in §1.2, but concave rather than linear: unlike Kyle's (1985) constant lambda, impact grows more slowly than order size. Treat the hedge flow triggered by a 1% move as such an order, and κ is the resulting impact as a fraction of that 1% — which makes D itself depend on the size of the flow, since a concave law has no single order-independent depth.

For the S&P 500, where futures alone trade on the order of $400 billion a day (Cboe, 2023), take dealer gamma of $5 billion per 1% — about what a hundred thousand at-the-money 30-day index contracts carry (§6.2, rounded). Under stated assumptions, the calibration gives:

Situation	Dealer GEX per 1%	Daily volume	Daily vol	$\lvert\kappa\rvert$ for Y = 0.5 to 1	Volatility multiplier
S&P 500, 30-day gamma, normal liquidity	$5bn	$400bn	1%	0.06–0.11	0.90–0.95 long; 1.06–1.13 short
The same contracts, one day before expiry	$30bn	$400bn	1%	0.14–0.28	0.78–0.88 long; 1.16–1.38 short
30-day gamma, stressed liquidity	$5bn	$150bn	2%	0.18–0.37	0.73–0.85 long; 1.22–1.58 short
Mid-cap stock, dealers short after call buying	−$200m	$1bn	3%	0.67–1.34	3× to unstable

These are illustrations of the arithmetic, not measurements, and they carry the model's assumptions: every dealer hedges fully, the hedge flow is not netted against other flows, and the square-root law applies to it. **[Simulated]** in the sense of calibrated. But the first row agrees with the best measurement available. Amaya, Garcia-Ares, Pearson and Vasquez (2025), using Cboe records of every SPX trade to construct market makers' actual positions from July 2020 to June 2023, estimate that the *largest* effect of market-maker gamma was to raise annualised daily volatility by 3.3 points and 30-minute volatility by 6.4 points, with a median effect of reducing daily volatility by 0.08 points. They conclude that these are not large next to ordinary day-to-day swings in realised volatility, whose standard deviation in their sample was 4.5 points.

Three things make κ larger than the headline row, and they are where the interesting episodes live.

1. **Expiry.** The same open interest carries several times the gamma in its final day (row 2), concentrated around a few strikes (§7.7).
2. **Stress.** Depth falls and volatility rises together (row 3). Worse, negative dealer gamma tends to arrive *after* declines, which push prices towards the strikes of the puts customers own, just as liquidity is deteriorating. The magnitude of κ tends to grow exactly when its sign is negative — the interaction the phrase "gamma fragility" in Barbon and Buraschi's title refers to. [Hypothesis]
3. **Single stocks.** Dealer gamma can be a large fraction of daily volume in a stock under concentrated call buying, and the calibration then crosses the singularity (row 4). That is the arithmetic of a squeeze (§7.10).

One further consequence of the square-root law is easy to miss. Because impact grows only with the square root of the hedge, the *relative* feedback is strongest for small moves and weakest for large ones: in the first row's market, the effective κ is about 0.35 for a 0.1% move but 0.11 for a 1% move and 0.07 for a 3% move. Hedging should therefore be most visible where moves are small — intraday chop, closing prices pulled onto strikes — and least decisive in the middle of a large repricing, unless liquidity itself collapses. [Hypothesis]

7.4 The long-gamma regime

When dealers are net long gamma — customers have, on balance, sold them options near the current price — the mechanics predict a recognisable market.

- Realised volatility below implied, as hedging absorbs part of each move.
- Negative intraday autocorrelation: moves that partially reverse.
- Narrow daily ranges, and closes gravitating towards large strikes into expiry (§7.7).

The evidence that this regime exists at all comes mostly from measured positions. Amaya and co-authors find index volatility falls when market-maker gamma is positive, which it usually was in SPX between 2020 and 2023, though the typical effect is small. Adams, Fontaine and Ornthanalai (2024) find index volatility 60 to 90 annualised basis points lower on days when 0DTE options trade, which they attribute to market makers' hedging of expiring positions. Ni and co-authors find the same direction in single stocks. SqueezeMetrics (2017) reported that S&P 500 daily returns had a standard deviation of 0.55% after days in the highest quartile of its GEX measure and 0.85% after days in the second-highest — a striking gap, from an interested party, with no controls.

That last point is the problem with the whole regime. **High gamma-exposure readings and low volatility are linked mechanically, whether or not hedging calms anything.** The same open interest produces higher at-the-money gamma when implied volatility is low (§3.5), and calm markets tend to be rising markets, in which the price has moved up towards call strikes and away from put strikes. A sign-convention GEX will therefore read high in calm markets *because* they are calm, and volatility clusters, so calm tends to continue. A study that finds high GEX followed by low volatility without controlling for current implied and realised volatility has not distinguished hedging from persistence. [Contested]

The same caution applies to a popular recent hypothesis: that the growth of funds selling index calls for income has left dealers structurally long gamma and suppressed index volatility. [Hypothesis] It is consistent with the mechanics of Case 2 (§5.4). I know of no study that separates it from everything else that happened to volatility over the same years.

7.5 The short-gamma regime

When dealers are net short gamma — customers have bought options near the price, above all puts in a falling market — the mechanics predict the mirror image.

- Realised volatility above what fundamentals alone would produce.
- Positive intraday autocorrelation: moves that extend, often into the close, when hedging concentrates.
- "Air pockets": declines that accelerate as they approach large put strikes.

Here the measured evidence is consistent in direction. Ni and co-authors find higher stock volatility and more frequent large moves when likely hedgers are net written; Amaya and co-authors find index volatility elevated when market-maker gamma turns negative; Baltussen and co-authors connect end-of-day momentum to short-gamma hedging demand.

The confound runs the other way from §7.4. Dealer gamma usually turns negative *after* a decline has pushed prices into the strikes of protective puts, by which time volatility is already elevated, implied volatility has jumped, and liquidity has thinned. Negative gamma and high volatility then appear together partly because the selloff caused both. Studies that use lagged gamma and control for lagged volatility — Amaya and co-authors estimate models with lagged squared returns; Ni and co-authors use stock-level panels — address part of this. The direction of the effect is established with measured positions; its size for broad indices, and how much of it is causal, is [Contested].

7.6 Crossing the flip

The commentary version of the gamma flip is a switch: above it the market is calm, below it the market is violent. The mechanics say something more interesting.

At the flip, dealer hedging does nothing. Gamma exposure is zero there, so κ is zero, and hedging neither dampens nor amplifies. Crossing the flip is not a jump from full dampening to full amplification. It is passing through neutrality into a region where amplification *grows* with distance below the flip — following the slope of the profile — until the profile's trough, typically near the largest put strikes, beyond which it fades as those puts go deep into the money.

What does change at the flip is the sign. A decline that starts above the flip is resisted by hedging; if it continues through the flip, every further decline adds hedging pressure instead of absorbing it. That asymmetry is why the flip can matter as a *level* even though the hedging effect at the level itself is zero. [Hypothesis]

The flip also moves while you cross it. A selloff raises implied volatility, which reshapes the profile, and prompts new put buying, which leaves dealers shorter still and pushes the flip *up* — towards, or above, a falling price. The regime can therefore deepen faster than the price falls. [Hypothesis]

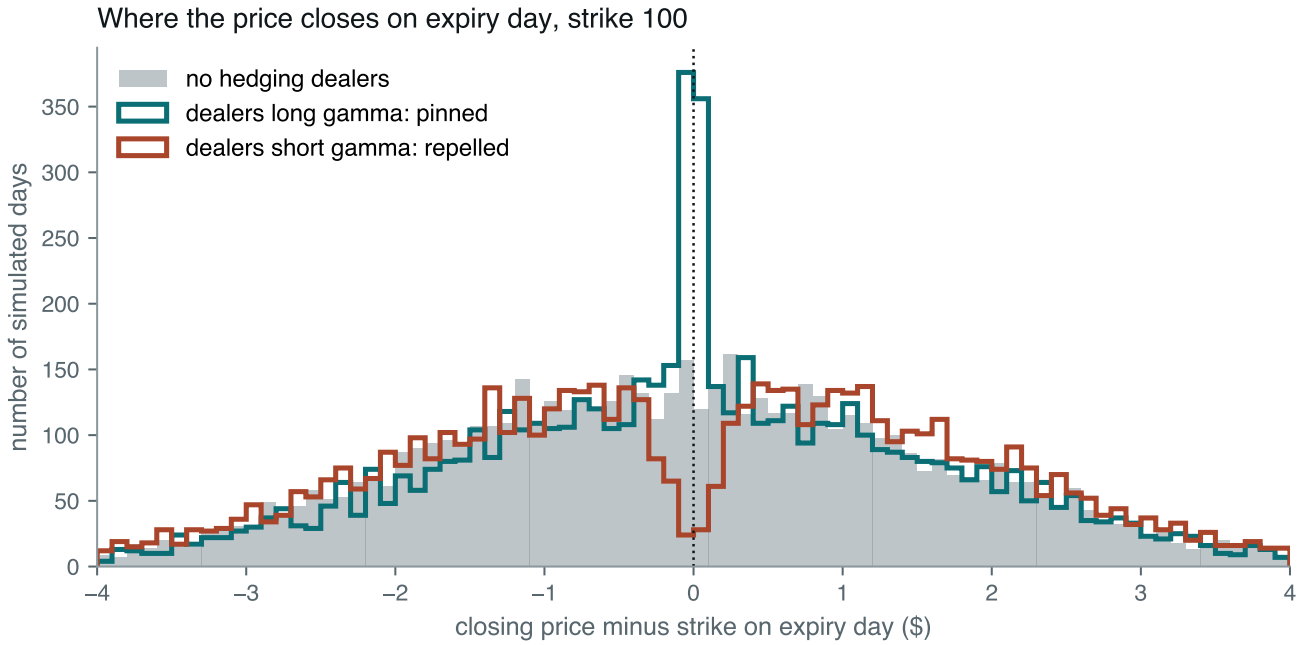
Two practical conclusions follow. First, because the flip's location is uncertain by several percent (§6.5), classify regimes with a buffer — for example, by the distance from the flip in units of daily volatility — rather than by which side of a point estimate the price sits. Second, the more informative quantity is usually not where the profile crosses zero but **how negative it becomes within one or two days' typical move below the price**: the depth and slope of the trough, which set how large κ can get if the decline continues. That is my view, not an established result.

7.7 Approaching expiry

Four things happen as options approach expiry, and together they explain most of what is distinctive about expiration days.

1. Gamma concentrates. At-the-money gamma grows like $1/\sqrt{\tau}$ while out-of-the-money gamma dies (§3.5). Open interest that was a gentle slope of gamma exposure a month earlier becomes, in the final days, a set of sharp spikes at the strikes nearest the price. A hundred thousand at-the-money SPX contracts carry $5.6 billion per 1% with 30 days left and $30.5 billion with one (§6.2).

2. Long-gamma strikes pin; short-gamma strikes repel. Suppose dealers are long a large amount of options at a strike close to the price on expiry day. Above the strike, their hedging sells into every uptick; below it, it buys every downtick; and as the hours pass, gamma at the strike grows. Charm reinforces the pull: above the strike the calls' deltas rise towards one as time passes, so dealers must sell more; below it the deltas drain towards zero, so dealers buy back hedges. Both push the price towards the strike. If dealers are short the strike, every sign reverses and the price is pushed away. The figure simulates an expiry day with a stock starting near a strike carrying large open interest.



With no hedging dealers, 4.6% of simulated days close within 10 cents of the strike. With dealers long the strike, 12.2% do; with dealers short it, 0.9%. **[Simulated]**, with liquidity-limited hedging and parameters set out in `figures/dh_pinning.py`.

The theory is due to Krishnan and Nelken (2001) and Avellaneda and Lipkin (2003), who wrote down the price dynamics with a drift towards the strike that becomes singular as expiry approaches; Jeannin, Iori and Samuel (2008) extended it to show that short positions repel. The evidence is among the best in this field. Ni, Pearson and Poteshman (2005) found that closing prices of optionable US stocks cluster at strike prices on expiration dates, altering the returns of optionable stocks by at least 16.5 basis points on each expiration date, and attributed the clustering to hedge rebalancing by option market makers and to deliberate manipulation by firm proprietary traders. [Fact] for single stocks. Golez and Jackwerth (2012) found S&P 500 futures pulled towards the at-the-money strike on days when options on the futures expire, and pushed *away* from it just before index options expire, which they attribute to hedge rebalancing and unwinding.

3. Pin risk makes dealers trade. A dealer short a physically settled strike near the price does not know whether it will be assigned (§2.2), so it tends to reduce or hedge the exposure in the final hours, adding flow at exactly the strikes that matter (§9.5). [Practice]

4. Roll-off. At expiry the options disappear, and so do the dealers' gamma and the hedges against it. Two consequences follow.

- **The regime can change with no price move.** In the synthetic chain of §6.5, the front expiry's roll-off moved the flip from 99.3 to 101.8 and turned $43.7 million of long gamma at a price of 100 into $55.3 million of short gamma. **[Simulated]**
- **Hedges must be unwound.** For cash-settled index options, dealers holding futures against in-the-money positions must trade them away around the settlement print — the opening auction for AM-settled monthlies, the close for PM-settled contracts. For physically settled options, exercise delivers stock that may be sold the next session.

The practitioner claim that volatility tends to rise in the days after large monthly expirations, as long-gamma positions roll off — sometimes called a "window of weakness" — follows from the mechanics when dealers were long gamma going in. The systematic evidence is anecdotal. [Practice] It is a testable event study, and §11.4 describes how to test it without fooling yourself.

The calendar of these events: the **monthly** expiry on the third Friday; the **quarterly** expiry in March, June, September and December, when index futures, index options and single-stock options expire together ("triple witching"); the expiry of options and futures on the **VIX** — Cboe's index of the S&P 500's expected volatility over the next 30 days, computed from a wide strip of index option quotes — on a Wednesday, 30 days before the following month's SPX expiry; and, since 2022, a **daily** SPX expiry.

7.8 Charm and vanna: flows without price moves

Gamma exposure describes only one reason a dealer's hedge changes. Write the change in the dealers' aggregate position delta over an interval, in shares, and the required hedge trade is minus that:

$$\text{hedge trade} \;\approx\; -\left(\Gamma_D\,\Delta S \;+\; \text{vanna}_D\,\Delta\sigma \;+\; \text{charm}_D\,\Delta t\right),$$

where Γ_D, vanna_D and charm_D are dealers' aggregate Greeks in shares. Gamma exposure is the first term. The second and third trade **without any move in the price**: when implied volatility changes, and simply because time passes.

Take a concrete book. Customers have bought protection: dealers are short 10,000 contracts of a 95-strike put, 30 days from expiry, with the stock at 100 and implied volatility at 20%. The dealers' position delta is +177,940 shares, so they hold a short hedge of 177,940 shares.

What changes	Dealers' new position delta	Hedge trade required
One day passes, nothing else (charm)	+174,108	**Buy 3,832 shares** ($0.4m)
One week passes, nothing else (charm)	+147,601	**Buy 30,339 shares** ($3.0m)
Implied volatility falls 3 points (vanna)	+140,783	**Buy 37,157 shares** ($3.7m)
Implied volatility rises 5 points (vanna)	+226,177	**Sell 48,237 shares** ($4.8m)
For comparison: the price falls 1% (gamma)	+227,242	**Sell 49,301 shares** ($4.9m)

A three-point fall in implied volatility forces almost as much buying as a 1% price move forces selling. From these mechanics, practitioners tell three stories. [Practice] throughout, with the mechanisms as stated and the market-level magnitudes untested.

- **Charm into expiry.** When dealers are short out-of-the-money puts, the passage of time steadily shrinks those puts' deltas, and dealers buy back hedges day after day — a supportive drift into a monthly expiry that stops when the puts expire.
- **The vanna rally.** When a feared event passes — a central-bank meeting, an inflation print — implied volatility falls, dealers short puts buy back hedges, the price rises, and because equity implied volatility tends to fall as prices rise, the loop reinforces itself.
- **The vanna selloff.** The mirror image: a decline lifts implied volatility, the puts' deltas grow, and dealers must sell more than gamma alone would require.

Two caveats keep these in proportion. Every sign reverses if dealers are *long* the out-of-the-money options — if customers sold the puts. And I am not aware of peer-reviewed work isolating aggregate charm or vanna flows in index markets; the concepts are exact for any hedged book, while their market impact is a [Hypothesis].

7.9 Zero-days-to-expiry options

Options that expire on the day they are traded — including older contracts in their final session — have gone from a curiosity to the largest segment of index options. Cboe reports that they grew from about 5% of SPX volume in 2016 to more than 40% after daily expirations were introduced in 2022 (Cboe, 2023), and reached a record 62% in August 2025 (Cboe, 2025).

For gamma exposure they matter for two reasons. First, their gamma per contract is enormous (§3.5): with hours left, a modest position at the money carries more hedging need than a large position a month out. Second, **most 0DTE positions are opened and closed within the day, so they never appear in the open interest published the next morning.** A gamma-exposure estimate built from that file misses, by construction, the part of the market with the most gamma. [Fact]

Whether 0DTE trading destabilises the index has been the most publicised debate in this area, and it has been argued with data.

Study	Data	Finding
Brogaard, Han and Won (2023)	0DTE share of index option volume; staggered introduction of weekly expirations as an instrument	Higher 0DTE trading raises index volatility: one standard deviation more 0DTE volume, about 9% more volatility relative to its mean
Dim, Eraker and Vilkov (2024)	Open-interest gamma and intraday volume in SPX 0DTEs	High 0DTE gamma does not propagate past volatility; volume shocks do not amplify recent returns
Adams, Fontaine and Ornthanalai (2024)	Days with and without 0DTE expiries before daily expirations existed	Index volatility 60–90 annualised basis points lower on 0DTE days; market makers' hedging of expiring positions dampens
Amaya, Garcia-Ares, Pearson and Vasquez (2025)	Cboe records of every SPX trade, market-maker positions each minute	Market-maker gamma usually positive; maximum effect on volatility modest
Cboe (2023)	Cboe's own capacity-coded SPX data	Market makers' net 0DTE gamma of $170 million to $670 million through the day, a fraction of a percent of S&P futures liquidity — from the exchange that lists the product

My read: [Contested], but the studies that measure market makers' actual positions agree that those positions are usually net long gamma and that the average effect is mild dampening. When they are short — which Amaya and co-authors find is not rare — the amplification they estimate is not large relative to ordinary swings in volatility. The strongest destabilising result relates volatility to 0DTE *volume*, which is not the same as dealers' gamma. And every study ends by 2023 or 2024, before 0DTE options passed 60% of SPX volume.

7.10 Single stocks and gamma squeezes

A **gamma squeeze** is the short-gamma loop of §7.1 in a single stock, driven by call buying. Customers buy large quantities of short-dated out-of-the-money calls; dealers who sold them are short gamma and buy stock to hedge; the price rises; the calls move towards the money, where their gamma is larger (§3.5); dealers must buy more. In a stock whose daily volume is small relative to dealers' hedging needs, the calibration of §7.3 puts κ near or past the singularity. Short sellers covering into the rally add to the same buying.

The mechanism is sound, and there is systematic evidence that it operates: Ni and co-authors' result that stock volatility rises when likely hedgers are net short options is the general form, and Zaynutdinova, Strong and Baig (2025), in a working paper, identify gamma-squeeze events followed by an average cumulative abnormal return of 5.13% over the next month.

The most famous case is also the best cautionary tale. In January 2021, GameStop's share price rose more than tenfold in a few weeks amid heavy retail options trading, and "gamma squeeze" became the popular explanation. The SEC staff's report (SEC, 2021) found no evidence of one: individual customers' options volume rose from $58.5 million on 21 January to $2.4 billion on 27 January, but the increase was driven more by the buying of *puts* than of calls, and market makers were *buying* calls rather than writing them. An ad hoc committee of academics (Mitts and co-authors, 2022) argued that the staff's analysis could not rule out short or gamma squeezes. [Contested] The lesson survives the dispute: **call buying followed by a rally is not evidence of a gamma squeeze.** A squeeze requires dealers to be net short gamma and hedging, and position data can show that they were not.

§7 Key takeaways

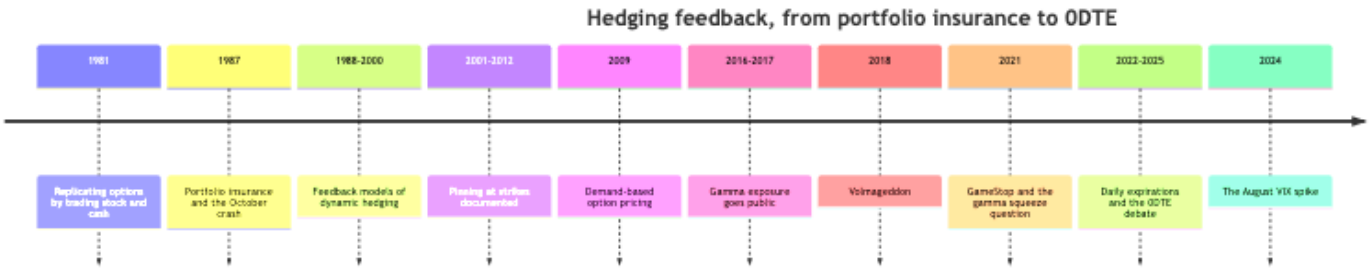
1. Dealer hedging multiplies moves by $1/(1+\kappa)$, where κ is gamma exposure divided by market depth. Long gamma dampens, short gamma amplifies, and $\kappa \to -1$ is a singularity: the model's crash.

2. With realistic lags the same effect appears as intraday autocorrelation — negative when dealers are long, positive when short — and as scaled volatility over the day.
3. Calibrated with the square-root impact law, 30-day index gamma of $5 billion per 1% moves volatility by only 5–13%. Measured market-maker gamma in SPX raised daily volatility by at most 3.3 points over 2020–2023.
4. κ grows near expiry, in stressed liquidity, and in single stocks under concentrated call buying, and its magnitude tends to grow when its sign is negative.
5. High gamma exposure coincides with calm markets partly by construction; negative gamma coincides with volatility partly because selloffs cause both. The direction of the hedging effect is established with measured positions; its size for broad indices is contested.
6. At the gamma flip, hedging does nothing; what changes is the sign of the feedback. The depth of the profile below the price matters more than the zero crossing.
7. Near expiry, gamma concentrates at strikes: long-gamma strikes pin, short-gamma strikes repel, and roll-off can change the regime with no price move. Pinning in single stocks is among the best-documented effects in this field.
8. Charm and vanna force hedging without price moves; a three-point fall in implied volatility can require almost as much buying as a 1% move. Their market-level effects are untested.
9. 0DTE options carry enormous gamma that end-of-day open interest cannot see; the studies that measure positions find mild dampening on average.
10. A rally with call buying is not proof of a gamma squeeze; GameStop's January 2021 data showed market makers buying calls.

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8. Episodes: what history does and does not show

History offers few clean experiments in this field. Positions are rarely observed, the events that make hedging interesting are the ones in which everything else is happening too, and narratives form within hours. Each episode below is worth knowing for what it rules in or out, and each is given the same four fields: what happened, the hedging mechanism proposed, what it taught, and what remains in dispute.



8.1 October 1987: portfolio insurance

What happened. On Monday 19 October 1987 the Dow Jones Industrial Average fell 22.6%, the largest one-day percentage decline in its history. [Fact]

The mechanism. Portfolio insurance, sold to pension funds through the 1980s, promised a floor under an equity portfolio without buying puts; Leland and Rubinstein (1988) recount how it was conceived and grew. It *replicated* a protective put by selling index futures as prices fell and buying them back as prices rose, following the logic of Rubinstein and Leland (1981) — Case 3 of §5.5 with no dealer in the middle. The Brady Commission's report (Presidential Task Force on Market Mechanisms, 1988) concluded that portfolio insurers' selling was central to the crash: on 19 October their futures sales amounted to the equivalent of about $4 billion of stock, more than 40% of futures volume excluding locals (independent floor traders dealing for their own account), and index arbitrage — buying the now-cheap futures and selling the equivalent basket of stocks to close the gap — carried that selling into the stock market, as Carlson (2007) summarises.

What it taught. Synthetic options behave like short gamma at market scale, and replication assumes a liquidity that disappears when many replicators trade the same way at once. The episode generated the theory of §7.1: Grossman (1988) on why synthetic puts are not equivalent to traded ones, Gennotte and Leland (1990) on how thin liquidity turns a little hedging into a discontinuity, and Jacklin, Kleidon and Pfleiderer (1992) on how a market that *underestimates* the amount of dynamic hedging can overprice stocks until the hedging reveals itself in a collapse.

Still disputed. How much portfolio insurance *caused* the decline, as opposed to accompanying and accelerating it. Shiller (1988), surveying investors at the time, argued that the crash reflected a shift in investor sentiment, of which portfolio insurance was one expression rather than an independent cause. [Contested]

8.2 2001–2012: pinning is documented

What happened. A sequence of papers established that option expirations leave marks on underlying prices: Krishnan and Nelken (2001) on single stocks, Avellaneda and Lipkin (2003) on the mechanism, Ni, Pearson and Poteshman (2005) on the clustering of US stock closes at strikes on expiration dates, and Golez and Jackwerth (2012) on pinning and anti-pinning in S&P 500 futures.

The mechanism. Delta hedging by dealers long gamma at strikes near the price, reinforced by charm near expiry (§7.7), plus — in the single-stock evidence — manipulation by proprietary traders with an interest in where the stock closed.

What it taught. Hedging effects are clearest where the square-root law says they should be (§7.3): at small scales, near expiry, in the final minutes, and around specific strikes. This is the most solid empirical ground in the subject. [Fact]

Still disputed. The split between hedging and manipulation in single stocks, and whether pinning in index markets has weakened as expirations multiplied. [Contested]

8.3 2009–2017: from demand pressure to gamma exposure

What happened. Gârleanu, Pedersen and Poteshman (2009) showed that dealers who cannot hedge perfectly charge more for options that end users demand, which explains part of the index skew, and documented that end users were net long index options and net short single-stock options. Chen, Joslin and Ni (2019) showed that the terms on which intermediaries absorb deep out-of-the-money put demand vary with their balance-sheet constraints. Meanwhile SqueezeMetrics released its gamma exposure white paper in 2016, revised in 2017, and the term entered market commentary, followed by commercial data services and bank strategists' notes.

The mechanism. Positioning, rather than prices alone, as the state variable: who holds the options determines both how they are priced (demand pressure) and how their hedging moves the underlying (gamma exposure).

What it taught. The academic and practitioner strands arrived at the same object from different directions. The academic strand measured positions and asked about prices of options; the practitioner strand assumed positions and asked about prices of the underlying.

Still disputed. Whether assumed-position gamma exposure has predictive content beyond implied and realised volatility (§7.4, §11.4). [Contested]

8.4 February 2018: Volmageddon

What happened. On 5 February 2018 the S&P 500 fell 4.1% and the VIX index (§7.7) more than doubled, from 17.31 to 37.32 — then the largest one-day rise in its history. Exchange-traded products that paid the inverse of VIX futures lost more than 90% of their value in a day, and the largest was shut down (Augustin, Cheng and Van den Bergen, 2021). [Fact]

The mechanism. Not options dealers, but the same arithmetic. An inverse volatility product must rebalance daily by *buying* VIX futures after volatility rises — the $L(L-1)$ rebalancing of §5.10 with $L = -1$ — and leveraged long-volatility products must buy too. Their combined rebalancing was large, predictable in timing, and concentrated in the final minutes of trading, in a futures market that could not absorb it.

What it taught. Hedging and rebalancing feedback operates wherever a large position must trade in proportion to recent moves, not only in dealers' option books; and predictable rebalancing is front-runnable, which pulls the price move forward rather than preventing it. Augustin and co-authors draw the parallel with portfolio insurance explicitly.

Still disputed. The role of anticipatory trading around the known rebalancing window. [Contested]

8.5 January 2021: GameStop and the gamma-squeeze question

What happened. GameStop's shares rose more than tenfold in a few weeks, amid record retail trading in the stock and its options, heavy short interest, and broker trading restrictions.

The mechanism proposed. A gamma squeeze (§7.10): retail call buying leaving dealers short gamma, their hedge purchases feeding the rally, combined with a short squeeze.

What it taught. That the data can contradict a compelling story. The SEC staff (SEC, 2021) found no evidence of a gamma squeeze: the surge in customer options volume was driven more by put buying than call buying, and market makers were buying calls rather than writing them.

Still disputed. An ad hoc academic committee (Mitts and co-authors, 2022) argued that the staff's data and methods could not rule out short or gamma squeezes. [Contested] The committee included the founder of a gamma-analytics firm, which is worth knowing and is not a reason to dismiss its argument.

8.6 2022–2025: daily expirations and the 0DTE debate

What happened. The addition of Tuesday and Thursday SPX expirations in 2022 completed a daily expiry cycle, and zero-days-to-expiry trading grew from a minority of SPX volume to more than 60% of it by 2025 (§7.9). Market commentary warned of a "Volmageddon 2.0" in which market makers short 0DTE gamma would amplify an intraday crash.

The mechanism proposed. Enormous same-day gamma, held short by dealers, hedged intraday into a market that could not absorb it.

What it taught. That measured positions settle questions that assumed positions cannot. The studies using market makers' actual positions (§7.9) find them usually long gamma, with modest average effects; the concern rested largely on assumed positions and on volume.

Still disputed. The size of rare negative-gamma episodes, and whether the conclusions survive the growth since 2023. [Contested]

8.7 August 2024: the VIX spike that was about quotes

What happened. On 5 August 2024, during the unwinding of yen carry trades — positions funded by borrowing cheaply in yen to hold higher-yielding assets (A.23) — the VIX recorded its largest one-day spike on record, exceeding its moves in 2008 and March 2020, touching levels above 60 before US markets opened. By the end of that week the S&P 500 had recovered its losses.

The mechanism proposed. In real time, dealer gamma was among the first explanations offered.

What it taught. Todorov and Vilkov (2024) found that the spike owed much to market makers' *quoting*: the VIX is computed from option quotes, not trades, and the widening of bid–ask spreads — especially on puts — as market makers protected themselves from an imbalanced book lifted the quote midpoints that enter the calculation. Volatility ETFs and dispersion trades — bets that index volatility is priced high relative to the volatility of its members (A.22) — were unlikely to have been the main driver. Dealers mattered, but through withdrawing liquidity rather than through delta hedging.

Still disputed. Little about the mechanism; the episode's lesson is methodological. Stress events invite gamma explanations, and the first explanation is often not the right one.

§8 Key takeaways

1. The 1987 crash established that synthetic option replication behaves like short gamma at market scale, and generated the feedback theory of §7.1; how much it caused the crash remains disputed.
2. Pinning at strikes near expiry is the most solidly documented hedging effect, in exactly the conditions the square-root law favours: small scale, near expiry, specific strikes.
3. Volmageddon showed the same feedback arithmetic in volatility products, with no options dealers involved.
4. GameStop showed that position data can contradict a compelling gamma-squeeze story.
5. The 0DTE debate showed that measured positions settle questions assumed positions cannot.
6. August 2024 showed dealers mattering through quotes and liquidity, not hedging — and that the first explanation of a stress event is often wrong.

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9. How dealers use this

A dealer does not need a gamma-exposure *estimate*: it knows its own positions exactly. What it needs is to turn those positions into risk it can see, prices that manage it, and hedges that keep it bounded. The market-wide picture matters to a

dealer in a narrower way — as information about crowding and liquidity. This section is necessarily inference about private practice from public sources and first principles. [Practice] throughout unless marked otherwise.

9.1 The risk report: Greeks, ladders and scenario grids

An options desk's daily risk report aggregates Greeks by underlying, by expiry bucket and by strike, in the share, dollar and per-1% units of §3.9. Three views do most of the work.

- **Gamma and vega by expiry.** Near-dated gamma behaves differently from far-dated gamma (§3.5), and vega in the front month responds to different events than vega a year out.
- **Strike ladders.** Gamma and open positions by strike around the current price, with the next expiry highlighted: this is where pin risk (§9.5) and pinning flows live.
- **Scenario grids.** The book's P&L revalued under joint shocks to the price and to implied volatility — the "slide". Greeks describe small moves; grids describe large ones, including the gaps no hedge can prevent (§4.9).

Here is the grid for the simplest interesting book, the collar book of §5.9: long 10,000 contracts of a 105-strike call and short 10,000 of a 95-strike put, 30 days to expiry, delta-hedged at a price of 100. P&L in millions of dollars:

Implied vol shock ↓ / price shock →	−10%	−5%	0%	+5%	+10%
−5 points	−1.44	+0.22	−0.03	−0.21	+1.42
unchanged	−1.73	−0.24	0.00	+0.31	+1.80
+5 points	−2.07	−0.61	+0.03	+0.74	+2.23
+10 points	−2.40	−0.90	+0.07	+1.09	+2.64
Dollar gamma, $m per 1%	−3.94	−5.09	+0.42	+5.79	+5.11

Read it the way a risk manager would. Small moves barely matter: the book is hedged and its gamma at 100 is almost zero. A 10% decline costs $1.7 million, and $2.4 million if implied volatility rises 10 points at the same time — which, in equity markets, it usually does. A 10% rally *makes* money. The bottom row is the book's gamma profile: short gamma below the price, long gamma above, the collar shape. A desk running this book knows before anything happens that in a selloff it will be a short-gamma hedger selling into the decline, and that the loss will be worst in exactly the scenario where everyone else's hedging and liquidity look the same.

9.2 Pricing the inventory: quotes as a risk tool

A stock market maker shades its quotes against its inventory (§1.2). An options market maker shades them against its *Greeks*. A dealer that has accumulated short downside gamma and vega raises both its bid and its offer on downside puts — making further put buying more expensive and put selling more attractive — until customers' trades bring the book back towards balance. Stoikov and Sağlam (2009) derive optimal option quotes under inventory risk, and show when inventory should matter: if the underlying could be traded continuously and costlessly, delta hedging would remove all risk and quotes would not depend on inventory; it is the unhedgeable residue — discrete trading, volatility risk, jumps — that makes inventory shape prices.

The market-level version is **demand-based option pricing**. Gârleanu, Pedersen and Poteshman (2009) show that when competitive dealers cannot hedge perfectly, end users' demand for an option raises its price in proportion to the variance of the part of the option that cannot be hedged, and raises the prices of other options in proportion to the covariance of their unhedgeable parts. End users' net demand for index puts then helps explain why index options are expensive and skewed. Muravyev (2016) finds that inventory risk has a first-order effect on option prices, with order imbalances attributable to inventory risk having a much larger price impact than earlier estimates suggested. [Fact] that dealer positioning is priced; [Contested] as to how much of the index skew it explains.

There is a lesson here for traders on the other side. **Dealer positioning is already partly in option prices.** When dealers are short and constrained, implied volatility and skew rise. A gamma-exposure signal that does not add information beyond implied volatility may simply be a noisier reading of something the options market has already priced. [Hypothesis]; §11.4 shows how to test it.

9.3 Hedging policy in practice

- **Hedge the net, not the parts.** Delta is aggregated across the whole book on an underlying and hedged once; offsetting options never generate a trade.

- **Use the cheapest instrument that tracks.** Index options are hedged mostly with index futures, which trade almost around the clock; ETF options with the ETF or the index future; single-stock options with the stock. Cross-hedging related underlyings — SPX, SPY and E-mini futures — is routine, and leaves basis risk.
- **Hedge in bands, with judgement about time.** The band logic of §4.8 sets how far delta may drift; schedules and events set when it is reviewed. Gamma is often reduced ahead of scheduled announcements.
- **Execute like any large trader.** Hedges are worked with execution algorithms that participate in volume rather than crossing the spread at once, because a dealer's own price impact is a hedging cost. Many electronic market makers hedge within moments of an option fill, while others let hedge needs accumulate.
- **Mind the close.** Positions are marked at the close and exposed overnight, so desks re-hedge towards the end of the session. That concentration is one reason the end-of-day momentum of Baltussen and co-authors (§7.2) is where hedging effects appear. [Hypothesis]

9.4 Hedging gamma with options, and what that does to the market

A dealer that is uncomfortably short gamma can do better than hedge it in the underlying: it can buy gamma back, by buying options. Buying back the same strikes, or cheaper out-of-the-money "wings" that protect against large moves, reduces the dealer's hedging needs and its exposure to gaps.

At the market level, this is subject to a conservation law that follows from §2.5. **Gamma bought from another hedger is gamma that hedger no longer has.** If one market maker buys puts from another, the first becomes less short gamma and the second more; the market-wide hedging flow does not change. Aggregate dealer gamma changes only when options move between hedgers and *non*-hedgers: when an end user who does not hedge — an overwriter, a pension fund selling puts — takes the other side. Persuading end users to do so takes price: higher implied volatility for the options dealers want to buy. This is the demand-pressure mechanism of §9.2 seen from the dealer's side, and it is why a market in which dealers are collectively short gamma tends to be a market in which implied volatility is rising.

9.5 Expiry, pin and dividend risk

- **Pin risk.** A dealer short a physically settled strike at which the stock is closing faces uncertain assignment (§2.2), and after-hours news can change holders' exercise decisions. Dealers reduce or re-hedge such positions during the final session, adding flow at exactly the strikes where pinning operates (§7.7).
- **Settlement risk.** AM-settled index options settle on a value computed from the opening prints of the index's component stocks, not on a price anyone can trade at the open. A dealer hedged into Thursday's close holds basis risk until Friday's settlement value is known.
- **Early exercise and dividends.** Holders of in-the-money American calls exercise before ex-dividend dates to capture the dividend. A dealer short those calls can find itself unexpectedly short stock across the dividend.
- **The roll.** At each expiry, hedges against expiring options must be unwound or rolled into positions against the next expiry, which is the unwinding flow of §7.7.

9.6 Tail risk, capital and intermediary constraints

The risks that hedging in the underlying cannot remove (§4.9) are managed with capital, margin and limits: stress-scenario losses like the corners of the §9.1 grid, exchange margin, and bank capital rules. Those constraints bind hardest in stress, and when they bind, dealers supply less.

- **Prices respond.** Chen, Joslin and Ni (2019) show that when intermediaries' constraints tighten, the terms on which they absorb public demand for crash protection worsen: options become more expensive, risk premia rise across markets, and broker-dealers deleverage. [Fact] for the relation they document.
- **Quotes widen.** In August 2024, market makers widened option spreads, particularly on puts, to avoid imbalanced books; because the VIX is computed from quotes, the index spiked (Todorov and Vilkov, 2024).
- **Liquidity falls exactly when κ matters.** Wider quotes and smaller size are a fall in the depth D of §7.1, arriving together with the negative dealer gamma that follows selloffs. The dealer's rational risk management and the market's fragility are the same event seen from two sides. [Hypothesis]

§9 Key takeaways

1. Dealers know their own positions; what they manage is Greeks by expiry and strike, and P&L under joint price and volatility shocks.

2. A collar-shaped book loses most in a selloff with rising volatility — the scenario in which every other hedger and liquidity provider is under the same pressure.
3. Dealers shade option quotes against their Greeks, and dealer positioning is priced: demand for options dealers cannot hedge raises their prices. A gamma signal must beat what implied volatility already says.
4. Real hedging nets across books, uses futures, runs in bands and concentrates towards the close.
5. Buying gamma from another hedger does not change market-wide hedging flow; only trades with non-hedging end users do, and they require paying up in implied volatility.
6. Constraints bind in stress: dealers widen quotes and reduce size, lowering market depth exactly when dealer gamma turns negative.

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10. How traders on the other side use this

10.1 What gamma information is, and what it is not

Before any use, the properties of the information set its limits. Everything in Parts II and III reduces to seven statements about a gamma-exposure estimate built from public data.

1. **It is about volatility, not direction.** Hedging chases or leans against moves in whichever direction they come (§4.4). A reading that "dealers need the market higher" misunderstands the mechanism.
2. **It rests on a positioning assumption that can be wrong-signed** (§6.3).
3. **It is stale.** Open interest describes the previous close, and the day's 0DTE positions — the largest concentrations of gamma — are invisible to it (§7.9).
4. **It is partial.** Over-the-counter positions and some venues are missing.
5. **Its effect on broad indices is modest** — a few volatility points at the extremes in measured data — and larger near expiry, in thin liquidity and in single stocks (§7.3).
6. **It is confounded with the volatility regime itself** (§7.4, §7.5): high readings coincide with calm, and negative readings with stress, partly by construction.
7. **It is public.** Many participants watch the same levels, and anticipatory trading can pull moves forward; but a hedge that *must* be executed does not disappear because it is anticipated. [Hypothesis]

What survives these limits is a **conditioning variable**: information that shifts the probability of calmer or wilder, choppier or trendier price action, most usefully in the situations where §7.3 says the effect is large.

10.2 Use 1: a conditional volatility forecast

The natural use is as an input to a forecast of realised volatility over the next session or the next few days, and the natural test is whether it adds anything to what a good baseline already knows. The standard baseline is a regression of realised volatility on its own recent values at daily, weekly and monthly horizons — the **HAR** model of Corsi (2009) — augmented with implied volatility such as the VIX. Candidate gamma features:

- the sign and magnitude of dealer gamma, normalised by liquidity (dollar gamma per 1% divided by average daily dollar volume, a proxy for κ);
- the distance from the price to the flip, in units of daily volatility;
- the share of gamma expiring within the forecast horizon;
- the depth of the profile within one or two days' typical move below the price (§7.6).

What should you expect? For broad indices, small incremental improvements, concentrated in particular regimes. Maurer (2026), a working paper, reports for example that dealer gamma adds forecasting information about the size of overnight gaps only when the VIX is low, and that pooling calm and stressed days hides the effect. [Hypothesis] — one unrefereed study, but its structure is exactly the one to expect if the baseline already captures stressed regimes. A forecast improved in this way is useful for sizing positions to a volatility target, for comparing implied volatility with expected realised volatility, and for setting stops.

10.3 Use 2: choosing between reversal and continuation

The second signature of hedging is autocorrelation (§7.2): partial reversals when dealers are long gamma, continuations when they are short. The corresponding tactics are to **fade moves and expect ranges when the estimated dealer position is long gamma**, and to **respect breakouts and expect extension — especially late in the session — when it is short**. The empirical anchors are Barbon and Buraschi's link between dealer gamma imbalance and intraday momentum or reversal,

and the last-half-hour momentum of Baltussen and co-authors. [Hypothesis] as a trading rule. Two cautions: the effect is small relative to the costs most traders pay, and the regime estimate is least reliable exactly where it changes sign.

10.4 Use 3: the expiry calendar

Expiry is where the mechanics are strongest and the evidence best.

- **Single stocks with large open interest at a strike near the price on expiration day** are more likely than usual to close at that strike (Ni, Pearson and Poteshman, 2005). A trader short options at that strike is exposed to pin risk; a trader betting on a late breakout through it is betting against a documented effect.
- **Index expiries** bring hedge unwinding around settlement — the opening for AM-settled monthlies, the close for PM-settled contracts — and pinning or anti-pinning in index futures around serial and index option expirations (Golez and Jackwerth, 2012).
- **After expiry**, the regime can change with no price move (§6.5). The practical habit is to recompute the gamma profile *excluding* the expiring contracts before the expiry happens, so that the post-expiry regime is known in advance rather than discovered.

10.5 Use 4: strike levels

Walls and high-gamma strikes mark where hedging pressure concentrates (§6.6), and traders commonly treat them as support and resistance. [Practice] The defensible way to use them is as **conditional expectations about volatility near those prices** — dampening near large long-gamma strikes, acceleration towards large short-gamma ones — rather than as levels that must hold. They are also the easiest features to fool yourself with, because strikes sit at round numbers and prices cluster at round numbers anyway. Any test of a "wall" effect needs round-number levels *without* large open interest as its control.

10.6 Use 5: choosing options strategies

A trader who sells options is short gamma exactly as the dealer in Cases 1 and 3 is (§5.3, §5.5), with the same identity governing the result. A market in which dealers are collectively short gamma is therefore the worst environment to be short gamma yourself: realised volatility is elevated and moves extend. The mirror image is that long-gamma regimes favour selling premium, *but* implied volatility is typically already low in those regimes, so the edge is not the regime label. It is the gap between implied volatility and the realised volatility you forecast (§10.2) — the variance risk premium, conditioned on the regime. A regime that flips on a shock turns a steady premium-selling strategy into Case 3 within hours.

10.7 Use 6: risk management

Much of the practical value of gamma information is defensive.

- Reduce leverage and widen stops when estimated dealer gamma is deeply negative and liquidity thin: moves and gaps will be larger than recent history suggests.
- Avoid supplying liquidity into accelerating moves in short-gamma regimes; the other side of your limit order may be a hedger who must keep trading.
- Treat long-gamma calm with suspicion ahead of large expiries: the long gamma that produced the calm may roll off on a known date.
- Add a dealer-gamma dimension to your own stress scenarios, alongside the price and volatility shocks of §9.1.

10.8 Anti-patterns

Anti-pattern	Why it fails	Do instead
Treating the flip as a precise trigger	It is a model output uncertain by several percent (§6.5)	Use a band; measure distance in units of volatility
Assuming dealers are long calls and short puts everywhere	Evidence differs for single stocks and has changed over time (§6.3)	Use participant-coded data; test several conventions
Using same-day open interest in a backtest	Open interest is published the next morning	Lag it; timestamp every input
Ignoring 0DTE	The largest gamma never appears in open interest (§7.9)	Use intraday flow-based estimates, or restrict claims accordingly

Anti-pattern	Why it fails	Do instead
Reading direction into hedging	Hedging is direction-neutral (§4.4)	Read it as a volatility and autocorrelation regime
Testing levels without a baseline	Strikes are round numbers	Compare with matched round levels without open interest
Crediting every selloff to gamma	Leveraged ETFs, trend followers, volatility targeters and liquidity withdrawal amplify too (§5.10, §8.7)	Attribute with position data, not narrative
Assuming the effect is large	Measured index effects are a few volatility points at most (§7.3)	Size accordingly; focus on expiry, stress and single stocks
Trusting a signal because everyone watches it	Crowding pulls moves forward and decays signals	Test out of sample; monitor decay after publication

§10 Key takeaways

1. Gamma information is a conditioning variable for volatility and autocorrelation, not a directional signal or a price target.
2. Its most defensible use is as an input to a volatility forecast, and it must beat a HAR-plus-implied-volatility baseline to be worth anything.
3. Tactically it suggests fading moves when dealers are long gamma and respecting extensions when they are short; the effect is small relative to most traders' costs.
4. Expiry is where the mechanics are strongest: pins in single stocks, unwinding around index settlement, and regime changes as contracts roll off, which can be computed in advance.
5. Walls are places where hedging pressure concentrates, not levels that must hold, and they must be tested against round numbers.
6. For option sellers, a dealer short-gamma regime is the worst environment to be short gamma; in long-gamma regimes the edge is still implied versus forecast realised volatility, not the label.
7. Much of the value is defensive: sizing, stops, and not supplying liquidity into hedging cascades.

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11. Measuring it yourself

11.1 What matters, in order

Readers building a gamma-exposure estimate for the first time tend to spend their effort in the reverse order of what matters. Here is the order I would spend it in.

Rank	Decision	Why it ranks here
1	**Positioning** — which side dealers hold in each series	Errors flip the sign of the answer (§6.3)
2	**Coverage and timing** — every product on the underlying, correct timestamps, the 0DTE blind spot	Missing products bias the magnitude; look-ahead invalidates every test
3	**The volatility surface** — implied volatility by series, and how it moves with the price	Moves gamma at hypothetical prices and therefore the flip (§3.7, §6.5)
4	**Hedging realism** — bands, lags, netting	Real flows are smaller and later than dollar gamma (§4.8)
5	**Normalisation** — by liquidity	Turns dollars into the κ that determines impact (§7.1)
6	**Presentation** — profiles, flips, walls	Easiest to build, least important

Most public tools invest heavily in rank 6 and assume rank 1. The worked example of §6.4 showed why that is backwards: the convention changed the answer from +$3.9 million to −$63 million per 1%.

11.2 Data

Input	Typical source	Timing and caveats
Open interest by series	OCC and exchanges, via data vendors	Published before the next session, as of the previous close
Option quotes and implied volatility	The consolidated options feed (OPRA); vendors; academic histories such as OptionMetrics	Compute implied volatility from midpoints consistently, or use one vendor's surface throughout
Participant-coded volume	Exchange open–close files, such as Cboe's, which record buys and sells by customer, professional customer, broker-dealer and market maker, opening and closing	End of day or intraday snapshots; each file covers only its own exchanges; Cboe's main exchange history starts in 2011
Trades and quotes	OPRA tick data	For trade-side inference, with its midpoint ambiguity (§6.3)
Underlying prices and volume	Stock, ETF and futures exchanges	For normalising by liquidity
Rates and dividends	Treasury curves; dividend forecasts or parity-implied carry	Matter for long-dated and deep in-the-money deltas

Two points about participant-coded data save a lot of confusion. First, **a participant class's net position changes by its buys minus its sells**, whatever the opening and closing flags say: an opening buy and a closing buy both add to the net position. The flags matter for open interest accounting, not for the net. Second, the net position in a series is the running sum of those changes **since the series was listed**, on **every exchange that lists it**. For SPX, which trades only on Cboe, one exchange's files suffice; for SPY or a single stock, options trade on many exchanges, and a single exchange's file measures only part of the flow.

11.3 A reference implementation

The code below computes a dealer gamma profile from an option chain, under any of three positioning inputs: the sign convention, the all-short bound, or measured dealer positions. It holds each series' implied volatility fixed as the hypothetical price moves — the sticky-strike assumption of §3.7 — and measures time to expiry in calendar years. Run against the five-strike chain of §6.4, it returns +$3.946 million per 1% at a price of 100 and a flip at 99.28, and −$63.2 million under the all-short convention.

```python
import math

import numpy as np
import pandas as pd

YEAR_SECONDS = 365.0 * 86400.0


def bs_gamma(spot: np.ndarray, strike: np.ndarray, tau: np.ndarray, vol: np.ndarray,
             rate: float = 0.0, div: float = 0.0) -> np.ndarray:
    d1 = (np.log(spot / strike) + (rate - div + 0.5 * vol**2) * tau) / (vol * np.sqrt(tau))
    numerator = np.exp(-div * tau - 0.5 * d1**2)
    denominator = math.sqrt(2 * math.pi) * spot * vol * np.sqrt(tau)
    return numerator / denominator


def dealer_contracts(chain: pd.DataFrame, convention: str) -> np.ndarray:
    oi = chain["open_interest"].to_numpy(dtype=float)
    is_call = chain["type"].eq("C").to_numpy()
    if convention == "long_calls_short_puts":
        return np.where(is_call, oi, -oi)
    if convention == "short_all":
        return -oi
    if convention == "measured":
        return chain["dealer_net"].to_numpy(dtype=float)
    raise ValueError(f"unknown convention: {convention}")


def gex_profile(chain: pd.DataFrame, spots: np.ndarray, asof: pd.Timestamp,
                convention: str = "long_calls_short_puts", multiplier: float = 100.0,
                rate: float = 0.0, div: float = 0.0) -> pd.Series:
    # dealer $ gamma per 1% move at each hypothetical spot, implied vols held fixed by series
    tau = (chain["expiry"] - asof).dt.total_seconds().to_numpy() / YEAR_SECONDS
    live = tau > 0
    c, tau = chain.loc[live], tau[live]
    n = dealer_contracts(c, convention) * multiplier
    k = c["strike"].to_numpy(dtype=float)
    vol = c["iv"].to_numpy(dtype=float)
    s = np.asarray(spots, dtype=float)[:, None]
    g = bs_gamma(s, k[None, :], tau[None, :], vol[None, :], rate, div)
    gex = (n[None, :] * g).sum(axis=1) * s[:, 0] ** 2 * 0.01
    return pd.Series(gex, index=np.asarray(spots, dtype=float), name="gex_usd_per_1pct")


def gamma_flip(profile: pd.Series, spot: float) -> float | None:
    x, y = profile.index.to_numpy(dtype=float), profile.to_numpy()
    crossings = np.flatnonzero(np.sign(y[:-1]) * np.sign(y[1:]) < 0)
    if crossings.size == 0:
        return None
    i = crossings[np.argmin(np.abs(x[crossings] - spot))]
    return float(x[i] - y[i] * (x[i + 1] - x[i]) / (y[i + 1] - y[i]))


def dealer_net_positions(flows: pd.DataFrame,
                         dealer_classes: tuple[str, ...] = ("market_maker",)) -> pd.Series:
    # flows: one row per series, date and participant class, with contracts bought and sold,
    # covering every exchange that lists the series from the day it was listed
    d = flows[flows["participant_class"].isin(dealer_classes)]
    change = (d["buy_qty"] - d["sell_qty"]).groupby([d["series"], d["date"]]).sum()
    return change.groupby(level="series").cumsum().rename("dealer_net")
```

A typical call, with `chain` holding one row per series (`type`, `strike`, `expiry`, `iv`, `open_interest`, and optionally `dealer_net`):

```
asof = pd.Timestamp("2026-09-11 16:00")
spots = np.round(np.arange(0.85, 1.15, 0.0005) * spot, 4)
profile = gex_profile(chain, spots, asof, convention="long_calls_short_puts")
flip = gamma_flip(profile, spot)
at_spot = profile.iloc[np.argmin(np.abs(profile.index - spot))]
gamma_to_volume = at_spot / avg_daily_dollar_volume
```

Five implementation notes, each of which changes answers.

1. **Combine products in dollars per 1%.** Compute a profile for each product on the same underlying risk — SPX, SPY, E-mini futures options — on a common grid of *percentage* moves, each with its own price and multiplier, then add.
2. **Decide what "live" means on expiry day.** The code drops expired series at `asof`. For a profile computed before the open on an expiry day, AM-settled contracts are effectively gone and PM-settled ones are at their most concentrated.
3. **Guard the surface.** Stale or crossed quotes produce absurd implied volatilities and, near expiry, absurd gammas. Filter by quote quality and cap implied volatility before computing.
4. **Know what the Greeks ignore.** Black–Scholes gamma for American options is a good approximation except for deep in-the-money options near early-exercise boundaries, such as calls before an ex-dividend date.
5. **Report ranges, not points.** Compute the profile under at least two positioning inputs and two surface assumptions, and publish the spread of the flip across them (§6.5).

11.4 Validating a gamma signal without fooling yourself

A gamma-exposure signal is exceptionally easy to backtest badly: it is built from data with awkward timestamps, it is mechanically correlated with volatility, and it comes with many free choices. This protocol is ordered from cheapest to most expensive, and each stage is a gate.

Stage 0 — reproduce what must be true. The computed profile should reproduce the worked example of §6.4. Swapping a call for a put at the same strike, with the delta adjusted, should leave gamma unchanged (§5.8). Historical gamma readings should correlate *negatively* with the level of implied volatility, for the mechanical reason of §3.5. If they do not, something is wrong with the data.

Stage 1 — define the target and horizon exactly. Candidates: next-session realised variance from intraday returns; the absolute size of the overnight gap; the lag-one autocorrelation of 30-minute returns; the predictability of the last half hour from the rest of the day. Each maps to a different claim in §7.

Stage 2 — fix information timing. Open interest dated t is usable only from the session after t. Implied volatilities come from the previous close. Participant-coded files are usable from their publication time, not their trade date. Every feature carries the timestamp at which it became known.

Stage 3 — build the baseline you must beat. A HAR model of realised volatility (Corsi, 2009) with daily, weekly and monthly lags, plus implied volatility, day-of-week and expiry-day indicators, and indicators for scheduled announcements such as central-bank decisions and inflation releases.

Stage 4 — add the gamma features. Sign and magnitude normalised by liquidity, distance to the flip in units of daily volatility, gamma expiring within the horizon, and the depth of the profile below the price (§10.2). Include interactions with liquidity, because §7.3 says the effect should scale with it.

Stage 5 — evaluate out of sample. Use rolling or expanding estimation windows. Compare volatility forecasts with a loss function that is robust to noise in the realised-volatility proxy, such as QLIKE (Patton, 2011), and test the difference in accuracy formally (Diebold and Mariano, 1995).

Stage 6 — attack the result. It must survive every reasonable positioning convention, not just the one that worked; separate products (SPX and SPY); sub-periods before and after daily expirations began in 2022; and placebos — shuffled positioning signs, or open interest shifted by a week. Count every convention, horizon and threshold you tried, and discount the best result for the number of trials.

Stage 7 — measure economic significance. Translate the forecast improvement into volatility points and into the after-cost P&L of a sizing rule that uses it. A statistically significant improvement of a tenth of a volatility point is not a trading signal.

Three specific claims deserve specific designs.

- **"Volatility rises after large expiries."** An event study around monthly expiries, comparing realised volatility before and after, split by the amount of dealer gamma that rolls off, with non-expiry Fridays as placebos and controls for the volatility term structure and the macroeconomic calendar.

- **"Walls act as support and resistance."** Price behaviour as the price approaches strikes with large estimated dealer gamma, against matched round-number levels without open interest (§10.5).
- **"Prices pin on expiry."** Distance from the close to the nearest large-open-interest strike on expiry days against non-expiry days, following Ni, Pearson and Poteshman (2005).

11.5 Pitfalls checklist

- ☐ Open interest lagged to its publication time; no same-day open interest in any feature.
- ☐ Every product on the underlying included, or the omission stated.
- ☐ The 0DTE blind spot acknowledged in every intraday claim.
- ☐ Results reported under at least two positioning inputs.
- ☐ Gamma expressed in dollars per 1% and normalised by liquidity.
- ☐ Flip reported as a band.
- ☐ Baseline includes implied volatility and HAR lags.
- ☐ Out-of-sample evaluation with a robust loss and a formal comparison.
- ☐ Number of trials counted.
- ☐ Economic significance after costs.

§11 Key takeaways

1. Spend effort on positioning first and presentation last; most tools do the reverse.
2. A participant class's net position is its cumulative buys minus sells since listing, on every exchange listing the series.
3. Compute profiles per product in dollars per 1% on a common percentage grid, then add; guard the implied volatility surface; report ranges across positioning and surface assumptions.
4. Validate in gated stages: reproduce what must be true, fix timing, beat a HAR-plus-implied-volatility baseline out of sample, attack the result with placebos and alternative conventions, and measure economic value after costs.

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12. Synthesis

12.1 The framework on one page

Everything in this document follows from a sign rule and three identities.

The sign rule. Options are in zero net supply, and the dealer holds the opposite of the customer. A position's gamma is positive if it is long options and negative if short, for calls and puts alike. So the sign of dealer gamma is a statement about *who is long*, never about the contracts (§2.5, §5.8).

Identity 1 — what a hedged option earns (§4.4):

$$\Delta\Pi \approx \tfrac{1}{2}\,\Gamma S^2 \left[\left(\frac{\Delta S}{S}\right)^2 - \sigma_i^2\,\Delta t\right].$$

This is why dealers hedge, why short gamma collects theta and fears large moves, why hedged dealers have no directional preference, and why selling index options earns a premium that is really a price for crash risk.

Identity 2 — what a hedge must trade (§5.1, §7.8):

$$\text{hedge trade} \approx -\left(\Gamma_D\,\Delta S + \text{vanna}_D\,\Delta\sigma + \text{charm}_D\,\Delta t\right).$$

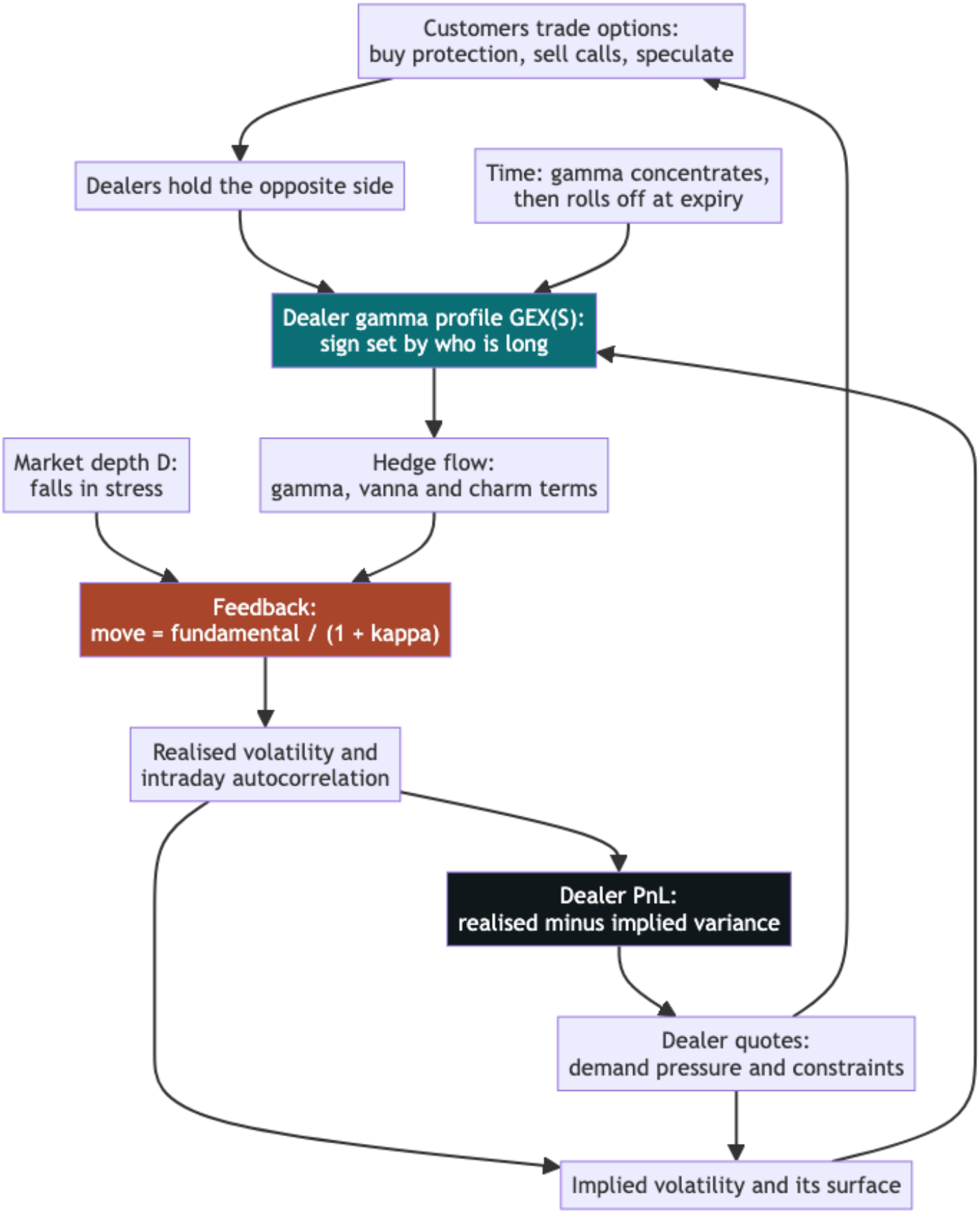
This is the quadrant, gamma exposure (the first term, aggregated), and the flows that arrive without price moves (the second and third).

Identity 3 — what the hedge does to the price (§7.1):

$$x = \frac{x_f}{1+\kappa}, \qquad \kappa = \frac{\mathrm{GEX}}{D}.$$

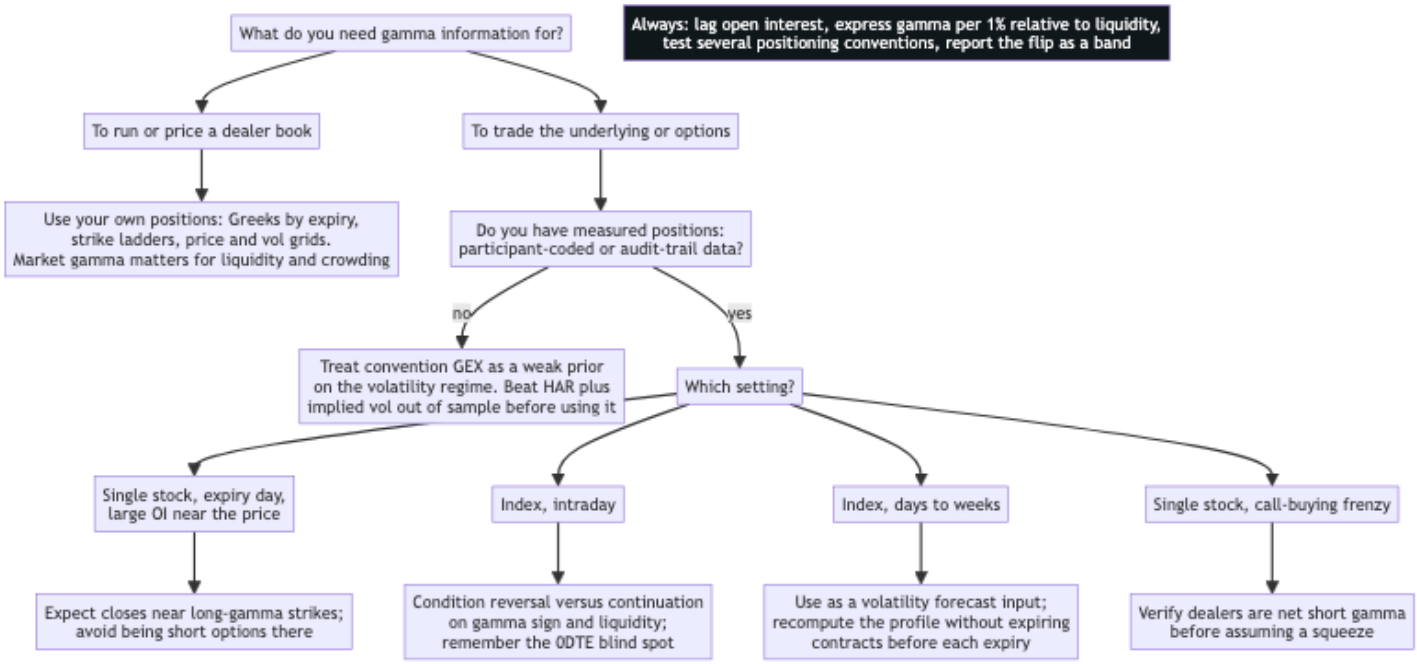
This is dampening, amplification, the singularity behind crashes and squeezes, the autocorrelation signature, pinning, and — through the calibration of §7.3 — the reason the effect on broad indices is modest except near expiry, in stress, and in single stocks.

The loop that connects them:



The loop has two places where information about positioning leaks into prices: through the hedge flow into the underlying, and through dealers' quotes into implied volatility. A trader who reads gamma exposure is trying to anticipate the first. The options market has often already priced part of it through the second (§9.2).

12.2 A decision tree



12.3 A staged build, with gates

The early stages are infrastructure, not the interesting part, and skipping them is the usual reason gamma signals fail.

Stage	Build	Gate before moving on
1	Black–Scholes Greeks and an implied volatility surface	Reproduce the §3 and §5 numbers; call and put gamma identical at a strike
2	Option chain and open interest ingestion with timestamps	Open interest totals reconcile with the clearing house; no feature uses data before its publication time
3	Positioning: conventions, plus participant-coded flow accumulation where available	Measured net positions never exceed open interest; results computed under at least two conventions
4	Gamma profile, flip and walls, across all products on the underlying	The §6.4 worked example reproduced exactly
5	Normalisation by liquidity	Implied κ of the order of §7.3 for the index; larger near expiry
6	Validation, as in §11.4	Out-of-sample improvement over HAR plus implied volatility, robust across conventions and placebos
7	Production: pre-open computation, intraday updates if flow data exist	Monitoring for decay after deployment

If stage 6 fails, stop. A gamma profile that does not improve a volatility forecast can still be a useful description of where hedging pressure sits, but it should not size a position.

12.4 Ten things I would tell someone starting today

1. **Learn the hedged P&L identity until you can do it in your head.** Nearly every dealer behaviour in this document follows from it.
2. **Gamma's sign is who is long, not call versus put.** Every gamma-exposure number is an assumption about who is long, whether or not it says so.
3. **Spend your effort on positioning, not on prettier profiles.** A wrong sign is not imprecision; it is the opposite answer.
4. **Put gamma in dollars per 1% and divide by liquidity before believing it matters.** Raw billions mean nothing without depth.
5. **Expect modest effects on broad indices.** Measured market-maker gamma moved SPX volatility by a few points at most over 2020–2023. Look for bigger effects where κ is large: expiry, stress, single stocks.

6. **Lag open interest by a day, and remember that 0DTE is invisible to it.**
7. **Treat the flip as a band, and look at the depth of the profile below the price** rather than at the zero crossing.
8. **Beat implied volatility and a HAR model before claiming a signal.** High gamma coincides with calm partly by construction.
9. **Do not read direction into hedging.** Dealers do not need the market anywhere; their hedging chases or leans against moves in whichever direction they come.
10. **When a stress event happens, wait for position data before accepting a gamma explanation.** GameStop and August 2024 both had better explanations than the first one offered.

12.5 What is known, what is not, and what I would bet on

Known. The mechanics are exact: the sign rules, the parity equivalence, the hedged P&L identity and the scaling of discrete hedging error are mathematics, not hypotheses. Stock prices cluster at strikes on expiration dates, and hedge rebalancing is part of the reason. Dealers were historically net short index options, especially downside puts, and net long single-stock options. Measured hedger positions relate to volatility in the direction the mechanics predict, in single stocks and in the index. Intraday momentum exists across many futures markets.

Not known. How much gamma exposure computed from *public* data adds to implied and realised volatility out of sample; whether aggregate charm and vanna flows move index prices measurably; whether volatility reliably rises after large expiries; how the market has changed since 0DTE options passed half of SPX volume, beyond the samples that end in 2023–2024; and how large the over-the-counter positions that no public data captures really are.

What I would bet on. That position-measured dealer gamma has small but real conditional effects on volatility, concentrated near expiry and in thin liquidity; that convention-based gamma exposure mostly repackages the level of implied volatility and the recent path of prices, and adds little beyond them for broad indices; that pinning is the most reliable effect to build on; and that genuine gamma squeezes in single stocks happen, but more rarely than they are claimed. These are my judgements, and §11.4 describes how to test every one of them.

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13. References

Grouped by kind. Where a free copy exists it is the link; paywalled-only entries are marked. Working papers are marked as such, and practitioner sources from parties with a commercial interest in the conclusion are flagged.

13.1 Option pricing and hedging

- **Stoll, H. R. (1969).** "The Relationship Between Put and Call Option Prices." *Journal of Finance* 24(5), 801–824. [[paywalled]] — Put–call parity, four years before Black–Scholes (§2.6).
- **Black, F. & Scholes, M. (1973).** "The Pricing of Options and Corporate Liabilities." *Journal of Political Economy* 81(3), 637–654. [[paywalled]] — The replication argument and the formula (§3.3).
- **Merton, R. C. (1973).** "Theory of Rational Option Pricing." *Bell Journal of Economics and Management Science* 4(1), 141–183. — The rigorous companion to Black–Scholes.
- **Boyle, P. P. & Emanuel, D. (1980).** "Discretely Adjusted Option Hedges." *Journal of Financial Economics* 8(3), 259–282. [[paywalled]] — The distribution of hedging error when rebalancing is discrete (§4.7).
- **Rubinstein, M. & Leland, H. E. (1981).** "Replicating Options with Positions in Stock and Cash." *Financial Analysts Journal* 37(4), 63–72. [[paywalled]] — The recipe portfolio insurance followed (§5.10, §8.1).
- **Leland, H. E. (1985).** "Option Pricing and Replication with Transactions Costs." *Journal of Finance* 40(5), 1283–1301. [[paywalled]] — Trading costs as an adjustment to volatility (§4.8).
- **Hodges, S. D. & Neuberger, A. (1989).** "Optimal Replication of Contingent Claims under Transactions Costs." *Review of Futures Markets* 8, 222–239. — Hedging under costs posed as utility maximisation. Link is a catalogue record.
- **Whalley, A. E. & Wilmott, P. (1997).** "An Asymptotic Analysis of an Optimal Hedging Model for Option Pricing with Transaction Costs." *Mathematical Finance* 7(3), 307–324. — The no-transaction band and its cube-root scaling (§4.8). Link is a working-paper version.
- **El Karoui, N., Jeanblanc-Picqué, M. & Shreve, S. E. (1998).** "Robustness of the Black and Scholes Formula." *Mathematical Finance* 8(2), 93–126. [[paywalled]] — Hedging with the wrong volatility, made rigorous (§4.4).
- **Carr, P. & Madan, D. (1998).** "Towards a Theory of Volatility Trading." In R. Jarrow (ed.), *Volatility: New Estimation Techniques for Pricing Derivatives*, Risk Books. — The P&L of a delta-hedged option as a gamma-weighted bet on variance (§4.4, §4.6).

- **Derman, E. (1999).** "When You Cannot Hedge Continuously: The Corrections to Black–Scholes." Goldman Sachs Quantitative Strategies research note; also published in *Risk*. — The $\sqrt{\pi/4}$ rule for hedging error (§4.7).
- **Bertsimas, D., Kogan, L. & Lo, A. W. (2000).** "When Is Time Continuous?" *Journal of Financial Economics* 55(2), 173–204. — Replication error under discrete trading, in general (§4.7).
- **Ahmad, R. & Wilmott, P. (2005).** "Which Free Lunch Would You Like Today, Sir? Delta Hedging, Volatility Arbitrage and Optimal Portfolios." *Wilmott Magazine*, November, 64–79. — Hedging at implied versus realised volatility, with the consequences for P&L paths (§4.6).

13.2 Market making and microstructure

- **Ho, T. & Stoll, H. R. (1981).** "Optimal Dealer Pricing under Transactions and Return Uncertainty." *Journal of Financial Economics* 9(1), 47–73. [[paywalled]] — Quotes shaded against inventory (§1.2).
- **Glosten, L. R. & Milgrom, P. R. (1985).** "Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders." *Journal of Financial Economics* 14(1), 71–100. — Why adverse selection alone produces a spread (§1.2).
- **Kyle, A. S. (1985).** "Continuous Auctions and Insider Trading." *Econometrica* 53(6), 1315–1335. [[paywalled]] — Price impact proportional to order size (§1.2).
- **Lee, C. M. C. & Ready, M. J. (1991).** "Inferring Trade Direction from Intraday Data." *Journal of Finance* 46(2), 733–746. [[paywalled]] — The standard trade-classification rule (§6.3).
- **Jameson, M. & Wilhelm, W. (1992).** "Market Making in the Options Markets and the Costs of Discrete Hedge Rebalancing." *Journal of Finance* 47(2), 765–779. [[paywalled]] — Rebalancing and volatility risk priced in option spreads (§1.3).
- **Easley, D., O'Hara, M. & Srinivas, P. S. (1998).** "Option Volume and Stock Prices: Evidence on Where Informed Traders Trade." *Journal of Finance* 53(2), 431–465. — Informed trading migrates to options (§1.3).
- **Savickas, R. & Wilson, A. J. (2003).** "On Inferring the Direction of Option Trades." *Journal of Financial and Quantitative Analysis* 38(4), 881–902. [[paywalled]] — How often trade-side rules misclassify option trades (§6.3).
- **Biais, B., Glosten, L. & Spatt, C. (2005).** "Market Microstructure: A Survey of Microfoundations, Empirical Results, and Policy Implications." *Journal of Financial Markets* 8(2), 217–264. — The survey to read first on microstructure (§1.2).
- **Avellaneda, M. & Stoikov, S. (2008).** "High-Frequency Trading in a Limit Order Book." *Quantitative Finance* 8(3), 217–224. — The reservation-price model of electronic market making (§1.2).
- **Stoikov, S. & Sağlam, M. (2009).** "Option Market Making under Inventory Risk." *Review of Derivatives Research* 12(1), 55–79. — When inventory should move option quotes (§9.2).
- **Tóth, B., Lempérière, Y., Deremble, C., de Lataillade, J., Kockelkoren, J. & Bouchaud, J.-P. (2011).** "Anomalous Price Impact and the Critical Nature of Liquidity in Financial Markets." *Physical Review X* 1, 021006. — The square-root impact law and why liquidity is fragile (§7.3).
- **Hendershott, T. & Menkveld, A. J. (2014).** "Price Pressures." *Journal of Financial Economics* 114(3), 405–423. — Direct evidence that intermediary inventory moves prices (§1.2).

13.3 Who holds options, and what demand does to prices

- **Bakshi, G. & Kapadia, N. (2003).** "Delta-Hedged Gains and the Negative Market Volatility Risk Premium." *Review of Financial Studies* 16(2), 527–566. — Buyers of delta-hedged index options lose on average (§4.5).
- **Bollen, N. P. B. & Whaley, R. E. (2004).** "Does Net Buying Pressure Affect the Shape of Implied Volatility Functions?" *Journal of Finance* 59(2), 711–753. — Demand for index puts moves the skew (§3.7).
- **Pan, J. & Poteshman, A. M. (2006).** "The Information in Option Volume for Future Stock Prices." *Review of Financial Studies* 19(3), 871–908. — Option order flow predicts stock returns (§1.3).
- **Lakonishok, J., Lee, I., Pearson, N. D. & Poteshman, A. M. (2007).** "Option Market Activity." *Review of Financial Studies* 20(3), 813–857. — Who holds which side of single-stock options; the prevalence of covered calls (§5.4, §6.3). Link is a working-paper version.
- **Gârleanu, N., Pedersen, L. H. & Poteshman, A. M. (2009).** "Demand-Based Option Pricing." *Review of Financial Studies* 22(10), 4259–4299. — End users long index options and short single-stock options; demand pressure priced through the unhedgeable part of the option. The single most useful paper for §6.3 and §9.2.
- **Carr, P. & Wu, L. (2009).** "Variance Risk Premiums." *Review of Financial Studies* 22(3), 1311–1341. [[paywalled]] — The variance risk premium measured directly (§4.5).
- **Muravyev, D. (2016).** "Order Flow and Expected Option Returns." *Journal of Finance* 71(2), 673–708. — Inventory risk is first-order in option prices (§9.2).
- **Chen, H., Joslin, S. & Ni, S. X. (2019).** "Demand for Crash Insurance, Intermediary Constraints, and Risk Premia in Financial Markets." *Review of Financial Studies* 32(1), 228–265. — Intermediary constraints and the price of crash protection (§6.3, §9.6).

13.4 Hedging feedback: theory

- **Grossman, S. J. (1988).** "An Analysis of the Implications for Stock and Futures Price Volatility of Program Trading and Dynamic Hedging Strategies." *Journal of Business* 61(3), 275–298. — Why synthetic options are not the same as traded ones (§7.1, §8.1).
- **Gennotte, G. & Leland, H. (1990).** "Market Liquidity, Hedging, and Crashes." *American Economic Review* 80(5), 999–1021. [[paywalled]] — A little hedging in a thin market as a discontinuity (§7.1).
- **Jacklin, C. J., Kleidon, A. W. & Pfleiderer, P. (1992).** "Underestimation of Portfolio Insurance and the Crash of October 1987." *Review of Financial Studies* 5(1), 35–63. [[paywalled]] — Crashes from underestimated hedging demand (§8.1).
- **Frey, R. & Stremme, A. (1997).** "Market Volatility and Feedback Effects from Dynamic Hedging." *Mathematical Finance* 7(4), 351–374. — How hedging demand transforms volatility (§7.1).
- **Platen, E. & Schweizer, M. (1998).** "On Feedback Effects from Hedging Derivatives." *Mathematical Finance* 8(1), 67–84. — Feedback as a source of smile-like implied volatility (§7.1).
- **Sircar, K. R. & Papanicolaou, G. (1998).** "General Black-Scholes Models Accounting for Increased Market Volatility from Hedging Strategies." *Applied Mathematical Finance* 5(1), 45–82. — The nonlinear pricing equation with feedback (§7.1).
- **Schönbucher, P. J. & Wilmott, P. (2000).** "The Feedback Effect of Hedging in Illiquid Markets." *SIAM Journal on Applied Mathematics* 61(1), 232–272. — Replication in illiquid markets (§7.1).
- **Krishnan, H. & Nelken, I. (2001).** "The Effect of Stock Pinning upon Option Prices." *Risk*, December. [[paywalled]] — Early statement of pinning (§7.7).
- **Avellaneda, M. & Lipkin, M. D. (2003).** "A Market-Induced Mechanism for Stock Pinning." *Quantitative Finance* 3(6), 417–425. — The pinning model with a singular drift at the strike (§7.7).
- **Jeannin, M., Iori, G. & Samuel, D. (2008).** "Modeling Stock Pinning." *Quantitative Finance* 8(8), 823–831. [[paywalled]] — Extends the model; short positions repel (§7.7).

13.5 Hedging feedback: evidence

- **Ni, S. X., Pearson, N. D. & Poteshman, A. M. (2005).** "Stock Price Clustering on Option Expiration Dates." *Journal of Financial Economics* 78(1), 49–87. — The best evidence in the field: closes cluster at strikes (§7.7).
- **Cheng, M. & Madhavan, A. (2009).** "The Dynamics of Leveraged and Inverse Exchange-Traded Funds." *Journal of Investment Management* 7(4). — The rebalancing arithmetic of leveraged ETFs (§5.10).
- **Golez, B. & Jackwerth, J. C. (2012).** "Pinning in the S&P 500 Futures." *Journal of Financial Economics* 106(3), 566–585. — Pinning and anti-pinning in index futures (§7.7).
- **Gao, L., Han, Y., Li, S. Z. & Zhou, G. (2018).** "Market Intraday Momentum." *Journal of Financial Economics* 129(2), 394–414. — Intraday momentum in the S&P 500 ETF, with non-hedging explanations (§7.2).
- **Ivanov, I. T. & Lenkey, S. L. (2018).** "Do Leveraged ETFs Really Amplify Late-Day Returns and Volatility?" *Journal of Financial Markets* 41, 36–56. — Fund flows offset much of the rebalancing (§5.10).
- **Barbon, A. & Buraschi, A. (2020).** "Gamma Fragility." Working paper, University of St. Gallen. — Dealer gamma imbalance, liquidity, and intraday momentum or reversal (§7.2).
- **Baltussen, G., Da, Z., Lammers, S. & Martens, M. (2021).** "Hedging Demand and Market Intraday Momentum." *Journal of Financial Economics* 142(1), 377–403. — Last-half-hour momentum across 60 futures markets, linked to short-gamma hedging (§7.2).
- **Barbon, A., Beckmeyer, H., Buraschi, A. & Moerke, M. (2021).** "Liquidity Provision to Leveraged ETFs and Equity Options Rebalancing Flows: Evidence from End-of-Day Stock Prices." Working paper, Swiss Finance Institute. — Separates leveraged-ETF and options hedging effects at the close (§5.10).
- **Ni, S. X., Pearson, N. D., Poteshman, A. M. & White, J. (2021).** "Does Option Trading Have a Pervasive Impact on Underlying Stock Prices?" *Review of Financial Studies* 34(4), 1952–1986. — Net positions of likely hedgers and stock volatility (§6.3, §7.2).
- **Zaynutdinova, G. R., Strong, C. & Baig, A. S. (2025).** "Seeking Gamma: Lessons from the Meme Frenzy." Working paper. — Identifies gamma-squeeze events and their returns (§7.10).
- **Maurer, M. (2026).** "Dealer Gamma Exposure and Overnight Gap Risk: Incremental Information in Low-Volatility Regimes." Working paper, unrefereed. — Dealer gamma's forecasting value conditional on the VIX regime (§10.2).

13.6 Zero-days-to-expiry options

- **Brogaard, J., Han, J. & Won, P. Y. (2023).** "Does 0DTE Options Trading Increase Volatility?" Working paper; earlier titled "How Does Zero-Day-to-Expiry Options Trading Affect the Volatility of Underlying Assets?" — The case that 0DTE volume raises index volatility (§7.9).
- **Dim, C., Eraker, B. & Vilkov, G. (2024).** "0DTEs: Trading, Gamma Risk and Volatility Propagation." Working paper. — 0DTE gamma does not propagate volatility (§7.9).

- **Adams, G., Fontaine, J.-S. & Ornthanalai, C. (2024).** "The Market for 0DTE: The Role of Liquidity Providers in Volatility Attenuation." Working paper. — Identification from days without 0DTE expiries; market makers dampen (§6.3, §7.9).
- **Amaya, D., Garcia-Ares, P. A., Pearson, N. D. & Vasquez, A. (2025).** "0DTE Index Options and Market Volatility: How Large is Their Impact?" Working paper. — Market-maker positions from every SPX trade; the maximum effect on volatility (§6.3, §7.3). Uses data provided by Cboe; hosted by Cboe.
- **Cboe (2023).** "Volatility Insights: Much Ado About 0DTEs — Evaluating the Market Impact of SPX 0DTE Options." Cboe Global Markets, by M. Xu. — Market makers' net 0DTE gamma from exchange data (§6.3, §7.9). Interested party: Cboe lists SPX options.
- **Cboe (2025).** "SPX 0DTE Options Jump to Record 62% Share in August." Cboe Global Markets. — 0DTE share of SPX volume (§7.9). Interested party.

13.7 Episodes and official reports

- **Presidential Task Force on Market Mechanisms (1988).** "Report of the Presidential Task Force on Market Mechanisms." US Government Printing Office. — The Brady report on October 1987 (§8.1).
- **Leland, H. E. & Rubinstein, M. (1988).** "The Evolution of Portfolio Insurance." In D. Luskin (ed.), *Portfolio Insurance: A Guide to Dynamic Hedging*, Wiley. — The history, by its inventors (§8.1).
- **Shiller, R. J. (1988).** "Portfolio Insurance and Other Investor Fashions as Factors in the 1987 Stock Market Crash." *NBER Macroeconomics Annual* 3, 287–297. — Survey evidence that sentiment, not only insurance, drove the crash (§8.1).
- **Carlson, M. (2007).** "A Brief History of the 1987 Stock Market Crash with a Discussion of the Federal Reserve Response." Finance and Economics Discussion Series 2007-13, Federal Reserve Board. — Accessible summary, including the Brady report's figures (§8.1).
- **Augustin, P., Cheng, I.-H. & Van den Bergen, L. (2021).** "Volmageddon and the Failure of Short Volatility Products." *Financial Analysts Journal* 77(3), 35–51. — February 2018 (§8.4).
- **SEC (2021).** "Staff Report on Equity and Options Market Structure Conditions in Early 2021." US Securities and Exchange Commission. — GameStop, and the best public description of US options market structure for individual investors (§1.5, §7.10, §8.5).
- **Mitts, J., Battalio, R. H., Brogaard, J., Cain, M. D., Glosten, L. R. & Kochuba, B. (2022).** "A Report by the Ad Hoc Academic Committee on Equity and Options Market Structure Conditions in Early 2021." Working paper. — The critique of the SEC report (§8.5). One author founded a gamma-analytics firm.
- **Hilliard, J. E. & Hilliard, J. (2023).** "The GameStop Short Squeeze: Put–Call Parity and the Effect of Frictions Before, During and After the Squeeze." *Journal of Futures Markets* 43(5), 635–661. — Put–call parity under short-selling frictions, in the most extreme case on record (§2.6).
- **Todorov, K. & Vilkov, G. (2024).** "Anatomy of the VIX Spike in August 2024." *BIS Bulletin* 95, Bank for International Settlements. — Quotes, not hedging, behind the spike (§8.7, §9.6).

13.8 Practitioner sources and data

- **SqueezeMetrics (2017).** "Gamma Exposure (GEX): Quantifying Hedge Rebalancing in SPX Options." White paper, March 2016, revised December 2017. — The document that popularised gamma exposure, with its four assumptions stated (§6.1, §6.3, §7.4). Interested party: the firm sells the data.
- **Cboe Open-Close Volume Summary.** Product description. — Participant-coded opening and closing volume for Cboe's options exchanges (§6.3, §11.2).

13.9 Books and econometric tools

- **Taleb, N. N. (1997).** *Dynamic Hedging: Managing Vanilla and Exotic Options.* Wiley. — The practitioner's book on what hedging actually involves. Dated in its products, not in its insight.
- **Harris, L. (2003).** "Trading and Exchanges: Market Microstructure for Practitioners." Oxford University Press. — Who trades, why, and how markets are organised.
- **Gatheral, J. (2006).** "The Volatility Surface: A Practitioner's Guide." Wiley. — How the surface is modelled and why it moves.
- **Sinclair, E. (2013).** "Volatility Trading." 2nd ed., Wiley. — Hedging, the variance premium and trade evaluation, for traders.
- **Natenberg, S. (2015).** "Option Volatility and Pricing." 2nd ed., McGraw-Hill. — The book trading firms give new options traders.
- **Bouchaud, J.-P., Bonart, J., Donier, J. & Gould, M. (2018).** "Trades, Quotes and Prices: Financial Markets Under the Microscope." Cambridge University Press. — Market impact and liquidity from the data up (§7.3).

- **Hull, J. C. (2021).** "Options, Futures, and Other Derivatives." 11th ed., Pearson. — The standard textbook reference for contract mechanics and pricing.
- **Diebold, F. X. & Mariano, R. S. (1995).** "Comparing Predictive Accuracy." *Journal of Business and Economic Statistics* 13(3), 253–263. — Testing whether one forecast beats another (§11.4).
- **Corsi, F. (2009).** "A Simple Approximate Long-Memory Model of Realized Volatility." *Journal of Financial Econometrics* 7(2), 174–196. — The HAR baseline every gamma signal must beat (§10.2, §11.4).
- **Patton, A. J. (2011).** "Volatility Forecast Comparison Using Imperfect Volatility Proxies." *Journal of Econometrics* 160(1), 246–256. — Which loss functions rank volatility forecasts correctly (§11.4).

13.10 If you only read eight things

In order:

1. **Gârleanu, Pedersen and Poteshman (2009)** — who holds options, and why it shows up in their prices.
2. **Derman (1999)** — six pages on what hedging really costs, with the rule of thumb for hedging error.
3. **SqueezeMetrics (2017)** — the practitioner framing, with its assumptions stated plainly enough to criticise.
4. **Ni, Pearson and Poteshman (2005)** — the strongest evidence that hedging moves prices.
5. **Ni, Pearson, Poteshman and White (2021)** — measured hedger positions and volatility, across stocks.
6. **Baltussen, Da, Lammers and Martens (2021)** — the intraday signature, across markets and decades.
7. **Amaya, Garcia-Ares, Pearson and Vasquez (2025)** — how large the effect is when positions are measured.
8. **SEC (2021), then Mitts and co-authors (2022)** — a gamma-squeeze story, the data, and the dispute.

Appendix A. Concepts and prerequisites

Everything the main text leans on without stopping to explain. The reader this is written for is mathematically comfortable but has not necessarily seen stochastic calculus, market microstructure or forecast-evaluation econometrics before: they hit "quadratic variation" in §4.4 or "QLIKE" in §11.4, want the idea and the precise definition rather than a citation, and would rather not leave the document to get them.

Entries are ordered by **dependency**, not alphabetically — a later entry uses an earlier one — and grouped into four parts that are themselves in dependency order, so the appendix reads as a build-up from probability, through market mechanics, to the volatility products and the forecasting toolkit the later sections use. Cross-references between entries are written "(A.3)"; references back to the main text are "§x.y" as everywhere else in the document.

Each entry gives the idea in words first, then the formal definition, then why it appears here, then where to go deeper. Notation follows the main text's notation block exactly; where a standard formula from another field would collide with it — the document's r is the risk-free rate, not a return; its P is a put's price, not a probability — the collision is flagged and a substitute symbol is introduced explicitly, rather than reusing the letter and hoping context disambiguates it.

Index. Where each concept first bites:

Concept	First used	Concept	First used
The lognormal distribution and fat tails	§3.1	Kyle's lambda and the square-root law	§1.2
Brownian motion and GBM	§3.2	Payment for order flow	§1.5
Quadratic variation and Itô's lemma	§4.4	Short selling and hard-to-borrow	§2.6
The law of Z^2 and $\sqrt{N}$ error growth	§4.7	Cost of carry and basis risk	§4.9
Self-financing strategies and replication	§3.2	Front-running and anticipatory trading	§8.4
Risk-neutral probability	§3.3	Variance swaps and the variance risk premium	§1.3
Convexity and one-sided super-hedging	§4.4	The VIX index	§7.7
Nonlinear PDEs and model blow-up	§7.1	Dispersion trading	§8.7

Concept	First used	Concept	First used
Asymptotic analysis	§4.8	The carry trade and its unwinds	§8.7
Autocorrelation, AR(1), the variance ratio	§7.2	The HAR model	§10.2
Volatility clustering	§7.4	The QLIKE loss	§11.4
Expected utility and risk aversion	§1.2	The Diebold–Mariano test	§11.4
Inventory models: Ho–Stoll, Avellaneda–Stoikov	§1.2	Event studies and placebo tests	§11.4
Glosten–Milgrom and adverse selection	§1.2	Multiple testing and data snooping	§11.4

Part I — Probability and stochastic calculus. The main text prices options and hedges them by an argument that is really a chain of eleven ideas from probability and stochastic calculus, compressed into a few lines at §3.2–§4.8. This part unpacks the chain in the order it is actually used: what kind of randomness a price has, what continuous trading does to it, the calculus that follows, and the statistics of the errors that remain once you try to trade it.

A.1 The lognormal distribution, and fat tails

The idea. A stock price cannot go negative, and a 100-point move means something different at a price of 100 than at a price of 10,000. Both problems disappear if you model not the price itself but its *ratio* from one day to the next, and treat the ratio's logarithm as the well-behaved, symmetric, additive quantity. That is the entire motivation for the lognormal model: it is the simplest distribution for a positive quantity whose *percentage* changes, not its absolute changes, are the stable, aggregatable thing.

Formally. A positive random variable S_T is **lognormally** distributed if $\ln(S_T/S_0)$ is normal. Writing the mean and variance of that log-return as $(\mu - \frac{1}{2}\sigma^2)\tau$ and $\sigma^2\tau$ — the reason for the $-\frac{1}{2}\sigma^2$ centring term is A.2 — gives $S_T = S_0 \exp\{(\mu - \frac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}\,Z\}$ for $Z \sim N(0,1)$. **Kurtosis** is $\mathbb{E}[(X - \mathbb{E}X)^4] / \operatorname{Var}(X)^2$; a normal distribution has kurtosis exactly 3, and **excess kurtosis** is kurtosis minus 3. A lognormal price implies log returns with zero excess kurtosis. Real daily returns do not: they show reliably positive excess kurtosis — more small, quiet days and more extreme days than the lognormal predicts, with less in between — a property inherited from **volatility clustering** (A.11) even when each day, conditional on its own local variance, is close to normal.

Why it appears here. §3.1 defines σ as the annualised standard deviation of log returns, which is only the *second* moment of a distribution the model otherwise assumes is lognormal; §3.3 lists "returns have fatter tails than the lognormal distribution allows" among the ways every Black–Scholes assumption is false, which is this entry's second paragraph in one clause.

Deeper. Hull (2021), in its chapter on the Black–Scholes–Merton model, for the lognormal assumption behind the pricing formula; Cont, R. (2001). "Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues." *Quantitative Finance* 1(2), 223–236, for the fat tails and the other stylised facts return series share.

A.2 Brownian motion and geometric Brownian motion

The idea. Shrink the up-or-down step of §3.2's binomial tree — a fixed percentage move over a fixed short interval — towards an instant, keeping the *variance per unit time* fixed rather than the step itself. What is left in the limit is a process that is continuous, has no memory beyond its current level, and whose increments over any interval are normal with variance proportional to the interval's length. That process is Brownian motion, and a stock price built from it — by feeding it through the same exponential that turned a random log return into a positive price in A.1 — is geometric Brownian motion (GBM). It is the continuous-time version of the tree, nothing more exotic.

Formally. Brownian motion W_t satisfies $W_0 = 0$, has independent increments, $W_t - W_s \sim N(0,\, t-s)$ for $t > s$, and continuous paths. **Geometric Brownian motion** solves the stochastic differential equation $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$, with solution $S_t = S_0 \exp\{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\}$ — A.1's lognormal price with $\tau = t$. The Cox–Ross–Rubinstein tree with n steps over $[0,\tau]$, up and down factors $u = e^{\sigma\sqrt{\tau/n}}$ and $d = 1/u$, converges as $n \to \infty$ to exactly this distribution. Note what is *not* in the main text's notation block: the real-world drift μ. It never needs a symbol, because A.5's replication argument prices the option without ever using it.

Why it appears here. §3.3: "Shrink the tree's steps towards zero, letting the price follow a random walk in log returns with constant volatility σ, and replication still works" is precisely this limit, stated in one clause; §3.2 is the tree it is the limit of.

Deeper. Cox, J. C., Ross, S. A. & Rubinstein, M. (1979). "Option Pricing: A Simplified Approach." *Journal of Financial Economics* 7(3), 229–263, for the tree and its continuous limit; Merton (1973) for the continuous-time model directly.

A.3 Quadratic variation, and Itô's lemma

The idea. Ordinary calculus has a rule of thumb: over a short interval, squared changes are negligible next to first-order changes, so $(dx)^2$ is dropped from any Taylor expansion. Brownian motion breaks that rule. Its path is so jagged that the sum of squared increments over a partition does not shrink to zero as the partition refines — it converges to the *length of the interval itself*, a finite, non-zero, deterministic number. That single fact is the reason a second-order term survives every hedging calculation in this document: what calculus calls second-order and negligible, Brownian motion delivers as first-order and unavoidable. Itô's lemma is simply the chain rule rewritten to keep that term.

Formally. For a partition of $[0, t]$ with mesh going to zero, the **quadratic variation** of Brownian motion is $\sum_k (W_{t_{k+1}} - W_{t_k})^2 \to t$, written $[W, W]_t = t$, or informally $(dW_t)^2 = dt$. For S_t following A.2's GBM and $f(S, t)$ twice differentiable in S, **Itô's lemma** gives

$$df = \left(\frac{\partial f}{\partial t} + \mu S \frac{\partial f}{\partial S} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} \right) dt + \sigma S \frac{\partial f}{\partial S}\, dW_t.$$

The extra term $\frac{1}{2}\sigma^2 S^2 f_{SS}\, dt$, absent from the ordinary chain rule, is exactly the quadratic-variation correction: $(dS)^2 = \sigma^2 S^2\, dt$ to leading order, not zero. (Here dS and dt are infinitesimals, distinct from the document's Greek Δ, exactly as the notation block already separates Δ from ΔS and Δt.)

Why it appears here. §4.4 derives $\Delta V \approx \Theta \Delta t + \Delta \cdot \Delta S + \frac{1}{2}\Gamma (\Delta S)^2$ — a finite-difference Itô expansion — and the whole point of the section is that the $\frac{1}{2}\Gamma(\Delta S)^2$ term does *not* vanish: "why the $(\Delta S)^2$ term survives" is this entry's first paragraph applied to $V(S, t)$.

Deeper. Shreve, S. E. (2004). *Stochastic Calculus for Finance II: Continuous-Time Models.* Springer, chapters 4–5 — the standard applied treatment, built around exactly this application.

A.4 The law of Z^2, and why summed errors grow like $\sqrt{N}$

The idea. Two separate facts, both needed for §4.7's arithmetic of hedging error. First: squaring a standard normal draw does not give you back something normal — it gives you a distribution concentrated near zero with a long right tail, because negative and positive draws of the same magnitude both land on the same positive square. Second: independent random errors do not add up in proportion to their number; their *variances* add, so their combined *size* (standard deviation) grows only with the square root of the number of them. Averaging N independent errors of fixed size shrinks the result by the same $\sqrt{N}$, the other side of the same fact.

Formally. If $Z \sim N(0, 1)$, then Z^2 follows the **chi-squared distribution with one degree of freedom**, χ^2_1, with $\mathbb{E}[Z^2] = 1$ and $\text{Var}(Z^2) = 2$. If $X_1, \ldots, X_N$ are independent with mean zero and variance v each, $\text{Var}(\sum_k X_k) = Nv$, so $\text{sd}(\sum_k X_k) = \sqrt{N} \cdot \sqrt{v}$, while $\text{sd}(N^{-1} \sum_k X_k) = \sqrt{v}/\sqrt{N}$.

Why it appears here. §4.7 writes a day's hedging error as $\frac{1}{2}\Gamma S^2 \sigma^2 \Delta t\, (Z^2 - 1)$: mean zero because $\mathbb{E}[Z^2] = 1$, and its scale is fixed by $\text{Var}(Z^2 - 1) = 2$. Summing N_h independent such errors over an option's life and using the second fact gives Derman's rule $\text{sd}(\Pi) \approx \sqrt{\pi/4} \cdot \nu \cdot \sigma / \sqrt{N_h}$ — the same $\sqrt{N}$ law that §3.1 already used, in its growing rather than shrinking direction, to scale daily volatility up to annual.

Deeper. Boyle and Emanuel (1980) — the paper that derives this distribution for discrete hedging error directly; Bertsimas, Kogan and Lo (2000) for the general case.

A.5 Self-financing strategies, and replication

The idea. §3.2's replication argument only works because the hedger never adds or removes outside cash: every share bought is paid for by borrowing more, and every share sold pays down the loan, both at the price prevailing *right then*. A trading strategy with that property — its value moves only because the things it holds change in value, never because money

is injected — is called self-financing, and it is the formal content of "manufacturing" a payoff by trading rather than by anything else.

Formally. A strategy holds ϕ_t shares and ψ_t units of a cash account B_t (with $B_t = e^{rt}$), worth $\Pi_t = \phi_t S_t + \psi_t B_t$ — a different letter from the document's V deliberately, since V already names the option's own value, and Π_t matches the document's use of Π for a hedged position's P&L, of which this is the running total. It is **self-financing** if $d\Pi_t = \phi_t\, dS_t + \psi_t\, dB_t$: value changes come only from price moves, not from rebalancing itself. A strategy **replicates** a claim paying V_T if $\Pi_T = V_T$ in every state; no-arbitrage then forces $V_t = \Pi_t$ for every earlier t, since a gap would let anyone buy the cheaper of the two and sell the dearer for a riskless profit — §3.2's argument exactly.

Why it appears here. §3.2 builds the two-state replicating portfolio "buy h shares and hold b dollars of cash" and rebalances it one step at a time without ever adding money; §4.4's identity is what that same rebalancing costs once the steps shrink to instants (A.3).

Deeper. Harrison, J. M. & Pliska, S. R. (1981). "Martingales and Stochastic Integrals in the Theory of Continuous Trading." *Stochastic Processes and Their Applications* 11(3), 215–260 — the rigorous foundation; Black and Scholes (1973) and Merton (1973) for the application.

A.6 Risk-neutral probability

The idea. §3.2's central surprise is that the probability of the stock rising never enters the option's price. That does not mean probability is irrelevant to pricing — it means the *real-world* probability is irrelevant, because pricing by replication implicitly uses a *different* set of probabilities: the ones under which every traded asset, once you strip out the time value of money, is expected to go nowhere. Those are not anyone's actual beliefs. They are a bookkeeping device — a change of measure that makes the replication argument's arithmetic come out as an expectation — and they are called risk-neutral because under them nobody needs to be compensated for risk: every asset already earns the riskless rate.

Formally. Write $\mathbb{P}$ for the real-world (physical) probability measure and $\mathbb{Q}$ for an equivalent measure (blackboard letters here specifically to avoid the document's own P, which names a put's price, not a probability). The **fundamental theorem of asset pricing** says a market is arbitrage-free essentially iff such a $\mathbb{Q}$ exists under which every discounted traded price is a martingale; a claim's no-arbitrage price is then $V_0 = e^{-r\tau}\,\mathbb{E}^{\mathbb{Q}}[V_T]$. Under $\mathbb{Q}$ the stock's drift becomes $r - q$, replacing whatever real-world μ it actually has (A.2) — which is why μ never needed a symbol in the first place. For the Black–Scholes call, $N(d_2) = \mathbb{Q}(S_T > K)$ exactly: the risk-neutral probability of finishing in the money.

Why it appears here. §3.3 states this outright and defers it here: "$N(d_2)$ … is the probability that the call finishes in the money under the **risk-neutral** probabilities — the probabilities that price assets as if no investor demanded compensation for risk, which Appendix A explains."

Deeper. Harrison, J. M. & Kreps, D. M. (1979). "Martingales and Arbitrage in Multiperiod Securities Markets." *Journal of Economic Theory* 20(3), 381–408 — the origin of the theorem; Cox, Ross and Rubinstein (1979) (A.2) for the concrete two-state version; Hull (2021), on binomial trees.

A.7 Convexity, and one-sided super-hedging

The idea. A hedger who does not know the true future volatility, only some guess, would like a guarantee that goes beyond "right on average": a guarantee that holds on *every* path, however the price actually moves. It turns out such a guarantee exists, and it costs nothing beyond guessing on the safe side. If you sell an option whose payoff curves the right way — a call or a put, not some exotic shape — and you *always* hedge using a volatility input at least as large as whatever volatility actually turns out to occur, moment to moment, you cannot lose, no matter what the price does. Guess low instead and you cannot win. The direction of the guess, not its accuracy, is what is guaranteed.

Formally. A payoff is **convex** in S if its second derivative is non-negative wherever it exists — true of both $\max(S - K, 0)$ and $\max(K - S, 0)$, so $\Gamma \geq 0$ for a long position in either. Let $\widehat{\sigma}_t$ be the (possibly time-varying, model-free) volatility used to compute hedging deltas, and let σ_t be the true instantaneous volatility actually realised along a given path, unknown and possibly not even following a fixed model. If $\widehat{\sigma}_t \geq \sigma_t$ for (almost) every t along that path, a **seller** who hedges at $\widehat{\sigma}_t$ has hedged P&L ≥ 0 *on that path* — not merely in expectation. If instead $\widehat{\sigma}_t \leq \sigma_t$ everywhere, the seller's hedged P&L is ≤ 0 on that path.

Why it appears here. This is §4.4 point 3, quoted directly: "for options with convex payoffs such as calls and puts, a seller who hedges with a volatility at or above the true one at every moment cannot lose on the hedged position, a seller who hedges below it cannot win" — a pathwise reading of the boxed identity $\Delta\Pi \approx \frac{1}{2}\Gamma S^2[(\Delta S/S)^2 - \widehat{\sigma}^2\Delta t]$, term by term, rather than only on average.

Deeper. El Karoui, Jeanblanc-Picqué and Shreve (1998) — the paper that makes this rigorous.

A.8 Nonlinear partial differential equations, and model blow-up

The idea. The Black–Scholes equation (§3.6) assumes the hedger's own trading has no effect on the price it is hedging. Once dealers as a group are large enough to move the market, that assumption breaks, and the equation governing the option's value has to feed the hedgers' own gamma back into the volatility they face. The pricing equation stops being *linear* in the option value: the effective volatility a position experiences now depends on the position itself, through how much of it needs to be hedged. Push that feedback hard enough — in a large enough position, or a thin enough market — and the equation can develop a genuine mathematical singularity: a point at which the model's own effective volatility is predicted to be infinite. That is not a numerical artefact; it is the model's way of saying a crash is coming.

Formally. Where the ordinary Black–Scholes equation is $\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta - rV = 0$ with constant σ, a feedback model replaces σ with an **effective volatility** $\hat{\sigma}(S, t, \Gamma)$ that increases with the hedging demand Γ generates, making the equation quasi-linear (the coefficient of the highest derivative, Γ itself, depends on the solution). In the models cited below, $\hat{\sigma}^2$ takes the schematic form σ^2 divided by a factor that shrinks towards zero as aggregate, liquidity-scaled hedging demand grows — so $\hat{\sigma} \to \infty$ at a finite level of that demand: a **blow-up**, the continuous-time twin of the document's own $\kappa \to -1$ singularity (§7.1).

Why it appears here. §7.1: "In each of them the effective volatility has a denominator that approaches zero as liquidity-scaled hedging demand grows, and that singularity is the model's crash. κ is the same object with the stochastic calculus removed" — this entry is the calculus κ was removed from.

Deeper. Frey and Stremme (1997); Schönbucher and Wilmott (2000) — both build the nonlinear equation and its singularity explicitly.

A.9 Asymptotic (small-parameter) analysis

The idea. Some problems have no clean closed-form solution but do have a small, natural parameter — a trading cost, a rebalancing interval — that vanishes in some limit where the problem *does* simplify. Asymptotic analysis expands the true solution as a power series in that small parameter around the solvable limit, keeping the first term or two and discarding the rest as negligible. The art is in the power: the right expansion is not always in the parameter itself, but sometimes in a fractional power of it, because that is the scaling at which the terms of the problem actually balance.

Formally. For a small parameter ε, seek a solution as $f(x;\varepsilon) = f_0(x) + \varepsilon^p f_1(x) + O(\varepsilon^{2p})$ for whatever power p makes the leading correction balance a term otherwise left out of f_0 — a **distinguished limit**. Whalley and Wilmott's hedging-band problem (§4.8) is solved this way in the small proportional cost c: balancing the cost of trading against the risk of not trading forces the band's half-width to scale as $c^{1/3}$, giving $w = (3c\Gamma^2 S/2a)^{1/3}$ — a *cube-root*, not linear, law, and a textbook example of a distinguished limit rather than a naive first-order expansion.

Why it appears here. §4.8's source is literally named for this technique: Whalley and Wilmott (1997) is titled "An Asymptotic Analysis of an Optimal Hedging Model for Option Pricing with Transaction Costs," and the section's own observation that "the band scales as the cube root of cost, so doubling costs widens it by only 26%" is exactly the distinguished-limit scaling in action.

Deeper. Whalley and Wilmott (1997) directly; Hodges and Neuberger (1989) for the utility-maximisation problem it is the small-cost limit of.

A.10 Autocorrelation, the AR(1) process, and the variance ratio

The idea. If today's return tells you something about tomorrow's — on average pulling it in the same direction (momentum) or the opposite one (reversal) — returns are **autocorrelated**, and market efficiency in its weakest form says they should not be, beyond what trading costs can exploit. The simplest model of a small departure from that ideal makes each return a fixed fraction of the previous one, plus fresh news; and a natural way to detect the departure without fitting that model directly is to compare how fast variance grows with the horizon, since independent returns must have variance grow exactly in proportion to time (A.4), while dependent ones do not.

Formally. An **AR(1) process** is $r_t = \phi\, r_{t-1} + e_t$, with e_t mean-zero and uncorrelated with r_{t-1}; its lag-1 autocorrelation is exactly ϕ, and its lag-k autocorrelation is ϕ^k, decaying geometrically. The **variance ratio** at horizon q is $\mathrm{VR}(q) = \mathrm{Var}(r_t^{(q)}) \big/ (q\,\mathrm{Var}(r_t))$, where $r_t^{(q)}$ is the q-period return; under an uncorrelated random walk $\mathrm{VR}(q) = 1$ for every q, while $\mathrm{VR}(q) > 1$ signals positive autocorrelation and $\mathrm{VR}(q) < 1$ negative.

Why it appears here. §7.2's lagged feedback $r_t = e_t - \kappa\, r_{t-1}$ is an AR(1) with $\phi = -\kappa$ exactly, which is why "consecutive returns now have correlation $-\kappa$" follows immediately; and the section's own ratio of daily to bar-level volatility (0.770 for $\kappa = 0.3$, matching the multiplier $1/(1+\kappa)$ of §7.1) is a variance ratio computed at a one-day horizon.

Deeper. Lo, A. W. & MacKinlay, A. C. (1988). "Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test." *Review of Financial Studies* 1(1), 41–66 — the variance-ratio test; Hamilton, J. D. (1994). *Time Series Analysis.* Princeton University Press, chapters 3–4, for AR processes generally.

A.11 Volatility clustering

The idea. Returns themselves are close to uncorrelated, as A.10 says they should be — but their *magnitudes* are not. A large move, of either sign, tends to be followed by more large moves, and a quiet spell tends to stay quiet, so that "how much" is predictable even when "which way" is not. This is volatility clustering, and it is one of the oldest and most robust facts about return data: it is why a single unconditional standard deviation is a poor description of risk on any given day, and why the natural model for variance is itself autoregressive, one level up from the returns.

Formally. Returns r_t show volatility clustering if r_t is close to serially uncorrelated while r_t^2 (or $|r_t|$) is positively autocorrelated over long lags. The canonical model is **GARCH(1,1)**: $r_t = \sigma_t Z_t$ with Z_t iid standard normal and $\sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2$, where $\omega > 0$ and $\alpha, \beta \geq 0$ with $\alpha + \beta < 1$ for stationarity; σ_t^2 here is the same conditional variance the document calls σ_r^2 once realised and annualised, just written with a time subscript to track how it evolves.

Why it appears here. §7.4's caution is exactly this fact used as a confound: "calm markets tend to be rising markets… and volatility clusters, so calm tends to continue. A sign-convention GEX will therefore read high in calm markets *because* they are calm… A study that finds high GEX followed by low volatility without controlling for current implied and realised volatility has not distinguished hedging from persistence" — clustering alone predicts exactly the pattern a naive gamma-exposure study would credit to hedging.

Deeper. Engle, R. F. (1982). "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation." *Econometrica* 50(4), 987–1007; Bollerslev, T. (1987). "A Conditionally Heteroskedastic Time Series Model for Speculative Prices and Rates of Return." *Review of Economics and Statistics* 69(3), 542–547.

Part II — Market microstructure and trading mechanics. §1 sketches why a market maker's spread exists and how price impact works, in a few paragraphs that lean on a specific, citable literature; §2, §4 and §8 add short-selling frictions, cost of carry and anticipatory trading. These eight entries give that literature's actual models, in the order the main text needs them: first the decision theory underneath every quote (A.12), then the two competing explanations for why a spread exists at all (A.13, A.14), then what happens once orders start moving prices (A.15), and finally the institutional plumbing — order flow, borrowing, arbitrage, anticipation — that the later sections assume.

A.12 Expected utility, and the risk-aversion coefficient

The idea. To say a market maker "dislikes risk" precisely, you need more than a preference ranking; you need a way to compare a certain amount of money with a risky one of the same average value, and to say by how much the risky one is worth less. Expected utility theory supplies that: an agent evaluates a risky prospect not by its average payoff but by the average of some concave function of the payoff, and concavity — the function bending down as wealth rises — is what makes a sure thing preferable to a fair bet of equal expectation. How sharply it bends is a single number, and that number is what "risk aversion" means formally.

Formally. An agent with utility $U(\cdot)$, increasing and concave ($U'' < 0$), ranks a random wealth outcome $\tilde{W}$ by $\mathbb{E}[U(\tilde{W})]$. The **Arrow–Pratt coefficient of absolute risk aversion** is $a(W) = -U''(W)/U'(W)$ — the document's own notation, since §4.8 calls it a directly. For a small risk $\tilde{x}$ with mean zero, the certainty equivalent falls short of $\mathbb{E}[\tilde{x}]$ by approximately $\frac{1}{2}\, a\, \mathrm{Var}(\tilde{x})$: the standard local approximation that both entries below build on.

Why it appears here. §1.2 introduces the Avellaneda–Stoikov reservation price as moving "against the inventory in proportion to the dealer's risk aversion, the variance of the price, and the time left in the trading session" (A.13); §4.8's hedging-band width depends on the same coefficient a (A.9). Both are the certainty-equivalent adjustment of this entry, applied to a market maker's inventory instead of a generic small risk.

Deeper. Pratt, J. W. (1964). "Risk Aversion in the Small and in the Large." *Econometrica* 32(1/2), 122–136 — defines the coefficient; von Neumann, J. & Morgenstern, O. (1944). *Theory of Games and Economic Behavior.* Princeton University Press, for the axiomatic foundation.

A.13 Inventory models of market making

The idea. A market maker holding unwanted shares faces a risk from every further minute it holds them, and the natural defence is to make holding them less likely: quote a less attractive price on the side that would add to the position, and a more attractive one on the side that would reduce it. Ho and Stoll formalised this first, for how *both* quotes should shade against inventory; Avellaneda and Stoikov later gave the idea a clean, usable form — center the quotes not on the market's mid-price but on a price that has already been shifted, by exactly the amount a risk-averse dealer with that inventory would accept, before spreading a bid and ask symmetrically around it.

Formally. Let y be the dealer's current signed inventory in shares (not q, which the document reserves for the dividend yield), s the mid-price, σ^2 the variance of the asset's return, a the risk-aversion coefficient (A.12), and $T - t$ the time left in the trading session — a session horizon, distinct from the document's option-expiry $\tau = T - t$. Avellaneda and Stoikov's **reservation price** is $R(s,t) = s - y\,a\,\sigma^2\,(T-t)$: below the mid when long ($y > 0$), above it when short, by more as inventory, risk aversion, variance or remaining time grow. Bid and ask are then set symmetrically (or asymmetrically, in the fuller model) around R, not around s.

Why it appears here. §1.2: "Ho and Stoll (1981) showed the natural defence: a dealer holding unwanted inventory shades *both* quotes downward… Avellaneda and Stoikov (2008) turned the same idea into the model most electronic market makers start from, in which the quotes are centred not on the mid but on a reservation price."

Deeper. Ho and Stoll (1981); Avellaneda and Stoikov (2008).

A.14 The Glosten–Milgrom model, and adverse selection

The idea. Even a market maker who is completely indifferent to risk still needs a spread, for a reason that has nothing to do with inventory: some of the people it trades with know more than it does. If a fraction of incoming orders come from traders who know the asset is worth more (or less) than the current price, and place orders accordingly, then a market maker who sets the same price for everyone will systematically overpay the informed sellers and underpay the informed buyers, funded by whatever it earns from the uninformed. The only defence is to charge every trader — informed and uninformed alike — a price that already reflects this: pushing the ask above, and the bid below, what the asset would be worth if nobody knew anything extra.

Formally. At each of a sequence of times, one trader arrives and buys, sells, or does not trade. With probability $\mu \in (0,1)$ the trader is **informed**: they know the asset's true value $V \in \{V_L, V_H\}$ and trade in the direction it implies. With probability $1 - \mu$ the trader is uninformed, buying or selling for reasons unrelated to value. A competitive, zero-profit market maker who cannot tell the two apart sets ask $= \mathbb{E}[V \mid \text{buy arrives}]$ and bid $= \mathbb{E}[V \mid \text{sell arrives}]$ by Bayes' rule; because a buy is relatively more likely when $V = V_H$, the ask exceeds the unconditional mean and the bid falls short of it, and a spread opens with no inventory motive at all.

Why it appears here. §1.2: "Glosten and Milgrom (1985) showed that if some fraction of order flow is informed, a competitive market maker must set a spread wide enough that what it earns from uninformed traders covers what it loses to informed ones" — this is that model, made explicit.

Deeper. Glosten and Milgrom (1985).

A.15 Price impact: Kyle's lambda and the square-root law

The idea. Once a market maker cannot tell an informed order from an uninformed one, the safest response to *any* order is to move the price a little in the direction it points, whether or not that particular trader knows anything — protection on average, even though it is wrong most of the time about any one trade. How much to move it, per unit traded, is price impact, and it turns out empirically not to be a fixed number: the price concession needed to trade a given fraction of a day's volume grows only with the *square root* of the size traded, not in proportion to it, so a ten-times-bigger order costs only about three times as much per share, not ten times. Two further distinctions matter alongside the shape of the curve: part of any order's impact is a temporary concession that fades once the order stops, and part is a permanent revision that persists because the order revealed something; and because the curve is concave rather than a straight line, "depth" — dollars per percent moved — is not a single number, but depends on how large a move you ask about.

Formally. Kyle's lambda posits a linear relation, $\Delta P = \lambda Q$ for signed net order flow Q — the original, tractable case. The better-supported empirical form is the **square-root law**: for an order of Q dollars executed over a day trading V dollars with daily volatility σ_d, impact $\approx Y\,\sigma_d\sqrt{Q/V}$, Y a constant of order one. Differentiating, the *marginal* impact of an extra dollar falls as Q grows — the source of the concavity — so an effective depth $D = Q/\text{impact}$ is itself a function of Q, not a constant.

Why it appears here. §1.2 introduces Kyle's lambda directly; §7.3 uses the square-root law to calibrate κ and states the concavity's consequence explicitly: "a concave law has no single order-independent depth," with the worked numbers showing effective κ falling from 0.35 at a 0.1% move to 0.07 at a 3% move. §7.2's finding that hedging-induced returns partially *reverse* at the next lag (negative autocorrelation under long gamma) is the temporary-impact component of this entry unwinding once the (uninformative, mechanical) hedging pressure passes.

Deeper. Kyle (1985); Tóth and co-authors (2011).

A.16 Payment for order flow, wholesalers, and price-improvement auctions

The idea. A retail broker's individual customer orders are, in aggregate, close to uninformed — no single retail order is likely to be driven by material non-public information — which makes that flow attractive to trade against. Rather than sending each order to an exchange, many US retail brokers route it to a wholesaler, who pays the broker for the right to fill it, either by trading against its own inventory (internalising it) or by running it through an auction that invites other liquidity providers to better the public quote. Both routes typically fill the retail customer at a price at least as good as the best publicly displayed price, which is the "price improvement" the arrangement is named for.

Formally. A **wholesaler** pays a retail broker a per-share (or per-contract) rebate for first refusal on that broker's marketable order flow. It then either **internalises** the order — executing against its own book at or better than the National Best Bid and Offer (NBBO) — or exposes it to a **price-improvement auction**, a brief competitive window in which other participants may execute at a price better than the NBBO. The wholesaler's profit is the spread it captures on internalised flow, net of the rebate paid and whatever it then costs to hedge the resulting position.

Why it appears here. §1.5's flow diagram routes retail orders through "a retail broker" to a "wholesaler or exchange auction… often for payment," and notes that "the same large firms are often both wholesalers and exchange market makers" (§1.4) — this entry is the arrangement that diagram compresses into one arrow.

Deeper. SEC (2021) — the report the main text calls "the best single public description of this structure."

A.17 Short selling, stock borrow, and "hard to borrow"

The idea. Selling a stock you do not own requires borrowing shares from someone who does, delivering the borrowed shares to the buyer, and eventually returning equivalent shares to the lender. That borrowing is not free, and its price is set by supply and demand for lendable shares like any other rental market: a stock most holders are happy to lend out is "easy to borrow" and nearly costless to short; a stock in high demand from short sellers relative to its lendable float is "hard to borrow," and the fee can be large enough to erode or reverse an otherwise-profitable short position, or to make the no-arbitrage relationships of §2.6 fail to hold exactly.

Formally. A short seller borrows shares through their broker, posts cash collateral (typically around 102% of the position's value), and pays an annualised **borrow fee** f on that value. The lender pays a **rebate rate** on the posted collateral equal to a reference rate minus f; when f exceeds the reference rate the rebate is negative and the short seller pays net cash to maintain the position. A stock is **hard to borrow (HTB)** when scarce lendable supply pushes f to a level that materially changes the economics of shorting it, or of positions — such as the reverse-conversion arbitrage behind put–call parity — that implicitly require a short stock position to enforce.

Why it appears here. §2.6: "prices can deviate from it visibly when the stock is hard to borrow. GameStop options in January 2021 are the extreme example; Hilliard and Hilliard (2023) find that the apparent violations there are largely accounted for by the costs of short selling rather than free money."

Deeper. Hilliard and Hilliard (2023); D'Avolio, G. (2002). "The Market for Borrowing Stock." *Journal of Financial Economics* 66(2–3), 271–306 — the standard reference on the stock-loan market itself.

A.18 Index arbitrage, the futures basis, cost of carry, and basis risk

The idea. A stock index future and the basket of stocks it tracks are two different ways to hold the same exposure, and the difference between their prices cannot exceed what it costs to convert one into the other — borrow cash and buy the basket instead of the future, or short the basket and lend the proceeds instead of buying the future — before someone does exactly that and captures the gap risk-free. The fair gap is set entirely by financing: the interest that must be paid (or is earned) holding the future's cash substitute for the time until it expires, net of the dividends the basket pays in the meantime that the future does not. Any hedge built from a related-but-not-identical instrument inherits a smaller version of the same problem: the hedge and the thing it hedges can drift apart even with no arbitrage opportunity involved, simply because they are not literally the same claim.

Formally. A futures contract's no-arbitrage (**cost-of-carry**) fair value is $F_0 = S_0\, e^{(r-q)\tau}$ — the document's own S, r, q and τ, since this is the same discounting that prices any forward claim. The **basis** is the difference between spot and futures prices (or its percentage form); **index arbitrage** is the trade that should hold it near its fair value, buying the cheaper side and selling the dearer. **Basis risk** is exposure to unexpected moves in the basis itself — the hedge and the hedged exposure decoupling — when a position in one instrument (SPX options, say) is hedged with a related but distinct one (E-mini futures, an ETF, a basket).

Why it appears here. §8.1's account of 19 October 1987 describes "index arbitrage — buying the now-cheap futures and selling the equivalent basket of stocks to close the gap"; §4.9 lists "basis" as a risk dynamic hedging cannot fix, since "the hedge does not track the option's underlying exactly"; §9.5 notes the AM-settlement basis risk a dealer holds "until Friday's settlement value is known."

Deeper. Hull (2021), on the determination of forward and futures prices, for cost-of-carry pricing; Carlson (2007) for its role in 1987.

A.19 Front-running, and anticipatory trading around predictable flows

The idea. If a large, price-moving trade is going to happen, and you can see it coming — because you were told, or because a mechanical rule makes its timing and direction public knowledge — you can trade ahead of it and let the trade itself pay for your position. Doing this does not stop the impact the big trade was always going to have; it mostly moves that impact earlier, spreading it across the period between when it became foreseeable and when it actually executes, and often making the anticipatory buying (or selling) a self-fulfilling accelerant in its own right.

Formally. Take A.15's linear-impact toy model: a flow of size Q known in advance to arrive at time 2 with impact λQ. If anticipatory traders believe this and trade ahead at time 1, their own buying (or selling) moves the price by some fraction of λQ *before* time 2, so the total impact λQ is realised earlier and smoothed across both periods rather than concentrated at time 2 — the total is essentially unchanged, but its timing is pulled forward, and if the anticipatory trade is itself large the pulling forward can overshoot.

Why it appears here. §8.4's account of Volmageddon: "predictable rebalancing is front-runnable, which pulls the price move forward rather than preventing it," with "the role of anticipatory trading around the known rebalancing window" flagged as still disputed; §10.1 makes the general point for gamma exposure itself: "a hedge that *must* be executed does not disappear because it is anticipated."

Deeper. Augustin and co-authors (2021).

Part III — Volatility products and strategies. §1.3, §7.7 and §8.7 assume the reader already knows what a variance swap is, how the VIX is built from one, and what a dispersion or carry trade does — assumptions this part makes good on, in the order the products actually nest: a variance swap first, since the VIX is a portfolio of the trades that replicate one; dispersion trading, which is a bet on the correlation embedded in the gap between index and single-stock variance; and the carry trade, whose August 2024 unwind shares this document's own feedback mechanism (§7.1) even though no options dealer is involved.

A.20 Variance swaps, and the variance risk premium

The idea. An option is a bet on volatility bundled together with a bet on direction, since its payoff depends on where the price ends up, not only on how much it moved. A variance swap strips the direction out: it is a contract whose only payoff is the difference between the variance that actually occurred and a variance level fixed in advance, with no strike, no moneyness, and (to a good approximation) no dependence on the path taken to get there. It exists because trading realised volatility directly, without the delta-hedging machinery of Part II of the main text, is exactly what many volatility traders and hedgers want, and because it can be built, and priced, from a portfolio of ordinary options.

Formally. A **variance swap** pays its long side $N_{\text{var}}(\sigma_r^2 - K_{\text{var}}^2)$ at maturity, where σ_r^2 is realised annualised variance over the swap's life (from a pre-agreed daily log-return sampling rule), K_{var} is the fixed **variance strike** set so the swap costs nothing to enter — a different symbol from the document's own strike K, since this is a level of variance, not of price — and N_{var} is the variance notional. No-arbitrage pricing under $\mathbb{Q}$ (A.6) gives $K_{\text{var}}^2 = \mathbb{E}^{\mathbb{Q}}[\sigma_r^2]$, which a classic replication argument shows equals a fixed, model-free weighted integral over option prices at every strike for that expiry — the fact A.21 turns into an index. The **variance risk premium** is $K_{\text{var}}^2 - \mathbb{E}^{\mathbb{P}}[\sigma_r^2]$: the gap between the priced-in and the physically expected variance, positive on average for equity indices, meaning sellers of variance are paid, on average, to bear crash risk.

Why it appears here. §4.5: "implied variance on the index has exceeded subsequently realised variance on average, by a margin that constitutes a large **variance risk premium** (Carr and Wu, 2009)"; §1.3 lists the variance-versus-realised gap as the first row of what an options market maker earns or loses relative to a stock market maker.

Deeper. Carr and Wu (2009); Carr and Madan (1998) for the replication argument itself.

A.21 The VIX index

The idea. If a variance swap's fair strike is a specific weighted average of option prices across every strike at one expiry (A.20), you can compute that average continuously from the options market without anyone actually trading a swap, and publish the result as an index. That is what the VIX is: not a forecast in the sense of a model's output, but a direct, formulaic reading of what a strip of real S&P 500 option quotes currently says a variance swap on the index would cost, converted to a volatility (a square root) and annualised to a 30-day horizon. Because it is built from *quotes* — bid–ask midpoints on resting orders — rather than from trades, anything that moves those quotes moves the index, whether or not a single contract changes hands.

Formally. The published construction is, schematically,

$$\text{VIX}^2 = \frac{2}{\tau}\sum_i \frac{\Delta K_i}{K_i^2}\, e^{r\tau}\, Q(K_i) \;-\; \frac{1}{\tau}\left(\frac{F}{K_0} - 1\right)^2,$$

summing out-of-the-money option prices $Q(K_i)$ across a wide strip of strikes K_i with spacing ΔK_i, F the forward index level, K_0 the first strike below F, and τ fixed at 30 calendar days — the document's own K, τ and r, since these are literally SPX option strikes, time to a target expiry, and the discount rate. The sum is precisely a discretisation of A.20's variance-swap replication integral.

Why it appears here. §7.7 defines the VIX as "Cboe's index of the S&P 500's expected volatility over the next 30 days, computed from a wide strip of index option quotes"; §8.7's account of August 2024 rests on exactly the "quotes, not trades" mechanism: "the widening of bid–ask spreads — especially on puts — as market makers protected themselves from an imbalanced book lifted the quote midpoints that enter the calculation," and §8.4's Volmageddon is a feedback loop in products written on this same index.

Deeper. Carr, P. & Wu, L. (2006). "A Tale of Two Indices." *Journal of Derivatives* 13(3), 13–29 — the academic account of the VIX as a variance-swap rate; Todorov and Vilkov (2024) for the quoting mechanism in practice.

A.22 Dispersion trading

The idea. An index's variance and the variance of its constituents are linked by a fixed accounting identity through the constituents' pairwise correlations: a basket of stocks that all move independently diversifies away much of its variance, while one where every stock moves together does not. Index and single-stock options are, separately, priced markets for each side of that identity, and they need not agree on the correlation implied by comparing the two. A dispersion trade takes a position on that gap — typically selling index variance and buying a weighted basket of single-stock variance, a bet that stocks will move together *less* than the two options markets jointly imply, or the reverse trade if the bet runs the other way.

Formally. With portfolio weights w_k and constituent volatilities σ_k for stocks $k = 1, \dots, n$ — subscripted by stock, not to be confused with the document's own σ_i for *implied* volatility, an unrelated use of the same letter "i" — index variance decomposes as

$$\sigma^2_{\text{index}} = \sum_k w_k^2\sigma_k^2 + \sum_{k\neq l} w_k w_l\, \sigma_k\sigma_l\, \rho_{kl}.$$

Assuming a single **implied correlation** ρ common to every pair and solving using each side's own quoted implied volatility gives a market-implied ρ; a dispersion trade is short (or long) index implied variance and long (or short) the weighted single-name implied variances, a direct bet that *realised* correlation will differ from this implied ρ.

Why it appears here. §8.7 lists dispersion trades, alongside volatility ETFs, among the mechanisms considered and largely ruled out for the August 2024 VIX spike: "Volatility ETFs and dispersion trades were unlikely to have been the main driver."

Deeper. Driessen, J., Maenhout, P. & Vilkov, G. (2009). "The Price of Correlation Risk: Evidence from Equity Options." *Journal of Finance* 64(3), 1377–1406.

A.23 The carry trade, and its unwinds

The idea. Borrow in a currency with a low interest rate, convert the proceeds, and invest in a currency (or any asset) with a higher one: if exchange rates did not move enough to offset the rate gap, this is free money, and on average, historically, they have not moved enough — a durable, if occasionally violent, anomaly against the theory that they should. The trade is usually run with leverage, which is fine as long as the funding currency stays cheap and the position's value stays comfortably above margin requirements. When the funding currency instead strengthens sharply, leveraged positions lose money and face margin calls at the same time, forcing exactly the deleveraging — sell the funded asset, buy back the funding currency — that pushes the funding currency up and the funded asset down further: the same forced, mechanical, self-reinforcing loop as §7.1's dealer hedging, with margin calls in the role dealers' gamma plays there.

Formally. Ignoring transaction costs, a carry trade funded in currency f at rate r_f and invested in currency d at rate r_d profits by approximately $(r_d - r_f) - \Delta e$, where Δe is the funding currency's appreciation over the holding period (subscripted rates here, distinct from the document's single risk-free r, because two currencies are in play). **Uncovered interest rate parity** predicts $\mathbb{E}[\Delta e]$ should exactly offset $r_d - r_f$ on average; the persistent, documented failure of this prediction is the "forward premium puzzle" the carry trade exploits. An **unwind** is the forced deleveraging described above, typically triggered by a sudden move in Δe against the trade.

Why it appears here. §8.7: "On 5 August 2024, during the unwinding of yen carry trades, the VIX recorded its largest one-day spike on record."

Deeper. Brunnermeier, M. K., Nagel, S. & Pedersen, L. H. (2008). "Carry Trades and Currency Crashes." *NBER Macroeconomics Annual* 23, 313–347; Todorov and Vilkov (2024) for the August 2024 episode specifically.

Part IV — Forecasting and evaluation. §10 and §11 ask whether a gamma signal is worth anything, and answering that honestly needs a specific econometric toolkit: a baseline forecast to beat, a loss function that ranks forecasts fairly even when the thing being forecast cannot be observed directly, a formal test of whether one forecast beats another, a design for testing a narrower causal claim, and a discipline for not fooling yourself after trying many variations. The five entries below are in the order §11.4 actually applies them.

A.24 The HAR model of realised volatility

The idea. The single best predictor of tomorrow's volatility is recent volatility, but "recent" is not one number: a trader who reacts to today's move, one who reacts to the week's, and one who reacts to the month's are all real and all price-relevant, and a forecast that only uses one horizon throws away what the others know. The HAR model's insight is that you do not need a sophisticated long-memory process to capture this — averaging realised variance over a few very different, simply chosen horizons and regressing on all of them at once reproduces most of the long-memory-like behaviour real volatility shows, with an ordinary linear regression.

Formally. With RV_t the realised variance on day t (the document's $\sigma^2_{r,t}$, annualised and dated),

$$RV_{t+1} = c + \beta_d\, RV_t + \beta_w\, \overline{RV}_{t-5,t} + \beta_m\, \overline{RV}_{t-22,t} + \varepsilon_{t+1},$$

where $\overline{RV}_{t-5,t}$ and $\overline{RV}_{t-22,t}$ are the averages of RV over the past 5 and 22 trading days (roughly a week and a month). All three coefficients are typically positive, reflecting persistence at every horizon simultaneously.

Why it appears here. §10.2: "The standard baseline is a regression of realised volatility on its own recent values at daily, weekly and monthly horizons — the **HAR** model of Corsi (2009), augmented with implied volatility such as the VIX," which §11.4 then makes the literal gate: "build the baseline you must beat."

Deeper. Corsi (2009).

A.25 The QLIKE loss, and robust forecast comparison

The idea. You cannot grade a volatility forecast against the truth, because the truth — the instantaneous variance actually driving the price at each moment — is never observed; you can only grade it against a *proxy*, such as variance estimated from a day's worth of intraday returns, which is itself noisy. An ordinary loss function like squared error can be fooled by that noise: a *worse* forecast can look better than a *better* one purely because of how the errors interact with the proxy's own measurement error. QLIKE is a loss function chosen specifically to avoid this trap — its ranking of two forecasts is provably preserved even when the proxy used to compute it is noisy, provided the noise is unbiased.

Formally. For a realised-variance proxy σ_r^2 and a forecast h of the same (both variances, not volatilities),

$$\mathrm{QLIKE}(\sigma_r^2, h) = \frac{\sigma_r^2}{h} - \ln\left(\frac{\sigma_r^2}{h}\right) - 1.$$

It is non-negative, zero only when $h = \sigma_r^2$, and asymmetric in a way that penalises under-prediction of large realised values more heavily than over-prediction — appropriate for a quantity, variance, that is itself non-negative and heavy-tailed.

Why it appears here. §11.4 Stage 5: "Compare volatility forecasts with a loss function that is robust to noise in the realised-volatility proxy, such as QLIKE (Patton, 2011)."

Deeper. Patton (2011).

A.26 The Diebold–Mariano test

The idea. Suppose one volatility forecast's average QLIKE loss (A.25) over a backtest is a little lower than another's. Is that a real, reliable difference, or could it easily have arisen from noise in the same sample? The Diebold–Mariano test answers exactly this, treating each period's *difference* in loss between the two forecasts as a time series in its own right and asking whether its mean is reliably different from zero — carefully accounting for the fact that forecast errors from overlapping or persistent volatility are themselves autocorrelated, so a naive standard error would understate the true uncertainty.

Formally. With loss differential $d_t = L(\sigma_{r,t}^2, h_{1,t}) - L(\sigma_{r,t}^2, h_{2,t})$ between two forecasts' losses at each $t = 1, \dots, T$,

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{avar}}(\bar{d})}} \xrightarrow{d} N(0,1) \quad \text{under } H_0 : \mathbb{E}[d_t] = 0,$$

where $\bar{d} = T^{-1}\sum_t d_t$ and the denominator is a heteroskedasticity- and autocorrelation-consistent standard error of the mean (not the document's market depth D, an unrelated use of a similar-looking letter). Rejecting H_0 says one forecast beats the other by more than sampling noise would produce.

Why it appears here. §11.4 Stage 5, immediately after QLIKE: "test the difference in accuracy formally (Diebold and Mariano, 1995)" — the gate a candidate gamma feature must clear against the HAR baseline (A.24).

Deeper. Diebold and Mariano (1995).

A.27 Event studies, and placebo tests

The idea. To test a claim like "volatility rises after large option expiries," it is not enough to show that volatility was high after expiries happened — volatility is high after all sorts of things, including nothing in particular. An event study isolates the claim by comparing an outcome around the events of interest against the same outcome around carefully matched non-events — dates that share every observable feature of a real event except the one mechanism under test. If the true events differ systematically from their placebos, the difference is evidence for the mechanism; if they do not, the original claim was picking up something else the calendar already explains.

Formally. Define an event date $\tau = 0$ for each occurrence (each monthly expiry, say), an outcome window such as realised volatility over $[\tau, \tau + 5]$ trading days, and a placebo set of pseudo-events sharing observable characteristics but lacking the mechanism (non-expiry Fridays, or the same dates shifted by a fixed offset). Compare the outcome's distribution across true events and placebos — typically a difference in means, or a regression with an event indicator and controls for known confounders such as day-of-week effects and the macro calendar — and attribute a significant gap to the mechanism unique to true events.

Why it appears here. §11.4's design for the expiry-volatility claim: "An event study around monthly expiries, comparing realised volatility before and after, split by the amount of dealer gamma that rolls off, with non-expiry Fridays as placebos and controls for the volatility term structure and the macroeconomic calendar."

Deeper. MacKinlay, A. C. (1997). "Event Studies in Economics and Finance." *Journal of Economic Literature* 35(1), 13–39 — the standard methodological survey.

A.28 Multiple testing, and data snooping

The idea. If you try twenty configurations of a trading rule against pure noise, at a significance threshold that should produce one false positive in twenty by chance, you should expect to *find* a "significant" result even though nothing real is there — and, worse, you will not be able to tell which one it is from the rest. Reporting only the best of many tried configurations, without saying how many were tried, silently swaps the honest question ("is this real?") for a biased one ("what is the largest number this noisy process happened to produce?"), and the gap between the two grows every time another configuration is added to the search.

Formally. If m independent trials (a lowercase m, distinct from the document's contract multiplier M) each test a true null at significance level α, the chance that *at least one* is falsely significant is $1-(1-\alpha)^m$, approaching 1 quickly as m grows. If each trial's statistic is, under the null, a draw from a standard normal, the *expected maximum* over m independent draws grows like $\sqrt{2\ln m}$ rather than staying at zero — so the benchmark "the best of m" must beat is this inflated quantity, not the single-trial null, and simple corrections (Bonferroni: require $p < \alpha/m$ for any one trial to count) or explicit deflation by $\sqrt{2\ln m}$ restore an honest test.

Why it appears here. §11.4 Stage 6: "It must survive every reasonable positioning convention, not just the one that worked… Count every convention, horizon and threshold you tried, and discount the best result for the number of trials" — the discipline this entry makes precise.

Deeper. White, H. (2000). "A Reality Check for Data Snooping." *Econometrica* 68(5), 1097–1126; Harvey, C. R., Liu, Y. & Zhu, H. (2016). "…and the Cross-Section of Expected Returns." *Review of Financial Studies* 29(1), 5–68 — the finance-specific treatment.

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